Algebraic Topology, Part I

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1 Homotopy, homotopy equivalence, and the spaces we work with

1.1 Homotopy and homotopy type

Fix the unit interval $I = [0, 1]$.

Definition 1.1. Homotopy and homotopy equivalence

Let X, Y be topological spaces.

1. Two continuous maps $f, g\colon X \to Y$ are **homotopic** if there exists a continuous map $H\colon X \times I \to Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$. In this case we write $f \simeq g$, and we call H a **homotopy** from f to g.
2. A map $f\colon X \to Y$ is **null-homotopic** if $f \simeq c$ for some constant map c.
3. Spaces X and Y are **homotopy equivalent** if there exist continuous maps $f\colon X \to Y$ and $g\colon Y \to X$ with $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. Such an f is called a **homotopy equivalence**, and we write $X \simeq Y$.
4. A space X is **contractible** if $X \simeq \{*\}$.

Proposition 1.2. Homotopy is an equivalence relation on maps

For fixed spaces X, Y, the relation $\simeq$ on $\mathrm{map}(X, Y)$ is reflexive, symmetric, and transitive.

Proof. Reflexivity is given by the constant homotopy $H(x, t) = f(x)$. Symmetry follows by reversing time: if H is a homotopy from f to g, then $H'(x, t) = H(x, 1 - t)$ is a homotopy from g to f. For transitivity, if H is a homotopy $f \simeq g$ and K is a homotopy $g \simeq h$, define

$$L(x, t) = \begin{cases} H(x, 2t) & 0 \le t \le \frac{1}{2}, \\ K(x, 2t - 1) & \frac{1}{2} \le t \le 1. \end{cases}$$

The two formulas agree at $t = \frac{1}{2}$ because $H(x, 1) = g(x) = K(x, 0)$, so L is continuous by the pasting lemma. Then L is a homotopy from f to h. □

Proposition 1.3. Homotopy respects composition

Let X, Y, Z be spaces.

1. If $f, g\colon X \to Y$ satisfy $f \simeq g$ and $u\colon Y \to Z$ is continuous, then $u \circ f \simeq u \circ g$.
2. If $f, g\colon X \to Y$ satisfy $f \simeq g$ and $v\colon W \to X$ is continuous, then $f \circ v \simeq g \circ v$.

Proof. If H is a homotopy from f to g, then $u \circ H\colon X \times I \to Z$ is a homotopy from $u \circ f$ to $u \circ g$. Similarly, $H \circ (v \times \mathrm{id}_I)\colon W \times I \to Y$ is a homotopy from $f \circ v$ to $g \circ v$. □

Proposition 1.4. Homotopy equivalence is an equivalence relation on spaces

The relation $\simeq$ on spaces is reflexive, symmetric, and transitive.

Proof. Reflexivity holds because id_X is a homotopy equivalence with inverse id_X. Symmetry is built into the definition. For transitivity, suppose $X \xrightarrow{f} Y \xrightarrow{h} Z$ are homotopy equivalences with homotopy inverses $g\colon Y \to X$ and $k\colon Z \to Y$. Then

$$(g \circ k) \circ (h \circ f) = g \circ (k \circ h) \circ f \simeq g \circ \mathrm{id}_Y \circ f = g \circ f \simeq \mathrm{id}_X,$$

and similarly

$$(h \circ f) \circ (g \circ k) = h \circ (f \circ g) \circ k \simeq h \circ \mathrm{id}_Y \circ k = h \circ k \simeq \mathrm{id}_Z.$$

So $h \circ f$ is a homotopy equivalence. □

Definition 1.5. Based maps and based homotopies

A **pointed space** is a pair (X, x_0) with $x_0 \in X$. A **based map** $f\colon (X, x_0) \to (Y, y_0)$ is a continuous map $f\colon X \to Y$ with $f(x_0) = y_0$.
Two based maps $f, g\colon (X, x_0) \to (Y, y_0)$ are **based-homotopic** if there exists a homotopy $H\colon X \times I \to Y$ from f to g such that $H(x_0, t) = y_0$ for all $t \in I$.

Basepoints enter naturally once one wants functorial invariants built from loops or spheres. Even for path-connected spaces, changing basepoints typically produces conjugate identifications rather than equal ones, so keeping track of basepoints is part of the structure.

1.2 Mapping cylinder and mapping cone as organizing tools

Definition 1.6. Mapping cylinder

Given a continuous map $f\colon X \to Y$, the **mapping cylinder** of f is the quotient space

$$M_f \;=\; (X \times I) \sqcup Y / \sim,$$

where $(x, 0) \sim f(x)$ for all $x \in X$. Let

$$i\colon Y \to M_f, \quad y \mapsto [y], \qquad j\colon X \to M_f, \quad x \mapsto [(x, 1)].$$

This construction is the pushout of $X \xrightarrow{f} Y$ and $X \xrightarrow{i_0} X \times I$ with $i_0(x) = (x, 0)$:

$$\begin{tikzcd} X \arrow[r, "i_0"] \arrow[d, "f"'] & X \times I \arrow[d] \\ Y \arrow[r, "i"'] & M_f \end{tikzcd}$$

Proposition 1.7. The target is a strong deformation retract of the mapping cylinder

There is a continuous map $r\colon M_f \to Y$ such that $r \circ i = \mathrm{id}_Y$. Moreover, $i \circ r \simeq \mathrm{id}_{M_f}$ through a homotopy that fixes $i(Y)$ pointwise. In particular, i and r are homotopy inverses, and $M_f \simeq Y$.

Proof. Define r on the disjoint union by

$$r(y) = y \quad \text{for } y \in Y, \qquad r(x, t) = f(x) \quad \text{for } (x, t) \in X \times I.$$

This respects the identification $(x, 0) \sim f(x)$, hence descends to a continuous map $r\colon M_f \to Y$ with $r \circ i = \mathrm{id}_Y$. Define a homotopy $H\colon M_f \times I \to M_f$ by

$$H([y], s) = [y] \quad \text{for } y \in Y, \qquad H([(x, t)], s) = [(x, (1 - s)t)] \quad \text{for } (x, t) \in X \times I.$$

At $s = 0$ this is the identity on M_f, and at $s = 1$ it collapses $X \times I$ onto the $t = 0$ end, which is identified with $i(Y)$. The formula keeps $[y] \in i(Y)$ fixed for all s, so this is a deformation of M_f onto $i(Y)$. Thus $i \circ r \simeq \mathrm{id}_{M_f}$ rel $i(Y)$. □

Proposition 1.8. Mapping cylinder factorization

With notation as above, the map f factors as

$$X \xrightarrow{j} M_f \xrightarrow{r} Y,$$

and $r \circ j = f$.

Proof. By definition, $r \circ j(x) = r([(x, 1)]) = f(x)$. □

Definition 1.9. Cone and mapping cone

The **(unreduced) cone** on a space X is

$$CX \;=\; (X \times I)/(X \times \{1\}),$$

where all points in $X \times \{1\}$ are identified to a single cone point. Given $f\colon X \to Y$, the **mapping cone** of f is

the quotient

$$C_f \;=\; M_f/(X \times \{1\}),$$

equivalently the pushout $C_f \cong Y \cup_f CX$.

The pushout description is often the cleanest way to see functoriality. It also matches the geometric picture that C_f is obtained by attaching a cone on X to Y along f.

Examples.
1. If $f = \mathrm{id}_X$, then C_f is homeomorphic to CX.
2. If $X = \varnothing$, then $M_f \cong Y$ and $C_f \cong Y$.
3. If f is the inclusion $S^{n-1} \hookrightarrow D^n$, then C_f is homeomorphic to S^n.

1.3 Two criteria for homotopy equivalence

Definition 1.10. Deformation retract

A subspace $A \subset X$ is a **strong deformation retract** of X if there exists a homotopy $H\colon X \times I \to X$ such that

$$H(x,0) = x, \qquad H(x,1) \in A, \qquad H(a,t) = a$$

for all $x \in X$, $a \in A$, and $t \in I$.

Proposition 1.11. Deformation retracts give homotopy equivalences

If A is a strong deformation retract of X, then the inclusion $A \hookrightarrow X$ is a homotopy equivalence.

Proof. Let $i\colon A \hookrightarrow X$ be the inclusion, and let $r\colon X \to A$ be the map $r(x) = H(x,1)$. Then $r \circ i = \mathrm{id}_A$ because $H(a,1) = a$ for $a \in A$. Also, the homotopy H itself shows $i \circ r \simeq \mathrm{id}_X$. □

The mapping cylinder provides a second criterion that reduces a general map to an inclusion.

Proposition 1.12. Two-out-of-three for homotopy equivalences

Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be continuous maps. If two among f, g, and $g \circ f$ are homotopy equivalences, then the third is a homotopy equivalence.

Proof. The argument is formal once one knows that being a homotopy equivalence means being invertible up to homotopy.
For instance, assume f and $g \circ f$ are homotopy equivalences. Choose $u\colon Y \to X$ and $v\colon Z \to X$ with $u \circ f \simeq \mathrm{id}_X$ and $f \circ u \simeq \mathrm{id}_Y$, and $v \circ (g \circ f) \simeq \mathrm{id}_X$ and $(g \circ f) \circ v \simeq \mathrm{id}_Z$. Define $w\colon Z \to Y$ by $w = f \circ v$. Then

$$g \circ w = g \circ f \circ v \simeq \mathrm{id}_Z.$$

Also

$$w \circ g = f \circ v \circ g \simeq f \circ u \simeq \mathrm{id}_Y,$$

where the middle homotopy uses $v \circ g \simeq u$, which follows from

$$v \circ g \circ f \simeq \mathrm{id}_X \simeq u \circ f$$

and postcomposition with a homotopy inverse of f. Thus g is a homotopy equivalence. The other cases are similar. □

Proposition 1.13. A mapping-cylinder criterion

Let $f\colon X \to Y$ be any map, and write $X \xrightarrow{j} M_f \xrightarrow{r} Y$ for the mapping cylinder factorization. Then f is a homotopy equivalence if and only if j is a homotopy equivalence.

Proof. By Proposition 1.7, the map r is a homotopy equivalence. Since $f = r \circ j$ by Proposition 1.8, two-out-of-three (Proposition 1.12) gives the claim. □

This criterion is useful because j is an inclusion into a space built functorially from f, and inclusions are exactly where extension and gluing statements tend to live.

1.4 Homotopy extension property as a working lemma

A recurring technical need is to extend a homotopy defined on a subspace to a homotopy on the ambient space, while keeping time 0 fixed.

Definition 1.14. Homotopy extension property

An inclusion $i\colon A \hookrightarrow X$ has the **homotopy extension property** if for every space Y, every map $F\colon X \to Y$, and every homotopy $H\colon A \times I \to Y$ satisfying $H(a,0) = F(i(a))$ for all $a \in A$, there exists a homotopy $\widetilde{H}\colon X \times I \to Y$ such that

$$\widetilde{H}(x,0) = F(x) \qquad \text{and} \qquad \widetilde{H}(i(a),t) = H(a,t)$$

for all $x \in X$, $a \in A$, and $t \in I$.

Remark. When an inclusion has the homotopy extension property, it behaves well with respect to forming quotients and attaching spaces. This is the reason it appears throughout constructions built from pushouts.

A standard reformulation of the homotopy extension property uses mapping cylinders.

Proposition 1.15. A universal test object for homotopy extension

Let $i\colon A \hookrightarrow X$ be an inclusion, and let

$$M_i = X \cup_i (A \times I)$$

be the mapping cylinder of i. Then i has the homotopy extension property if and only if the canonical map

$$j\colon M_i \to X \times I$$

that restricts to $x \mapsto (x,0)$ on X and to $(a,t) \mapsto (i(a),t)$ on $A \times I$ admits a retraction $r\colon X \times I \to M_i$ with $r \circ j = \mathrm{id}_{M_i}$.

Proof. Assume i has the homotopy extension property. Apply it to the diagram where $Y = M_i$, F is the inclusion $X \to M_i$, and H is the inclusion $A \times I \to M_i$. The resulting extension $\widetilde{H}\colon X \times I \to M_i$ is a retraction of j.

Conversely, assume such a retraction r exists. Given F and H as in Definition 1.14, the maps F and H glue to a single map $\Phi\colon M_i \to Y$ by the pushout property of M_i. Then $\widetilde{H} = \Phi \circ r\colon X \times I \to Y$ is the required extension. □

The mapping cone is designed so that it behaves well even when a naive quotient is inconvenient. When the inclusion has the homotopy extension property, the two constructions agree up to homotopy.

Proposition 1.16. When homotopy cofibers agree with quotients

Let $i\colon A \hookrightarrow X$ be an inclusion that has the homotopy extension property. Form the mapping cone

$$C_i = X \cup_i (A \times I)/(A \times \{1\}),$$

and let $q\colon X \to X/A$ be the quotient map. Then there is a natural map $\phi\colon X/A \to C_i$ induced by $X \to C_i$, and ϕ is a homotopy equivalence.

Proof. Consider the mapping cylinder $M_i = X \cup_i (A \times I)$ and the quotient map

$$\psi\colon C_i \longrightarrow C_i/CA \cong X/A,$$

where CA denotes the image of $A \times I$ with $A \times \{1\}$ collapsed. By Proposition 1.15, there is a retraction $r\colon X \times I \to M_i$ of the canonical inclusion $j\colon M_i \to X \times I$. Embed X into $X \times I$ as $X \times \{1\}$, compose with r, and then pass to the quotient that collapses $A \times \{1\}$. This produces a map $\phi\colon X/A \to C_i$.

The composite $\psi \circ \phi$ is homotopic to $\mathrm{id}_{X/A}$ by collapsing along the I-coordinate in $X \times I$. For the other composite, use the retraction r to deform C_i into the image of ϕ, keeping the cone direction in CA under control. Concretely, one combines the homotopy $j \circ r \simeq \mathrm{id}_{X \times I}$ with the fact that r restricts to the identity on $A \times I$. This yields a homotopy $\phi \circ \psi \simeq \mathrm{id}_{C_i}$. Therefore ϕ is a homotopy equivalence. □

The main source of inclusions with the homotopy extension property will be CW-type constructions. That will be proved when CW complexes are introduced. Until then, whenever an attachment or quotient is meant to have a controlled homotopy type, the homotopy extension property is the condition that makes the resulting statements

correct with minimal extra hypotheses.

2 CW complexes and cellular constructions

2.1 Cell attachments and relative CW pairs

Let $D^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ and $S^{n-1} = \partial D^n$.

Definition 2.1. Attaching an n-cell

Let X be a space and let $\varphi\colon S^{n-1} \to X$ be continuous. The **adjunction space** obtained by attaching an n-cell to X along φ is the quotient

$$X \cup_\varphi D^n \;=\; (X \sqcup D^n)/\sim,$$

where $x \sim x$ for $x \in X$ and $u \sim \varphi(u)$ for $u \in S^{n-1} \subset D^n$. The image of $D^n \setminus S^{n-1}$ in $X \cup_\varphi D^n$ is called the **open n-cell**.

It is convenient to keep track of the quotient map

$$q\colon X \sqcup D^n \to X \cup_\varphi D^n$$

and the restriction $q|_{D^n}\colon D^n \to X \cup_\varphi D^n$, called the **characteristic map** of the attached cell.

Proposition 2.2. A pushout description of cell attachment

There is a pushout square

$$\begin{tikzcd} S^{n-1} \arrow[r, hook] \arrow[d, "\varphi"'] & D^n \arrow[d] \\ X \arrow[r] & X \cup_\varphi D^n \end{tikzcd}$$

in the category of topological spaces. Equivalently, a map $F\colon X \cup_\varphi D^n \to Y$ is the same as a pair of maps $f\colon X \to Y$ and $g\colon D^n \to Y$ satisfying $g|_{S^{n-1}} = f \circ \varphi$.

Proof. The quotient $X \sqcup D^n \to X \cup_\varphi D^n$ identifies precisely the points that must be identified in the pushout. The universal property is immediate from the fact that any map out of the quotient is determined by its restrictions to the two pieces, and the restrictions must agree on the identified subset S^{n-1}. □

Definition 2.3. CW complexes and skeleta

A **CW complex** is a space X together with an increasing sequence of subspaces

$$X^0 \subset X^1 \subset X^2 \subset \cdots \subset X$$

such that:

1. X^0 is a discrete space.
2. For each $n \geq 1$, the space X^n is obtained from X^{n-1} by attaching a (possibly empty) family of n-cells. Equivalently, there exist attaching maps $\varphi_\alpha\colon S^{n-1} \to X^{n-1}$ and a pushout square

$$\begin{tikzcd} \coprod_\alpha S^{n-1} \arrow[r, hook] \arrow[d, "\coprod \varphi_\alpha"'] & \coprod_\alpha D^n \arrow[d] \\ X^{n-1} \arrow[r] & X^n. \end{tikzcd}$$

3. (**Closure-finite**) The image of each characteristic map $\chi_\alpha\colon D^n \to X^n$ meets only finitely many open cells in X.
4. (**Weak topology**) A subset $U \subset X$ is open if and only if $U \cap \chi_\alpha(D^n)$ is open in $\chi_\alpha(D^n)$ for every characteristic map χ_α of every cell.

The subspace X^n is the **n-skeleton** of X.

The closure-finite condition makes the filtration compatible with local finiteness properties of cells. The weak

topology says that continuity and openness can be checked cell by cell.

Definition 2.4. Relative CW complexes and CW pairs

A **relative CW complex** (X, A) is an inclusion $A \subset X$ together with a filtration

$$A = X^{-1} \subset X^0 \subset X^1 \subset \cdots \subset X$$

such that each X^n is obtained from X^{n-1} by attaching n-cells, and the closure-finite and weak topology conditions hold. A **CW pair** is an inclusion $A \subset X$ where X is a CW complex and A is a subcomplex in the sense of Definition 2.10.

Proposition 2.5. Basic properties of skeleta

Let X be a CW complex.

1. Each skeleton X^n is closed in X.
2. If $f\colon X \to Y$ is continuous, then f is determined by its restrictions $f|_{X^n}$, and f is continuous if and only if each restriction $f|_{\chi_\alpha(D^k)}$ is continuous for every cell characteristic map χ_α.

Proof. (1) Let $x \in X \setminus X^n$. Then x lies in the image of some open k-cell with $k > n$. Inside that k-cell there is an open neighborhood disjoint from X^n, so $X \setminus X^n$ is open.
(2) The first statement is formal. For the second, the weak topology in Definition 2.3 says that openness in X may be tested on each cell image, and the same test characterizes continuity. □

2.2 Examples and standard constructions

Definition 2.6. Cellular maps

Let X and Y be CW complexes. A continuous map $f\colon X \to Y$ is **cellular** if $f(X^n) \subset Y^n$ for all $n \geq 0$. More generally, for relative CW complexes (X, A) and (Y, B), a map $f\colon (X, A) \to (Y, B)$ is cellular if $f(A) \subset B$ and $f(X^n) \subset Y^n$ for all n.

Examples.

1. The sphere S^n is obtained from a point by attaching one n-cell, so it has a CW structure with one 0-cell and one n-cell.
2. The real projective space $\mathbb{R}P^n$ admits a CW structure with one cell in each dimension $0, 1, \ldots, n$, attached via the antipodal map on spheres.
3. The complex projective space $\mathbb{C}P^n$ admits a CW structure with one cell in each even dimension $0, 2, \ldots, 2n$.

Wedge sums, products, and cones are naturally expressed in the cellular language. Only the constructions used frequently later are recorded here.

Proposition 2.7. Wedges and cones are CW complexes

Let $\{X_i\}_{i \in I}$ be CW complexes, each with a chosen basepoint that is a 0-cell.

1. The wedge $\bigvee_{i \in I} X_i$ is a CW complex whose cells are the cells of the X_i away from the basepoints.
2. The cone CX of a CW complex X is a CW complex.
3. The suspension $\Sigma X = CX/X$ is a CW complex, and if X has a k-cell then ΣX has a $(k+1)$-cell arising from it.

Proof. (1) Start with the disjoint union $\bigsqcup_i X_i$ and identify all chosen 0-cells to a point. This identification happens in the 0-skeleton, and each higher skeleton is obtained by attaching the same cells as before, so the CW axioms are preserved.
(2) Give $X \times I$ the product cell structure where each cell of X is crossed with I. Collapsing $X \times \{1\}$ to a point identifies a subcomplex to a point, and the resulting quotient carries a CW structure.
(3) The quotient $\Sigma X = CX/X$ collapses the base $X \times \{0\}$ to a point. This again collapses a subcomplex to a

point and yields a CW structure. □

Proposition 2.8. Products of CW complexes

If X and Y are CW complexes, then $X \times Y$ admits a CW structure such that:

1. the n-skeleton $(X \times Y)^n$ is contained in $\bigcup_{p+q \le n} X^p \times Y^q$,
2. each cell of $X \times Y$ is contained in a product $e^p \times e^q$ of an open p-cell of X and an open q-cell of Y.

Proof. A standard construction uses a subdivision of $D^p \times D^q$ into cells compatible with the boundary decomposition, applied cellwise to characteristic maps. One obtains a filtration whose successive quotients are wedges of spheres, hence arise by attaching cells. Closure-finiteness follows from closure-finiteness in each factor, and the weak topology follows because a set in $X \times Y$ is open precisely when it is open on each product of cell images. □

Mapping cylinders and mapping cones from the previous section are compatible with CW structures once the map respects skeleta.

Proposition 2.9. Mapping cylinders and cones in the cellular setting

Let $f\colon X \to Y$ be a cellular map of CW complexes.

1. The mapping cylinder M_f admits a CW structure such that the inclusions $Y \hookrightarrow M_f$ and $X \hookrightarrow M_f$ are cellular, and $M_f \simeq Y$.
2. The mapping cone C_f admits a CW structure with $C_f \cong Y \cup_f CX$ as a cellular attachment.

Proof. Write $M_f = Y \cup_f (X \times I)$. Give $X \times I$ the product CW structure where $X^n \times I$ is the n-skeleton. Since f is cellular, the attachment along $X \times \{0\}$ respects skeleta, so the pushout description attaches cells in each dimension exactly as in $X \times I$ together with the cells of Y. The deformation retraction $M_f \simeq Y$ is the one from the mapping cylinder construction, and it respects the CW filtration because it collapses only along the I-direction.

For the cone, collapse $X \times \{1\}$ to a point inside M_f. This collapses a subcomplex, so the quotient inherits a CW structure, and the pushout description $Y \cup_f CX$ identifies the resulting filtration. □

2.3 Subcomplexes and a cofibration viewpoint

Definition 2.10. Subcomplex

Let X be a CW complex. A subset $A \subset X$ is a **subcomplex** if it is a union of open cells of X, equivalently if for every cell characteristic map $\chi\colon D^n \to X$, the preimage $\chi^{-1}(A)$ is either empty or a subcomplex of D^n containing S^{n-1} whenever it meets the boundary.

Proposition 2.11. Subcomplexes inherit CW structures

If $A \subset X$ is a subcomplex, then:

1. A is a CW complex with skeleta $A^n = A \cap X^n$,
2. A is closed in X,
3. each inclusion $A^n \hookrightarrow X^n$ is a closed inclusion.

Proof. Because A is a union of cells, its intersection with each skeleton X^n is obtained from A^{n-1} by attaching exactly those n-cells of X that lie in A. Closure-finiteness and the weak topology are inherited from X. The closedness follows from Proposition 2.5(1) applied to A and to skeleta differences. □

To control attachments up to homotopy, it is useful to record that inclusions of skeleta behave like the inclusions used in the homotopy extension property.

Definition 2.12. Cofibration

An inclusion $i\colon A \hookrightarrow X$ is a **cofibration** if it has the homotopy extension property from Definition 1.14.

Lemma 2.13. Cell attachments produce cofibrations Let X be a space and form $X' = X \cup_\varphi D^n$ by attaching an n-cell along $\varphi\colon S^{n-1} \to X$. Then the inclusion $X \hookrightarrow X'$ is a cofibration.

Proof. Consider the pushout square from Proposition 2.2. The inclusion $S^{n-1} \hookrightarrow D^n$ has the homotopy extension property because S^{n-1} is a deformation retract of a neighborhood in D^n. The pushout description allows extensions on D^n and on X to be glued uniquely. Concretely, given $F\colon X' \to Y$ and a homotopy $H\colon X \times I \to Y$ with $H(x,0) = F(x)$ for $x \in X$, extend the restriction $F|_{D^n}$ along the boundary homotopy $H \circ (\varphi \times \mathrm{id}_I)$ using the homotopy extension property for (D^n, S^{n-1}). The resulting homotopy on $D^n \times I$ glues with H along $S^{n-1} \times I$ to a homotopy on $X' \times I$ extending H. □

Proposition 2.14. Skeleta inclusions are cofibrations

If X is a CW complex, then for each $n \geq 0$ the inclusion $X^n \hookrightarrow X^{n+1}$ is a cofibration. If $A \subset X$ is a subcomplex, then $A \hookrightarrow X$ is a cofibration.

Proof. The inclusion $X^n \hookrightarrow X^{n+1}$ is obtained by attaching $(n+1)$-cells to X^n. By Lemma 2.13, each such attachment yields a cofibration, and a finite union of cells may be handled by iterating the homotopy extension property and gluing along pushouts. For a general CW complex, closure-finiteness allows one to localize to finitely many cells near a given compact subset and apply the same argument, then use the weak topology to glue the resulting extensions. For $A \hookrightarrow X$, build X from A by attaching the cells of X that are not in A skeletally. At each stage the inclusion is a cofibration, hence the composite is a cofibration. □

Proposition 2.15. Quotients by subcomplexes remain CW

Let X be a CW complex and let $A \subset X$ be a subcomplex. Then the quotient X/A is a CW complex whose cells are the cells of X not contained in A, together with a single 0-cell given by the image of A.

Proof. Collapse A to a point skeletally. At the 0-skeleton, all 0-cells of A are identified to a single 0-cell. Inductively, if X^n/A^n is a CW complex, then X^{n+1}/A^{n+1} is obtained from X^n/A^n by attaching the $(n+1)$-cells of X not in A, since the cells in A collapse into the basepoint. Closure-finiteness and the weak topology pass to the quotient because openness may be tested on cell images. □

2.4 Cellular approximation, statement and first consequences

The cellular filtration is compatible with homotopies in a strong sense.

Theorem 2.16. Cellular approximation

Let X and Y be CW complexes, and let $f\colon X \to Y$ be continuous. Then f is homotopic to a cellular map $f'\colon X \to Y$.
More generally, if $A \subset X$ is a subcomplex and $f|_A$ is cellular, then there exists a homotopy $H\colon X \times I \to Y$ such that $H(\cdot,0) = f$, $H(\cdot,1) = f'$, and $H(a,t) = f(a)$ for all $a \in A$ and $t \in I$.

The proof is deferred until the homotopy extension property is used systematically with skeleta. The first consequences already organize many constructions.

Corollary 2.17 (Maps into low-dimensional CW complexes depend on low skeleta)**.** Let X be a CW complex and let Y be an n-dimensional CW complex, meaning $Y = Y^n$. Then every map $f\colon X \to Y$ is homotopic to a map that factors through the n-skeleton X^n. In particular, if $g, h\colon X \to Y$ agree on X^n, then after replacing each by a cellular approximation they are equal as cellular maps.

Proof. By Theorem 2.16, choose a cellular map $f' \simeq f$. Since $Y = Y^n$, one has $f'(X) = f'(X^n)$ because $f'(X^k) \subset Y^k = Y^n$ for $k \geq n$ and X is the union of its skeleta. Thus f' factors through the inclusion $X^n \hookrightarrow X$. The final statement follows because a cellular map is determined by its restrictions to cells, hence by its restriction to X^n once the target has no cells above dimension n. □

Corollary 2.18 (Cellular maps are built skeletally)**.** Let X and Y be CW complexes, and let $f\colon X \to Y$ be cellular. Then for each n the restriction $f|_{X^n}\colon X^n \to Y^n$ determines $f|_{X^{n+1}}$ up to homotopy relative to X^n by specifying the maps on the characteristic disks of the $(n+1)$-cells.

Proof. Write X^{n+1} as a pushout of X^n with a disjoint union of $(n+1)$-disks along their boundaries. A cellular map sends X^n into Y^n and each $(n+1)$-cell into Y^{n+1}. By the universal property of the pushout, extending $f|_{X^n}$ to X^{n+1} is equivalent to choosing, for each $(n+1)$-cell, a map $D^{n+1} \to Y^{n+1}$ whose boundary agrees with the attaching map composed with $f|_{X^n}$. Two such extensions are homotopic relative to X^n if the corresponding disk maps are homotopic relative to their boundaries. □

The cellular approximation theorem makes CW complexes usable as a default setting. Maps and homotopies can be moved into the skeletal world, and attachments can be tracked through the pushout descriptions without leaving the category of spaces built from cells.

3 The fundamental group

3.1 Loops, concatenation, and functoriality

Fix a pointed space (X, x_0).

Definition 3.1. Paths, loops, and path homotopy

A **path** in X is a continuous map $\alpha\colon I \to X$. Its **initial point** and **terminal point** are $\alpha(0)$ and $\alpha(1)$. A **loop** based at x_0 is a path $\alpha\colon I \to X$ with $\alpha(0) = \alpha(1) = x_0$.
Two paths $\alpha, \beta\colon I \to X$ with the same endpoints are **path-homotopic rel endpoints** if there exists a continuous map $H\colon I \times I \to X$ such that

$$H(t,0) = \alpha(t), \qquad H(t,1) = \beta(t), \qquad H(0,s) = \alpha(0), \qquad H(1,s) = \alpha(1)$$

for all $t, s \in I$.

Lemma 3.2. Path homotopy rel endpoints is an equivalence relation For fixed endpoints $x, y \in X$, path-homotopy rel endpoints is an equivalence relation on the set of paths $\alpha\colon I \to X$ with $\alpha(0) = x$ and $\alpha(1) = y$.

Proof. Reflexivity and symmetry are proved exactly as in Proposition 1.2, keeping the boundary conditions at $t = 0$ and $t = 1$ fixed. For transitivity, paste two homotopies along $s = \frac{1}{2}$ and use the pasting lemma; the endpoint conditions persist because each homotopy fixes the endpoints for every s. □

Definition 3.3. Concatenation and inverses of paths

Let $\alpha, \beta\colon I \to X$ be paths with $\alpha(1) = \beta(0)$. Their **concatenation** $\alpha \cdot \beta$ is the path

$$(\alpha \cdot \beta)(t) = \begin{cases} \alpha(2t) & 0 \le t \le \frac{1}{2}, \\ \beta(2t-1) & \frac{1}{2} \le t \le 1. \end{cases}$$

If α is a path, its **reverse** α^{-1} is defined by $\alpha^{-1}(t) = \alpha(1-t)$.

Lemma 3.4. Concatenation respects path homotopy Suppose $\alpha \simeq \alpha'$ rel endpoints and $\beta \simeq \beta'$ rel endpoints, with $\alpha(1) = \beta(0)$. Then $\alpha \cdot \beta \simeq \alpha' \cdot \beta'$ rel endpoints.

Proof. Let H be a homotopy rel endpoints from α to α', and let K be one from β to β'. Define $L\colon I \times I \to X$ by

$$L(t,s) = \begin{cases} H(2t,s) & 0 \le t \le \frac{1}{2}, \\ K(2t-1,s) & \frac{1}{2} \le t \le 1. \end{cases}$$

At $t = \frac{1}{2}$ the two formulas agree because $H(1,s) = \alpha(1) = \beta(0) = K(0,s)$ for all s. Thus L is continuous by the pasting lemma. The endpoint conditions follow from those for H and K, so L is a path homotopy rel endpoints. □

Lemma 3.5. Reparametrization does not change the path class Let $\alpha\colon I \to X$ be a path and let $\rho\colon I \to I$ be continuous with $\rho(0) = 0$ and $\rho(1) = 1$. Then $\alpha \circ \rho \simeq \alpha$ rel endpoints.

Proof. Define $H\colon I \times I \to X$ by

$$H(t, s) = \alpha((1 - s)\rho(t) + st).$$

The map $(t, s) \mapsto (1 - s)\rho(t) + st$ is continuous and takes values in I, so H is continuous. At $s = 0$ it equals $\alpha \circ \rho$, and at $s = 1$ it equals α. Moreover $H(0, s) = \alpha(0)$ and $H(1, s) = \alpha(1)$ for all s. □

Lemma 3.6. Associativity holds up to path homotopy For composable paths α, β, γ one has

$$(\alpha \cdot \beta) \cdot \gamma \;\simeq\; \alpha \cdot (\beta \cdot \gamma)$$

rel endpoints.

Proof. Both paths traverse α, then β, then γ, differing only by the time reparametrization. There exists a piecewise linear homeomorphism $\rho\colon I \to I$ with $\rho(0) = 0$, $\rho(1) = 1$, such that

$$(\alpha \cdot (\beta \cdot \gamma)) = ((\alpha \cdot \beta) \cdot \gamma) \circ \rho.$$

For instance, choose ρ sending the breakpoints $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ of $((\alpha \cdot \beta) \cdot \gamma)$ to the breakpoints $\frac{1}{2}, \frac{3}{4}$ of $\alpha \cdot (\beta \cdot \gamma)$, and linear on each subinterval. Then apply Lemma 3.5. □

Lemma 3.7. Inverses hold up to path homotopy If α is a path from x to y, then $\alpha \cdot \alpha^{-1}$ is path-homotopic rel endpoints to the constant path at x, and $\alpha^{-1} \cdot \alpha$ is path-homotopic rel endpoints to the constant path at y.

Proof. Define $H\colon I \times I \to X$ by

$$H(t, s) = \begin{cases} \alpha((1 - s) \cdot 2t) & 0 \le t \le \frac{1}{2}, \\ \alpha((1 - s) \cdot (2 - 2t)) & \frac{1}{2} \le t \le 1. \end{cases}$$

At $s = 0$ this is $\alpha \cdot \alpha^{-1}$, while at $s = 1$ it is the constant path at $\alpha(0) = x$. The endpoints remain fixed for all s. The second statement is analogous. □

Definition 3.8. Fundamental group

The **fundamental group** $\pi_1(X, x_0)$ is the set of path-homotopy classes rel endpoints of loops $\alpha\colon I \to X$ based at x_0. The product is defined by

$$[\alpha] \cdot [\beta] \;=\; [\alpha \cdot \beta],$$

the identity is the constant loop at x_0, and the inverse of $[\alpha]$ is $[\alpha^{-1}]$.

Theorem 3.9. $\pi_1(X, x_0)$ is a group

With the operation in Definition 3.8, the set $\pi_1(X, x_0)$ is a group.

Proof. By Lemma 3.4, the product is well-defined on homotopy classes. Associativity follows from Lemma 3.6. The constant loop at x_0 is a two-sided identity by a reparametrization homotopy as in Lemma 3.5. Inverses are provided by Lemma 3.7. □

Definition 3.10. Induced homomorphism on π_1

Let $f\colon (X, x_0) \to (Y, y_0)$ be a based map, so $f(x_0) = y_0$. Define

$$f_*\colon \pi_1(X, x_0) \to \pi_1(Y, y_0), \qquad f_*([\alpha]) = [f \circ \alpha].$$

Proposition 3.11. Functoriality of π_1

For based maps $(X, x_0) \xrightarrow{f} (Y, y_0) \xrightarrow{g} (Z, z_0)$ one has

$$(g \circ f)_* = g_* \circ f_*, \qquad (\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X, x_0)}.$$

Moreover, each f_* is a group homomorphism.

Proof. The identities follow directly from Definition 3.10. To see f_* is a homomorphism, compute

$$f_*([\alpha]\cdot[\beta]) = f_*([\alpha\cdot\beta]) = [f\circ(\alpha\cdot\beta)] = [(f\circ\alpha)\cdot(f\circ\beta)] = [f\circ\alpha]\cdot[f\circ\beta]$$

using the definition of concatenation. □

Proposition 3.12. Based homotopies induce the same map on π_1

If $f, g\colon (X, x_0) \to (Y, y_0)$ are based maps and f is based-homotopic to g, then $f_* = g_*$.

Proof. Let $H\colon X \times I \to Y$ be a based homotopy from f to g. For a based loop $\alpha\colon I \to X$, define a map $\widetilde{H}\colon I \times I \to Y$ by $\widetilde{H}(t, s) = H(\alpha(t), s)$. Then $\widetilde{H}$ is a homotopy rel endpoints from $f \circ \alpha$ to $g \circ \alpha$ because $H(x_0, s) = y_0$. Hence $[f \circ \alpha] = [g \circ \alpha]$ in $\pi_1(Y, y_0)$. □

3.2 Basepoint change and conjugation

A based loop is a rigid object: the basepoint is part of the data. For a path-connected space, different basepoints yield isomorphic groups, but the identifications are not canonical.

Definition 3.13. Change of basepoint

Let $\gamma\colon I \to X$ be a path with $\gamma(0) = x_1$ and $\gamma(1) = x_0$. Define

$$c_\gamma\colon \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad c_\gamma([\alpha]) = [\gamma\cdot\alpha\cdot\gamma^{-1}],$$

where α is a loop based at x_0.

Proposition 3.14. Basepoint change is an isomorphism

For any path γ as in Definition 3.13, the map c_γ is a group isomorphism with inverse $c_{\gamma^{-1}}$. If γ and γ' are path-homotopic rel endpoints, then $c_\gamma = c_{\gamma'}$.

Proof. The inverse statement follows from the relations

$$\gamma^{-1}\cdot\gamma \simeq \mathrm{const}_{x_0}, \qquad \gamma\cdot\gamma^{-1} \simeq \mathrm{const}_{x_1},$$

which are instances of Lemma 3.7. Thus

$$c_{\gamma^{-1}}(c_\gamma([\alpha])) = [\gamma^{-1}\cdot(\gamma\cdot\alpha\cdot\gamma^{-1})\cdot\gamma] \simeq [\alpha],$$

and similarly $c_\gamma \circ c_{\gamma^{-1}} = \mathrm{id}$.
If $\gamma \simeq \gamma'$ rel endpoints, then $\gamma\cdot\alpha\cdot\gamma^{-1} \simeq \gamma'\cdot\alpha\cdot(\gamma')^{-1}$ rel endpoints by Lemma 3.4, so the induced classes coincide. □

Proposition 3.15. Different basepoint-change paths differ by conjugation

Let $\gamma, \gamma'\colon (I, \{0, 1\}) \to (X, \{x_1, x_0\})$ be two paths from x_1 to x_0. Set $\delta = \gamma'\cdot\gamma^{-1}$, which is a loop based at x_1. Then for all $[\alpha] \in \pi_1(X, x_0)$,

$$c_{\gamma'}([\alpha]) \;=\; [\delta]\cdot c_\gamma([\alpha])\cdot[\delta]^{-1}.$$

Proof. Compute

$$c_{\gamma'}([\alpha]) = [\gamma'\cdot\alpha\cdot(\gamma')^{-1}] = [(\gamma'\cdot\gamma^{-1})\cdot(\gamma\cdot\alpha\cdot\gamma^{-1})\cdot(\gamma\cdot(\gamma')^{-1})].$$

Since $\gamma\cdot(\gamma')^{-1} = (\gamma'\cdot\gamma^{-1})^{-1}$, the right-hand side is $[\delta]\cdot c_\gamma([\alpha])\cdot[\delta]^{-1}$. □

Remark. For a path-connected space X, one may speak of **the** fundamental group up to non-canonical isomorphism. For constructions that involve maps between spaces, the basepoint is part of the functorial data, and the ambiguity in Proposition 3.15 is exactly what one expects.

Free homotopy of loops (allowing the basepoint to move) naturally leads to conjugacy classes in π_1.

Proposition 3.16. Free homotopy classes correspond to conjugacy classes

Let X be path-connected and fix $x_0 \in X$. Two based loops α, β at x_0 are freely homotopic (as maps $S^1 \to X$) if and only if their classes in $\pi_1(X, x_0)$ are conjugate.

Proof. A free homotopy between α and β allows the basepoint to travel along a path γ traced by the image of a chosen point of S^1 during the homotopy. Interpreting the homotopy on a cylinder shows that β is homotopic rel endpoints to $\gamma \cdot \alpha \cdot \gamma^{-1}$. Conversely, a conjugation $\beta \simeq \gamma \cdot \alpha \cdot \gamma^{-1}$ can be realized by a homotopy on a cylinder that moves the basepoint along γ. □

3.3 First computations

Proposition 3.17. Homotopy invariance of π_1

If $f\colon (X, x_0) \to (Y, y_0)$ is a homotopy equivalence, then f_* is an isomorphism. In particular, if $X \simeq Y$ as pointed spaces, then $\pi_1(X) \cong \pi_1(Y)$ (after identifying basepoints via Definition 3.13 when needed).

Proof. Let $g\colon (Y, y_0) \to (X, x_0)$ be a homotopy inverse, so $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$ through based homotopies. Then Proposition 3.12 gives

$$g_* \circ f_* = (g \circ f)_* = (\mathrm{id}_X)_*, \qquad f_* \circ g_* = (f \circ g)_* = (\mathrm{id}_Y)_*,$$

so f_* is an isomorphism with inverse g_*. □

Corollary 3.18 (Contractible spaces have trivial fundamental group)**.** If X is contractible, then $\pi_1(X, x_0)$ is the trivial group for every $x_0 \in X$.

Proof. A contractible space is homotopy equivalent to a point, and $\pi_1(\{*\})$ is trivial. Apply Proposition 3.17. □

Proposition 3.19. Fundamental group of a product

Let (X, x_0) and (Y, y_0) be pointed spaces. Then the map

$$\pi_1(X \times Y, (x_0, y_0)) \longrightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0), \qquad [\alpha] \mapsto ([\mathrm{pr}_X \circ \alpha], [\mathrm{pr}_Y \circ \alpha])$$

is an isomorphism.

Proof. Given loops α in X and β in Y, the map $t \mapsto (\alpha(t), \beta(t))$ is a loop in $X \times Y$, defining a homomorphism in the opposite direction. The two constructions are inverse to each other on the nose. To see injectivity, note that a homotopy of loops in $X \times Y$ is the same as a pair of homotopies in X and Y after composing with the coordinate projections. □

Graphs provide a family of spaces where the cell structure is visible and deformations can be constructed explicitly.

Definition 3.20. Graphs as 1-dimensional CW complexes

A **(finite) graph** is a CW complex G with $G = G^1$ and with finitely many cells. Thus G is obtained from finitely many points by attaching finitely many 1-cells. A **tree** is a connected finite graph with no embedded circles.

Lemma 3.21. A finite tree deformation retracts to a point If T is a finite tree and v is a vertex of T, then T admits a strong deformation retraction onto $\{v\}$.

Proof. If T has a single vertex there is nothing to show. Otherwise T has a leaf vertex ℓ (a vertex meeting exactly one edge). Let $e \cong [0, 1]$ be the unique edge incident to ℓ, with ℓ corresponding to 1 and the other endpoint corresponding to 0. Collapsing e linearly onto $\{0\}$ while fixing $T \setminus (e \setminus \{0\})$ defines a strong deformation retraction from T onto the subgraph obtained by removing the open edge $e \setminus \{0\}$ and the vertex ℓ. Iterating this procedure finitely many times yields a strong deformation retraction onto any chosen vertex. □

Theorem 3.22. A connected finite graph is a wedge of circles up to homotopy

Let G be a connected finite graph with V vertices and E edges. Then G is homotopy equivalent to a wedge of $r = E - V + 1$ circles.

Proof. Choose a maximal tree $T \subset G$ containing all vertices. Then T is a subcomplex of G and is a finite tree. By Lemma 3.21, the inclusion $\{v\} \hookrightarrow T$ is a homotopy equivalence for any vertex v of T. Collapsing T inside G can be done by collapsing its edges one at a time, each collapse being a deformation retraction on a neighborhood of that edge and fixing the rest of the graph. Composing these deformation retractions produces a homotopy equivalence

$$G \simeq G/T,$$

where G/T is obtained by contracting the tree T to a point.
In the quotient G/T, every edge of T disappears, and each edge of G not in T becomes a loop attached at the single point $[T]$. Thus G/T is a wedge of as many circles as there are edges outside T. Since T has $V-1$ edges, the number of remaining edges is $E-(V-1) = E-V+1$. □

Remark. Once the fundamental group of a wedge of circles is identified, Theorem 3.22 turns computations on graphs into computations in free groups. The next lecture provides the necessary tool.

3.4 Induced maps and degree-one intuition

The functoriality of π_1 is often visible in concrete terms when the target is S^1.

Theorem 3.23. The fundamental group of the circle

For any basepoint $x_0 \in S^1$, there is an isomorphism

$$\pi_1(S^1, x_0) \cong \mathbb{Z}.$$

Remark. With an identification $\pi_1(S^1) \cong \mathbb{Z}$ fixed, every based map $f\colon (S^1, x_0) \to (S^1, x_0)$ induces multiplication by an integer:

$$f_*(1) = \deg(f) \in \mathbb{Z}.$$

The integer $\deg(f)$ depends only on the based homotopy class of f and satisfies $\deg(g \circ f) = \deg(g)\deg(f)$.

Proposition 3.24. Degree one forces an isomorphism on $\pi_1(S^1)$

Assume an identification $\pi_1(S^1, x_0) \cong \mathbb{Z}$ is fixed. If $f\colon (S^1, x_0) \to (S^1, x_0)$ satisfies $\deg(f) = \pm 1$, then f_* is an isomorphism.

Proof. Under $\pi_1(S^1, x_0) \cong \mathbb{Z}$, the induced map is multiplication by $\deg(f)$. Multiplication by ± 1 is an isomorphism of $\mathbb{Z}$. □

The circle will later serve as a basic target for detecting nontrivial loops. The same philosophy applies to many other spaces once their fundamental groups are understood via gluing methods.

4 van Kampen theorem and applications

4.1 Free products and amalgamated products as targets

Definition 4.1. Free product

Let G and H be groups. A **free product** of G and H is a group $G * H$ equipped with homomorphisms

$$\iota_G\colon G \to G * H, \qquad \iota_H\colon H \to G * H$$

such that for every group K and homomorphisms $f\colon G \to K$ and $g\colon H \to K$, there exists a unique homomorphism $\Phi\colon G * H \to K$ satisfying $\Phi \circ \iota_G = f$ and $\Phi \circ \iota_H = g$.

Proposition 4.2. Uniqueness of the free product

If $(G * H, \iota_G, \iota_H)$ and $(G * H, \iota'_G, \iota'_H)$ are free products of G and H, then there is a unique isomorphism $\theta\colon G * H \to G * H$ such that $\theta \circ \iota_G = \iota'_G$ and $\theta \circ \iota_H = \iota'_H$.

Proof. Apply the universal property of $(G * H, \iota_G, \iota_H)$ to the maps ι'_G, ι'_H to obtain a unique homomorphism $\theta\colon G * H \to G * H$ with the stated property. Similarly obtain $\theta'\colon G * H \to G * H$ with $\theta' \circ \iota'_G = \iota_G$ and $\theta' \circ \iota'_H = \iota_H$. Uniqueness in the universal property gives $\theta' \circ \theta = \mathrm{id}$ and $\theta \circ \theta' = \mathrm{id}$. □

There is a concrete model for $G * H$ in terms of reduced words that alternate between G and H, but the universal property is the only feature that will be used.

Definition 4.3. Amalgamated product

Let C, G, H be groups and let $i\colon C \to G$ and $j\colon C \to H$ be homomorphisms. An **amalgamated product** of G and H over C is a group $G *_C H$ together with homomorphisms

$$\iota_G\colon G \to G *_C H, \qquad \iota_H\colon H \to G *_C H$$

such that $\iota_G \circ i = \iota_H \circ j$, and such that for every group K and homomorphisms $f\colon G \to K$, $g\colon H \to K$ satisfying $f \circ i = g \circ j$, there exists a unique homomorphism $\Phi\colon G *_C H \to K$ with $\Phi \circ \iota_G = f$ and $\Phi \circ \iota_H = g$.

Proposition 4.4. Amalgamated products as a pushout

The diagram

$$\begin{tikzcd} C \arrow[r, "i"] \arrow[d, "j"'] & G \arrow[d, "\iota_G"] \\ H \arrow[r, "\iota_H"'] & G *_C H \end{tikzcd}$$

is a pushout in the category of groups. In particular, $G *_C H$ is determined up to unique isomorphism by this universal property.

Proof. This is a restatement of Definition 4.3. Uniqueness up to unique isomorphism is proved as in Proposition 4.2. □

A useful construction is the following quotient model.

Lemma 4.5. A quotient model for the amalgamated product Let N be the normal subgroup of $G * H$ generated by the set

$$\{\, \iota_G(i(c))\, \iota_H(j(c))^{-1} \;:\; c \in C \,\}.$$

Then $(G * H)/N$ satisfies the universal property of $G *_C H$.

Proof. Let $Q = (G * H)/N$ and let $\bar{\iota}_G, \bar{\iota}_H$ be the composites of ι_G, ι_H with the quotient map. By construction, $\bar{\iota}_G(i(c)) = \bar{\iota}_H(j(c))$ for all $c \in C$. Now let $f\colon G \to K$ and $g\colon H \to K$ satisfy $f \circ i = g \circ j$. By Definition 4.1, there is a unique $\Phi\colon G * H \to K$ with $\Phi \circ \iota_G = f$ and $\Phi \circ \iota_H = g$. For $c \in C$ one has $\Phi(\iota_G(i(c))\, \iota_H(j(c))^{-1}) = f(i(c))\, g(j(c))^{-1} = 1$. Hence $N \subset \ker(\Phi)$, so Φ factors uniquely through Q. This gives the desired universal property. □

4.2 van Kampen statement and proof in the classical form

Let X be a space and let $U, V \subset X$ be open subsets with $X = U \cup V$. Set $W = U \cap V$. Choose a basepoint $x_0 \in W$ and write $i_U\colon W \hookrightarrow U$, $i_V\colon W \hookrightarrow V$ for inclusions, as well as $j_U\colon U \hookrightarrow X$, $j_V\colon V \hookrightarrow X$.

Theorem 4.6. van Kampen

Assume U, V, and W are path-connected. Then the homomorphisms $(j_U)_*\colon \pi_1(U, x_0) \to \pi_1(X, x_0)$ and $(j_V)_*\colon \pi_1(V, x_0) \to \pi_1(X, x_0)$ satisfy

$$(j_U)_* \circ (i_U)_* \;=\; (j_V)_* \circ (i_V)_*.$$

Moreover, the induced map

$$\Phi\colon \pi_1(U, x_0) *_{\pi_1(W, x_0)} \pi_1(V, x_0) \longrightarrow \pi_1(X, x_0)$$

is an isomorphism.

Proof. The equality $(j_U)_* \circ (i_U)_* = (j_V)_* \circ (i_V)_*$ is functoriality from Proposition 3.11. By Proposition 4.4, there is a unique homomorphism Φ with

$$\Phi \circ \iota_U = (j_U)_*, \qquad \Phi \circ \iota_V = (j_V)_*,$$

where ι_U and ι_V denote the structure maps into the amalgamated product.
It remains to show Φ is bijective. Surjectivity is proved by cutting loops into pieces that lie in U or V.

Step 1. Decomposing a loop. Let $\alpha\colon I \to X$ be a loop based at x_0. Since $\alpha(I)$ is compact and $\{U, V\}$ is an open cover of $\alpha(I)$, there exists $\delta > 0$ such that for any $t, t' \in I$ with $|t - t'| < \delta$, the points $\alpha(t)$ and $\alpha(t')$ lie in a common member of $\{U, V\}$. Choose a subdivision $0 = t_0 < t_1 < \cdots < t_m = 1$ with $t_{k+1} - t_k < \delta$. Then each subpath $\alpha_k = \alpha|_{[t_k, t_{k+1}]}$ lies entirely in U or entirely in V.
For each intermediate point $p_k = \alpha(t_k)$ with $1 \le k \le m-1$, choose a path λ_k in W from x_0 to p_k. This is possible since W is path-connected. Define based loops β_k in U or V by

$$\beta_k \;=\; \lambda_k \cdot \alpha_k \cdot \lambda_{k+1}^{-1},$$

where λ_0 and λ_m are taken to be the constant path at x_0. Each β_k is a loop based at x_0 that lies in U if α_k lies in U, and lies in V if α_k lies in V.
Concatenating these loops, the product $\beta_0 \cdot \beta_1 \cdots \beta_{m-1}$ is path-homotopic rel endpoints to α. Indeed, in the concatenation the segment $\lambda_{k+1}^{-1} \cdot \lambda_{k+1}$ appears between successive factors and is path-homotopic rel endpoints to the constant path by Lemma 3.7. Using Lemma 3.4 and Lemma 3.6, the class $[\alpha] \in \pi_1(X, x_0)$ equals the product of the classes $[j_U \circ \beta_k]$ or $[j_V \circ \beta_k]$. Hence $(j_U)_*(\pi_1(U, x_0))$ and $(j_V)_*(\pi_1(V, x_0))$ generate $\pi_1(X, x_0)$, so Φ is surjective.

Step 2. Constructing an inverse map. Define a map

$$\Psi\colon \pi_1(X, x_0) \to \pi_1(U, x_0) *_{\pi_1(W, x_0)} \pi_1(V, x_0)$$

as follows. Given $[\alpha] \in \pi_1(X, x_0)$, choose a subdivision and connecting paths λ_k as in Step 1, producing loops β_k in U or V. Set

$$\Psi([\alpha]) \;=\; \iota_U([\beta_{k_1}])\, \iota_V([\beta_{k_2}]) \cdots$$

in the amalgamated product, reading the factors in order and using ι_U or ι_V according to where β_k lives. If two consecutive pieces lie in the same set, combine them using the multiplication in that group.
This prescription depends on choices, so it must be checked that the resulting element is well-defined. There are three sources of choices.

(a) Refining the subdivision. If an interval is subdivided further, one replaces a factor $[\beta_k]$ by a product of two factors in the same group. This does not change the value in the amalgamated product.

(b) Changing the connecting paths in W. If λ_k is replaced by another path λ'_k, then $\lambda'_k \cdot \lambda_k^{-1}$ is a loop in W based at x_0. The corresponding loops β_{k-1} and β_k are modified by multiplying on the right and on the left by the image of this loop. These modifications cancel in the amalgamated product because the images of $\pi_1(W, x_0)$ in $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$ are identified there.

(c) Replacing α by a path-homotopic loop. Suppose α and α' are path-homotopic rel endpoints in X via $H\colon I \times I \to X$. Choose a subdivision of $I \times I$ into small squares so that the image of each square lies in U or lies in V. Such a subdivision exists by compactness of $I \times I$ and the openness of U and V. Reading H along the boundary of each square gives a relation in $\pi_1(U, x_0)$ or in $\pi_1(V, x_0)$, and when the image of the square meets both U and V its boundary lies in W. In the amalgamated product these boundary relations become trivial because the corresponding element from W is identified in both factors. Pasting the squares back together shows that $\Psi([\alpha]) = \Psi([\alpha'])$.
Therefore Ψ is well-defined. It is also a homomorphism because concatenation of loops corresponds to concatenation of the resulting words, up to the same relations used above.

Step 3. Verifying Φ and Ψ are inverses. By construction, for a loop β contained in U one has $\Psi([j_U \circ \beta]) = \iota_U([\beta])$, and similarly for loops in V. Hence $\Psi \circ (j_U)_* = \iota_U$ and $\Psi \circ (j_V)_* = \iota_V$. By the universal property of the amalgamated product, this implies $\Psi \circ \Phi = \mathrm{id}$.
On the other hand, given $[\alpha] \in \pi_1(X, x_0)$, Step 1 expresses $[\alpha]$ as a product of elements from $(j_U)_*(\pi_1(U, x_0))$ and $(j_V)_*(\pi_1(V, x_0))$. Applying Ψ and then Φ returns the same product in $\pi_1(X, x_0)$, hence $\Phi \circ \Psi = \mathrm{id}$. Thus Φ is an isomorphism. $\square$

4.3 Applications to CW complexes and graphs

Corollary 4.7 (Wedge sums and free products)**.** Let (X, x_0) and (Y, y_0) be pointed spaces. Assume X and Y are path-connected and admit open neighborhoods of their basepoints that deformation retract onto the basepoint. Then

$$\pi_1(X \vee Y, *) \;\cong\; \pi_1(X, x_0) * \pi_1(Y, y_0),$$

where $X \vee Y$ denotes the wedge obtained by identifying x_0 with y_0.

Proof. Write $X \vee Y$ as the union of U and V where U is X together with a small neighborhood of the wedge point in Y, and V is Y together with a small neighborhood of the wedge point in X. Then $U \simeq X$ and $V \simeq Y$ by deformation retraction, while $W = U \cap V$ deformation retracts to the wedge point and is path-connected. Apply Theorem 4.6 with $\pi_1(W, *)$ trivial. □

Corollary 4.8 (A wedge of circles has a free fundamental group)**.** Let $\bigvee_{k=1}^{n} S^1$ be the wedge of n circles. Then $\pi_1(\bigvee_{k=1}^{n} S^1, *)$ is the free group on n generators.

Proof. Apply Corollary 4.7 inductively and use Theorem 3.23. □

Graphs admit an efficient description using van Kampen together with the cellular viewpoint from the previous lecture.

Proposition 4.9. Attaching a 1-cell adds a free generator

Let X be path-connected and let $X' = X \cup_\varphi D^1$ be obtained by attaching a 1-cell. Assume the endpoints of D^1 attach to points of X that lie in the same path-component of X. Then $\pi_1(X')$ is isomorphic to a free product of $\pi_1(X)$ with a free group of rank 0 or 1. More precisely, if the endpoints attach to the same point, then $\pi_1(X') \cong \pi_1(X) * \mathbb{Z}$.

Proof. Let e be the open 1-cell. Choose a small open neighborhood N of e in X' that deformation retracts onto e together with small arcs in X near the endpoints. Let $U = X \cup N$ and let V be an open neighborhood of e that deformation retracts onto a circle if the endpoints coincide, or onto an interval if they do not. Then $W = U \cap V$ deformation retracts onto a union of arcs in X, hence is path-connected and has trivial fundamental group. Apply Theorem 4.6 and use that $\pi_1(V)$ is either trivial or $\mathbb{Z}$. □

Theorem 4.10. A connected finite graph has a free fundamental group

Let G be a connected finite graph with V vertices and E edges. Then $\pi_1(G)$ is a free group of rank $E - V + 1$.

Proof. Choose a maximal tree $T \subset G$ containing all vertices. By Lemma 3.21, T is contractible, so $\pi_1(T)$ is trivial. Build G from T by attaching the $E - (V - 1)$ edges not in T, one at a time. Each attachment is a 1-cell whose endpoints lie in T. At each step, Proposition 4.9 shows that π_1 changes by free product with $\mathbb{Z}$. Hence $\pi_1(G)$ is free of rank $E - (V - 1) = E - V + 1$. □

The same method extends to 2-dimensional CW complexes and produces presentations.

Proposition 4.11. A presentation from a 2-dimensional CW complex

Let X be a connected CW complex with $X = X^2$. Choose a maximal tree T in the 1-skeleton X^1 and collapse it to a point, forming X^1/T. Let $e_1, \ldots, e_m$ be the 1-cells of X not in T, viewed as loops in X^1/T. Then $\pi_1(X)$ admits a presentation with generators $x_1, \ldots, x_m$ corresponding to these 1-cells and relations given by the attaching maps of the 2-cells written as words in the x_i.

Proof. By Theorem 4.10 applied to the graph X^1, collapsing the maximal tree does not change π_1 and identifies $\pi_1(X^1) \cong F(x_1, \ldots, x_m)$ as a free group on the non-tree edges. Now attach the 2-cells one at a time. Attaching

a 2-cell means forming a pushout

$$\begin{tikzcd} S^1 \arrow[r, hook] \arrow[d, "\varphi"'] & D^2 \arrow[d] \\ X^1 \arrow[r] & X^1 \cup_\varphi D^2 \end{tikzcd}$$

with $\varphi\colon S^1 \to X^1$ the attaching map. Take an open cover by $U = X^1 \cup N$ where N is a small neighborhood of $\varphi(S^1)$, and V an open neighborhood of the attached 2-cell that deformation retracts onto S^1. Then $W = U \cap V$ deformation retracts onto a neighborhood of $\varphi(S^1)$, so the map on π_1 induced by $W \to U$ identifies $\pi_1(W)$ with the cyclic subgroup generated by $[\varphi]$ in $\pi_1(X^1)$. Applying Theorem 4.6 shows that attaching the 2-cell forces $[\varphi]$ to become trivial in π_1. Iterating over the 2-cells gives the stated presentation. □

4.4 A brief groupoid upgrade when it saves pain

The hypothesis that $W = U \cap V$ is path-connected is convenient, but it is not always satisfied. There is a version that removes this hypothesis by replacing groups with groupoids.

Definition 4.12. Fundamental groupoid

Let X be a space. The **fundamental groupoid** $\Pi_1(X)$ has as objects the points of X. A morphism from x to y is a path-homotopy class rel endpoints of paths $\alpha\colon I \to X$ with $\alpha(0) = x$ and $\alpha(1) = y$. Composition is induced by concatenation of paths.

For a fixed object x_0, the automorphism group $\mathrm{Aut}_{\Pi_1(X)}(x_0)$ is $\pi_1(X, x_0)$.

Theorem 4.13. van Kampen for groupoids

Let $X = U \cup V$ with U and V open. Let $A \subset U \cap V$ be a set that meets every path-component of $U \cap V$. Then the diagram of groupoids

$$\begin{tikzcd} \Pi_1(U \cap V)\big|_A \arrow[r] \arrow[d] & \Pi_1(U)\big|_A \arrow[d] \\ \Pi_1(V)\big|_A \arrow[r] & \Pi_1(X)\big|_A \end{tikzcd}$$

is a pushout. Here $\Pi_1(-)\big|_A$ denotes the full subgroupoid on objects A.

Remark. When $U \cap V$ is path-connected, one may take $A = \{x_0\}$ and recover Theorem 4.6. When $U \cap V$ has several components, choosing one basepoint in each component allows the same gluing method to proceed without forcing artificial paths between components.

5 Covering spaces: lifting properties

5.1 Definition and basic examples

Definition 5.1. Covering map and evenly covered neighborhoods

A continuous map $p\colon Y \to X$ is a **covering map** (and Y is a **covering space** of X) if p is surjective and for every $x \in X$ there exists an open neighborhood $U \subset X$ of x such that

$$p^{-1}(U) = \bigsqcup_{\alpha \in A} U_\alpha$$

is a disjoint union of open sets in Y, and for each α the restriction

$$p|_{U_\alpha}\colon U_\alpha \longrightarrow U$$

is a homeomorphism. Such a U is called **evenly covered**, and the sets U_α are called the **sheets** over U.

Definition 5.2. Trivial covering

Let X be a space and let F be a discrete space. The projection

$$\pi\colon X \times F \longrightarrow X$$

is a covering map, called the **trivial covering** with fiber F.

Definition 5.3. Lifts

Let $p\colon Y \to X$ be a map and let $f\colon Z \to X$ be a map. A **lift** of f along p is a map $\widetilde{f}\colon Z \to Y$ such that $p \circ \widetilde{f} = f$. If $Z = I$ and $f = \gamma$ is a path, then a lift $\widetilde{\gamma}$ is called a **lift of the path**. If $Z = I \times I$ and $f = H$ is a homotopy, then a lift $\widetilde{H}$ is called a **lift of the homotopy**.

Examples.

1. The exponential map $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, is a covering map. More generally, $S^1 \to S^1$, $z \mapsto z^n$, is an n-sheeted covering for $n \geq 1$.
2. The quotient map $S^n \to \mathbb{R}P^n$ is a 2-sheeted covering.
3. If G is a connected graph (a 1-dimensional CW complex), then a covering of graphs in the usual combinatorial sense is a covering map of spaces.

5.2 Unique path lifting and homotopy lifting

The key feature of covering maps is that paths and homotopies lift uniquely once a starting point is chosen.

Lemma 5.4. Path lifting for a trivial covering Let $\pi\colon X \times F \to X$ be a trivial covering with F discrete. Let $\gamma\colon I \to X$ be a path with $\gamma(0) = x_0$, and choose $(x_0, f_0) \in X \times F$. Then there exists a unique lift $\widetilde{\gamma}\colon I \to X \times F$ with $\widetilde{\gamma}(0) = (x_0, f_0)$.

Proof. Define $\widetilde{\gamma}(t) = (\gamma(t), f_0)$. This is continuous and satisfies $\pi \circ \widetilde{\gamma} = \gamma$ and $\widetilde{\gamma}(0) = (x_0, f_0)$.
For uniqueness, let $\eta\colon I \to X \times F$ be another lift with $\eta(0) = (x_0, f_0)$. Write $\eta(t) = (\gamma(t), \sigma(t))$ where $\sigma = \mathrm{pr}_F \circ \eta\colon I \to F$. Since I is connected and F is discrete, σ is constant, hence $\sigma(t) = f_0$ for all t. Therefore $\eta = \widetilde{\gamma}$. □

Theorem 5.5. Unique path lifting

Let $p\colon Y \to X$ be a covering map. Let $\gamma\colon I \to X$ be a path with $\gamma(0) = x_0$, and choose $y_0 \in p^{-1}(x_0)$. Then there exists a unique lift $\widetilde{\gamma}\colon I \to Y$ such that

$$p \circ \widetilde{\gamma} = \gamma, \qquad \widetilde{\gamma}(0) = y_0.$$

Proof. *Uniqueness.* Suppose $\widetilde{\gamma}$ and η are lifts of γ with $\widetilde{\gamma}(0) = \eta(0) = y_0$. Let

$$S = \{\, t \in I : \widetilde{\gamma}(t) = \eta(t) \,\}.$$

Then S is closed by continuity. To see S is open, fix $t_0 \in S$ and set $y = \widetilde{\gamma}(t_0) = \eta(t_0)$ and $x = \gamma(t_0) = p(y)$. Choose an evenly covered neighborhood U of x, and let U_α be the sheet containing y. Since γ is continuous, there exists an interval J around t_0 with $\gamma(J) \subset U$. For $t \in J$, both $\widetilde{\gamma}(t)$ and $\eta(t)$ lie in $p^{-1}(U)$ and map to the same point $\gamma(t) \in U$. Because $p|_{U_\alpha}\colon U_\alpha \to U$ is a homeomorphism and both lifts pass through $y \in U_\alpha$ at t_0, they must remain in U_α on J. Hence $\widetilde{\gamma}(t) = \eta(t)$ for $t \in J$, so $J \subset S$. Thus S is open, and since I is connected and $0 \in S$, one has $S = I$.
Existence. For each $t \in I$, choose an evenly covered neighborhood U_t of $\gamma(t)$. The sets $\gamma^{-1}(U_t)$ form an open cover of I, so by compactness there exist $t_1, \ldots, t_m$ with

$$I = \bigcup_{k=1}^{m} \gamma^{-1}(U_{t_k}).$$

Let $\delta > 0$ be a Lebesgue number for this cover: every subinterval of length $< \delta$ is contained in some $\gamma^{-1}(U_{t_k})$. Choose a subdivision $0 = s_0 < s_1 < \cdots < s_N = 1$ with $s_{j+1} - s_j < \delta$. Then for each j there exists an evenly covered U_j such that $\gamma([s_j, s_{j+1}]) \subset U_j$.

We now build the lift on each subinterval. Assume inductively that we have constructed a lift $\widetilde{\gamma}$ on $[0, s_j]$ with $\widetilde{\gamma}(0) = y_0$. Let $y_j = \widetilde{\gamma}(s_j)$ and $x_j = \gamma(s_j)$. Choose the unique sheet $U_{j,\alpha} \subset p^{-1}(U_j)$ containing y_j. Since $p|_{U_{j,\alpha}} \colon U_{j,\alpha} \to U_j$ is a homeomorphism and $\gamma([s_j, s_{j+1}]) \subset U_j$, define

$$\widetilde{\gamma}(t) \;=\; (p|_{U_{j,\alpha}})^{-1}(\gamma(t)) \qquad \text{for } t \in [s_j, s_{j+1}].$$

This is continuous on $[s_j, s_{j+1}]$ and matches the previously defined lift at s_j. Gluing the finitely many pieces gives a continuous lift on all of I. Uniqueness has already been proved, so the lift is the unique one with the given starting point. □

Theorem 5.6. Homotopy lifting

Let $p \colon Y \to X$ be a covering map. Let $H \colon I \times I \to X$ be a homotopy and choose $y_{00} \in p^{-1}(H(0,0))$. Then there exists a unique lift $\widetilde{H} \colon I \times I \to Y$ such that

$$p \circ \widetilde{H} = H, \qquad \widetilde{H}(0,0) = y_{00}.$$

In particular, if $H(0,s)$ is constant in s, then $\widetilde{H}(0,s)$ is constant in s.

Proof. Cover $I \times I$ by open sets of the form $H^{-1}(U)$ where U ranges over evenly covered neighborhoods in X. By compactness, extract a finite subcover. Choose $\delta > 0$ such that every rectangle of the form

$$[a, a+\delta] \times [b, b+\delta] \subset I \times I$$

is contained in one of these inverse images. Subdivide I into $0 = s_0 < s_1 < \cdots < s_N = 1$ with $s_{j+1} - s_j < \delta$ and likewise $0 = t_0 < t_1 < \cdots < t_M = 1$ with $t_{k+1} - t_k < \delta$. Then each small rectangle

$$R_{jk} = [s_j, s_{j+1}] \times [t_k, t_{k+1}]$$

satisfies $H(R_{jk}) \subset U_{jk}$ for some evenly covered $U_{jk} \subset X$.

We now construct $\widetilde{H}$ by induction over the grid. Start at $(0,0)$ with the chosen y_{00}. Assume $\widetilde{H}$ has been constructed continuously on the union of rectangles in the region

$$\bigcup_{j'<j} \bigcup_{k} R_{j'k} \quad \cup \quad \bigcup_{k'<k} R_{jk'}.$$

In particular, $\widetilde{H}$ is already defined on the left and bottom edges of R_{jk}. Pick a point $(s_j, t_k) \in R_{jk}$; its image under $\widetilde{H}$ has already been determined. Let $U_{jk,\alpha} \subset p^{-1}(U_{jk})$ be the sheet containing that point. Since $p|_{U_{jk,\alpha}} \colon U_{jk,\alpha} \to U_{jk}$ is a homeomorphism and $H(R_{jk}) \subset U_{jk}$, define

$$\widetilde{H}(u,v) \;=\; (p|_{U_{jk,\alpha}})^{-1}(H(u,v)) \qquad \text{for } (u,v) \in R_{jk}.$$

This is continuous on R_{jk} and agrees with the existing lift on the shared edges, because on overlaps the lift is forced by uniqueness of path lifting applied along line segments inside the rectangle. Proceeding across the grid defines a continuous lift on all of $I \times I$.

Uniqueness follows by the same connectedness argument as in Theorem 5.5: if two lifts agree at one point, the set of points where they agree is both open and closed in $I \times I$, hence all of it. The final statement is immediate: if $H(0,s)$ is constant, then $\widetilde{H}(0,s)$ is a lift of a constant path starting at y_{00}, hence constant by uniqueness. □

Corollary 5.7 (Injectivity on π_1)**.** Let $p \colon (Y, y_0) \to (X, x_0)$ be a covering map of pointed spaces. Then the induced homomorphism

$$p_* \colon \pi_1(Y, y_0) \longrightarrow \pi_1(X, x_0)$$

is injective.

Proof. Suppose $[\alpha] \in \pi_1(Y, y_0)$ satisfies $p_*([\alpha]) = 1$. Then $p \circ \alpha$ is null-homotopic rel basepoint, so there exists $H \colon I \times I \to X$ with

$$H(t,0) = p(\alpha(t)), \qquad H(t,1) = x_0, \qquad H(0,s) = H(1,s) = x_0.$$

Lift H to $\widetilde{H} \colon I \times I \to Y$ with $\widetilde{H}(0,0) = y_0$ using Theorem 5.6. Then the path $t \mapsto \widetilde{H}(t,0)$ is a lift of $p \circ \alpha$ starting at y_0, hence equals α by uniqueness of path lifting. Also, the path $t \mapsto \widetilde{H}(t,1)$ is a lift of the constant loop at x_0 starting at y_0, hence is the constant loop at y_0. Therefore α is homotopic rel endpoints to the constant loop, so $[\alpha] = 1$ in $\pi_1(Y, y_0)$. □

5.3 Coverings as local homeomorphisms plus discreteness

Proposition 5.8. A covering map is a local homeomorphism

If $p\colon Y \to X$ is a covering map, then for every $y \in Y$ there exists an open neighborhood V of y such that $p|_V\colon V \to p(V)$ is a homeomorphism onto an open subset of X.

Proof. Let $x = p(y)$ and choose an evenly covered neighborhood U of x. Let U_α be the sheet containing y. Then U_α is open in Y, $p(U_\alpha) = U$ is open in X, and $p|_{U_\alpha}\colon U_\alpha \to U$ is a homeomorphism. Taking $V = U_\alpha$ gives the claim. □

Proposition 5.9. Fibers of a covering are discrete

If $p\colon Y \to X$ is a covering map and $x \in X$, then the fiber $p^{-1}(x)$ is a discrete subspace of Y.

Proof. Choose an evenly covered neighborhood U of x and write $p^{-1}(U) = \bigsqcup_\alpha U_\alpha$. For each $y \in p^{-1}(x)$, the point y lies in a unique sheet U_α, and within U_α the restriction $p|_{U_\alpha}$ is injective. Hence $U_\alpha \cap p^{-1}(x) = \{y\}$. Since U_α is open in Y, this shows $\{y\}$ is open in the subspace topology on the fiber, so the fiber is discrete. □

Lemma 5.10. Sheets are the components over evenly covered sets under local path connectedness Let $p\colon Y \to X$ be a covering map. Assume $U \subset X$ is evenly covered and is path-connected. Then each sheet $U_\alpha \subset p^{-1}(U)$ is a path-component of $p^{-1}(U)$, and $p|_{U_\alpha}\colon U_\alpha \to U$ is a homeomorphism.

Proof. Each U_α is open by definition and maps homeomorphically onto the path-connected space U, so U_α is path-connected. If a path in $p^{-1}(U)$ starts in U_α, its image under p lies in U, and by uniqueness of path lifting the path cannot jump between sheets. Hence the path remains in U_α, so U_α is a path-component. □

5.4 Monodromy intuition

Path lifting produces transport maps between fibers.

Definition 5.11. Transport along a path

Let $p\colon Y \to X$ be a covering map. Let $\gamma\colon I \to X$ be a path with $\gamma(0) = x_0$ and $\gamma(1) = x_1$. For $y \in p^{-1}(x_0)$, let $\widetilde{\gamma}_y$ be the unique lift of γ with $\widetilde{\gamma}_y(0) = y$. Define the **transport map** along γ by

$$T_\gamma\colon p^{-1}(x_0) \longrightarrow p^{-1}(x_1), \qquad T_\gamma(y) = \widetilde{\gamma}_y(1).$$

Proposition 5.12. Transport depends only on the homotopy class of the path

If γ and γ' are homotopic rel endpoints, then $T_\gamma = T_{\gamma'}$. Moreover, if $\gamma\colon x_0 \to x_1$ and $\eta\colon x_1 \to x_2$, then

$$T_{\gamma\cdot\eta} \;=\; T_\eta \circ T_\gamma.$$

Proof. Let $H\colon I \times I \to X$ be a homotopy rel endpoints from γ to γ'. Fix $y \in p^{-1}(x_0)$ and lift H to $\widetilde{H}$ with $\widetilde{H}(0,0) = y$ using Theorem 5.6. Then $t \mapsto \widetilde{H}(t,0)$ is the lift of γ starting at y, while $t \mapsto \widetilde{H}(t,1)$ is the lift of γ' starting at the same y. Since the endpoints $t = 1$ are joined by the path $s \mapsto \widetilde{H}(1,s)$ in the fiber over x_1, and the fiber is discrete by Proposition 5.9, this path must be constant. Hence $\widetilde{H}(1,0) = \widetilde{H}(1,1)$, which is exactly $T_\gamma(y) = T_{\gamma'}(y)$.

For the composition law, fix $y \in p^{-1}(x_0)$. Let $\widetilde{\gamma}_y$ be the lift of γ starting at y, and set $y_1 = \widetilde{\gamma}_y(1) = T_\gamma(y) \in p^{-1}(x_1)$. Let $\widetilde{\eta}_{y_1}$ be the lift of η starting at y_1. Then the concatenation $\widetilde{\gamma}_y \cdot \widetilde{\eta}_{y_1}$ is a lift of $\gamma \cdot \eta$ starting at y. By uniqueness of path lifting,

$$T_{\gamma\cdot\eta}(y) = (\widetilde{\gamma}_y \cdot \widetilde{\eta}_{y_1})(1) = \widetilde{\eta}_{y_1}(1) = T_\eta(T_\gamma(y)).$$

□

Definition 5.13. Monodromy action

Let $p\colon Y \to X$ be a covering map and fix a basepoint $x_0 \in X$. For a loop γ based at x_0, the transport map T_γ is a permutation of the fiber $p^{-1}(x_0)$. The assignment

$$\rho\colon \pi_1(X, x_0) \longrightarrow \mathrm{Sym}(p^{-1}(x_0)), \qquad \rho([\gamma]) = T_\gamma$$

is called the **monodromy action** of $\pi_1(X, x_0)$ on the fiber.

Proposition 5.14. Monodromy is a group action

The map ρ in Definition 5.13 is a well-defined group homomorphism.

Proof. Well-definedness follows from Proposition 5.12 because a homotopy of loops rel basepoint is in particular a homotopy rel endpoints. The identity element of $\pi_1(X, x_0)$ is represented by the constant loop, whose lift from any $y \in p^{-1}(x_0)$ is constant, so $\rho(1)$ is the identity permutation. For $[\gamma], [\eta] \in \pi_1(X, x_0)$, the product $[\gamma] \cdot [\eta]$ is represented by the concatenated loop $\gamma \cdot \eta$. By Proposition 5.12,

$$\rho([\gamma] \cdot [\eta]) = T_{\gamma\cdot\eta} = T_\eta \circ T_\gamma = \rho([\eta]) \circ \rho([\gamma]),$$

which is the homomorphism law with the standard convention that permutations compose right-to-left. □

Proposition 5.15. Connected coverings give transitive monodromy

Let $p\colon Y \to X$ be a covering map and assume Y is path-connected. Then the monodromy action of $\pi_1(X, x_0)$ on the fiber $p^{-1}(x_0)$ is transitive.

Proof. Let $y, y' \in p^{-1}(x_0)$. Choose a path $\widetilde{\gamma}$ in Y from y to y'. Then $\gamma = p \circ \widetilde{\gamma}$ is a loop in X based at x_0. The path $\widetilde{\gamma}$ is a lift of γ starting at y, hence by uniqueness it must equal $\widetilde{\gamma}_y$, the distinguished lift used in Definition 5.11. Therefore $T_\gamma(y) = \widetilde{\gamma}_y(1) = \widetilde{\gamma}(1) = y'$. So y' lies in the orbit of y. □

6 Covering spaces: classification and group actions

6.1 Universal cover and existence criteria

Throughout this lecture we work in the following class of spaces.

Definition 6.1. Standing hypotheses on spaces

A space X is called **good for covering theory** if it is path-connected, locally path-connected, and semilocally simply connected.

Definition 6.2. Semilocally simply connected

A space X is **semilocally simply connected** if for every $x \in X$ there exists an open neighborhood U of x such that every loop in U based at x is null-homotopic in X. Equivalently, the map $\pi_1(U, x) \to \pi_1(X, x)$ induced by inclusion is the trivial homomorphism.

Definition 6.3. Universal covering

A covering map $p\colon \widetilde{X} \to X$ is a **universal covering** if $\widetilde{X}$ is simply connected.

Proposition 6.4. Universal covers are unique up to unique isomorphism over X

Let $p\colon \widetilde{X} \to X$ and $p'\colon \widetilde{X}' \to X$ be universal coverings, and choose basepoints $\widetilde{x}_0 \in \widetilde{X}$ and $\widetilde{x}'_0 \in \widetilde{X}'$ with $p(\widetilde{x}_0) = p'(\widetilde{x}'_0) = x_0$. Then there exists a unique homeomorphism $F\colon \widetilde{X} \to \widetilde{X}'$ such that

$$p' \circ F = p \qquad \text{and} \qquad F(\widetilde{x}_0) = \widetilde{x}'_0.$$

Proof. Consider the map $p\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$. Since $\widetilde{X}$ is simply connected, the map p lifts uniquely through p' once a starting point in the fiber is prescribed. Concretely, apply the lifting criterion of Theorem 6.7 below to the map p and the covering p'. This yields a unique lift $F\colon \widetilde{X} \to \widetilde{X}'$ with $F(\widetilde{x}_0) = \widetilde{x}'_0$ and $p' \circ F = p$. Similarly, p' lifts uniquely through p with prescribed basepoint, giving $G\colon \widetilde{X}' \to \widetilde{X}$ such that $p \circ G = p'$ and $G(\widetilde{x}'_0) = \widetilde{x}_0$. Then $F \circ G$ and $\mathrm{id}_{\widetilde{X}'}$ are both lifts of p' with the same value at $\widetilde{x}'_0$, so they agree by uniqueness of lifting. Likewise $G \circ F = \mathrm{id}_{\widetilde{X}}$. Hence F is a homeomorphism and is uniquely determined by the basepoint condition. □

Theorem 6.5. Existence of universal covers

If X is good for covering theory, then X admits a universal covering $p\colon \widetilde{X} \to X$.

Remark. A standard construction takes $\widetilde{X}$ to be the set of path-homotopy classes rel endpoints of paths starting at a fixed basepoint, with endpoint projection. Local path connectedness and semilocal simple connectivity ensure that this endpoint map is a covering. The point-set details will be used only through the consequences recorded below.

6.2 Classification via subgroups of π_1

The classification rests on a lifting criterion phrased in terms of fundamental groups.

Definition 6.6. Based lifts of maps

Let $p\colon (Y, y_0) \to (X, x_0)$ be a covering map and let $f\colon (Z, z_0) \to (X, x_0)$ be a based map. A **based lift** of f is a map $\widetilde{f}\colon (Z, z_0) \to (Y, y_0)$ such that $p \circ \widetilde{f} = f$.

Theorem 6.7. Lifting criterion for coverings

Let $p\colon (Y, y_0) \to (X, x_0)$ be a covering map, and assume Z is path-connected and locally path-connected. Let $f\colon (Z, z_0) \to (X, x_0)$ be continuous. Then a based lift $\widetilde{f}\colon (Z, z_0) \to (Y, y_0)$ exists if and only if

$$f_*(\pi_1(Z, z_0)) \subset p_*(\pi_1(Y, y_0)) \subset \pi_1(X, x_0).$$

If a based lift exists, it is unique.

Proof. *Uniqueness.* Suppose $\widetilde{f}$ and $\widetilde{f}'$ are based lifts. Let

$$A = \{z \in Z : \widetilde{f}(z) = \widetilde{f}'(z)\}.$$

Then A is closed. To see A is open, take $z \in A$ and set $y = \widetilde{f}(z) = \widetilde{f}'(z)$ and $x = f(z) = p(y)$. Choose an evenly covered neighborhood U of x. Since f is continuous, there exists a neighborhood W of z in Z with $f(W) \subset U$. Because $p^{-1}(U)$ is a disjoint union of sheets mapping homeomorphically onto U, and both lifts send z to the same point y in a specific sheet, they must agree on W. Thus A is open, and since Z is connected and $z_0 \in A$, one has $A = Z$.

Necessity. If $\widetilde{f}$ exists, then $f = p \circ \widetilde{f}$, hence $f_* = p_* \circ \widetilde{f}_*$ and the inclusion of images follows.

Sufficiency. Assume $f_*\pi_1(Z, z_0) \subset p_*\pi_1(Y, y_0)$. For each $z \in Z$, choose a path α in Z from z_0 to z. The path $f \circ \alpha$ in X lifts uniquely to a path $\widetilde{\alpha}$ in Y starting at y_0 by unique path lifting from Lecture 5. Define $\widetilde{f}(z) = \widetilde{\alpha}(1)$.

We must check that $\widetilde{f}(z)$ is independent of the choice of α. If α and β are two paths from z_0 to z, then $\alpha \cdot \beta^{-1}$ is a loop at z_0. Its image under f is the loop $(f \circ \alpha) \cdot (f \circ \beta)^{-1}$ based at x_0. By assumption, its class lies in $p_*\pi_1(Y, y_0)$, so there exists a loop λ in Y based at y_0 with

$$p_*([\lambda]) = [f \circ \alpha \cdot (f \circ \beta)^{-1}].$$

Lift the homotopy in X that identifies $p \circ \lambda$ with $f \circ \alpha \cdot (f \circ \beta)^{-1}$ to Y starting at y_0. By uniqueness of path lifting, the lift of $f \circ \alpha$ starting at y_0 ends at the same point as the lift of $f \circ \beta$ starting at y_0. Hence $\widetilde{\alpha}(1) = \widetilde{\beta}(1)$, so $\widetilde{f}$ is well-defined.

To see $\widetilde{f}$ is continuous, fix $z \in Z$ and choose an evenly covered neighborhood U of $f(z)$. Since Z is locally path-connected, after shrinking there is a path-connected neighborhood W of z with $f(W) \subset U$. Let U_α be the sheet containing $\widetilde{f}(z)$. For any $w \in W$, choose a path in W from z to w. Its image under f lies in U, so

its lift starting at $\widetilde{f}(z)$ stays inside U_α by uniqueness of lifting. This shows $\widetilde{f}(W) \subset U_\alpha$ and on W one has

$$\widetilde{f} = (p|_{U_\alpha})^{-1} \circ f,$$

hence $\widetilde{f}$ is continuous. Finally $\widetilde{f}(z_0) = y_0$ by construction. □

Definition 6.8. Associated subgroup

Let $p\colon (Y, y_0) \to (X, x_0)$ be a covering map. The subgroup

$$H(p, y_0) \;=\; p_*(\pi_1(Y, y_0)) \;\subset\; \pi_1(X, x_0)$$

is called the **subgroup associated to the based covering**.

Definition 6.9. Isomorphism of coverings

Two coverings $p\colon Y \to X$ and $p'\colon Y' \to X$ are **isomorphic over** X if there exists a homeomorphism $F\colon Y \to Y'$ such that $p' = p \circ F^{-1}$, equivalently $p' \circ F = p$. If basepoints are specified, we also require $F(y_0) = y_0'$.

Theorem 6.10. Classification of connected coverings

Assume X is good for covering theory and fix $x_0 \in X$. Let $p\colon (Y, y_0) \to (X, x_0)$ be a path-connected covering.

1. The based isomorphism class of p is determined by the subgroup $H(p, y_0) \subset \pi_1(X, x_0)$.
2. Every subgroup $H \subset \pi_1(X, x_0)$ arises as $H(p, y_0)$ for some path-connected covering $p\colon (Y, y_0) \to (X, x_0)$.
3. If y_0 is replaced by another point y_1 in the same fiber, then $H(p, y_1)$ is conjugate to $H(p, y_0)$ in $\pi_1(X, x_0)$. Consequently, unbased isomorphism classes of path-connected coverings correspond to conjugacy classes of subgroups of $\pi_1(X, x_0)$.

Proof. Let $\widetilde{p}\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a universal cover, which exists by Theorem 6.5.
Step 1. A universal cover dominates every connected cover. Apply Theorem 6.7 to the map $\widetilde{p}\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ and the covering $p\colon (Y, y_0) \to (X, x_0)$. Since $\pi_1(\widetilde{X}, \widetilde{x}_0) = 1$, the subgroup condition is automatic. Hence there exists a unique based lift

$$q\colon (\widetilde{X}, \widetilde{x}_0) \to (Y, y_0) \qquad \text{with} \qquad p \circ q = \widetilde{p}.$$

We claim q is a covering map. Fix $\widetilde{x} \in \widetilde{X}$ and choose an evenly covered neighborhood U of $\widetilde{p}(\widetilde{x})$ for p, so that $p^{-1}(U) = \bigsqcup_\alpha U_\alpha$ with $p|_{U_\alpha}\colon U_\alpha \to U$ a homeomorphism. Also choose U evenly covered for $\widetilde{p}$, so that $\widetilde{p}^{-1}(U) = \bigsqcup_\beta \widetilde{U}_\beta$ with $\widetilde{p}|_{\widetilde{U}_\beta}$ a homeomorphism. Then

$$\widetilde{p}|_{\widetilde{U}_\beta} \;=\; p \circ q|_{\widetilde{U}_\beta}.$$

Since $\widetilde{p}|_{\widetilde{U}_\beta}$ is a homeomorphism onto U and $p|_{U_\alpha}$ is a homeomorphism onto U, it follows that $q(\widetilde{U}_\beta) \subset U_\alpha$ for a unique α, and on $\widetilde{U}_\beta$ we have

$$q|_{\widetilde{U}_\beta} \;=\; (p|_{U_\alpha})^{-1} \circ \widetilde{p}|_{\widetilde{U}_\beta},$$

hence $q|_{\widetilde{U}_\beta}$ is a homeomorphism onto an open subset of U_α. This shows that q is a local homeomorphism with discrete fibers and is surjective because Y is connected and $p \circ q$ is surjective. Thus q is a covering map.
Step 2. Recovering Y as a quotient of $\widetilde{X}$. Define a subgroup

$$H \;=\; \{\, \varphi \in \mathrm{Deck}(\widetilde{p}) : q \circ \varphi = q \,\},$$

where $\mathrm{Deck}(\widetilde{p})$ denotes deck transformations of the universal cover, defined precisely in the next subsection. Then H acts on $\widetilde{X}$ by homeomorphisms and q is constant on H-orbits. Hence q factors through a continuous map

$$\overline{q}\colon \widetilde{X}/H \longrightarrow Y.$$

The map $\overline{q}$ is a homeomorphism over X, because locally $\widetilde{X} \to X$ looks like a disjoint union of copies of an open set and the subgroup H acts by permuting these copies in a way compatible with q. More explicitly, fix $\widetilde{x} \in \widetilde{X}$ and choose U evenly covered for all maps $\widetilde{p}$, p, and q. Over U, the map q identifies one chosen sheet of $\widetilde{p}^{-1}(U)$ homeomorphically with one chosen sheet of $p^{-1}(U)$, and any other sheet mapping to the same sheet of $p^{-1}(U)$ differs by a unique element of H. This gives a local description showing $\overline{q}$ is bijective and locally a homeomorphism, hence a homeomorphism.

Step 3. Identifying the subgroup with $p_*\pi_1(Y, y_0)$. Let $\widetilde{x}_0$ map to y_0 under q. A loop γ in X based at x_0 lifts to a path in $\widetilde{X}$ starting at $\widetilde{x}_0$, and its endpoint is $\varphi_\gamma(\widetilde{x}_0)$ for a unique deck transformation φ_γ of $\widetilde{p}$. This gives an identification $\mathrm{Deck}(\widetilde{p}) \cong \pi_1(X, x_0)$, recorded in Theorem 6.15 below. Under this identification, the subgroup H corresponds exactly to $p_*\pi_1(Y, y_0)$. Indeed, $q \circ \varphi_\gamma = q$ holds if and only if the lift of γ to Y starting at y_0 is a loop, which by the lifting criterion from Lecture 5 is equivalent to $[\gamma] \in p_*\pi_1(Y, y_0)$.
Combining Steps 1–3 proves (1) and (2): given $H \subset \pi_1(X, x_0)$, form $\widetilde{X}/H$ and obtain a covering whose associated subgroup is H, and any connected covering is isomorphic to such a quotient.
Step 4. Basepoint change gives conjugation. Let $y_1 \in p^{-1}(x_0)$ and choose a path $\widetilde{\gamma}$ in Y from y_0 to y_1. Then $\gamma = p \circ \widetilde{\gamma}$ is a loop in X based at x_0. By the basepoint change formula from Lecture 3, the isomorphism $\pi_1(Y, y_0) \to \pi_1(Y, y_1)$ given by conjugation with $\widetilde{\gamma}$ is intertwined by p_* with conjugation by γ in $\pi_1(X, x_0)$. Hence $H(p, y_1)$ is conjugate to $H(p, y_0)$. This yields (3). □

6.3 Deck transformations and normal coverings

Definition 6.11. Deck transformation

Let $p\colon Y \to X$ be a covering map. A **deck transformation** of p is a homeomorphism $\varphi\colon Y \to Y$ such that

$$p \circ \varphi = p.$$

The set of all deck transformations forms a group under composition, denoted $\mathrm{Deck}(p)$.

Lemma 6.12. A deck transformation is determined by one point in a connected cover Let $p\colon Y \to X$ be a covering map with Y path-connected. If $\varphi, \psi \in \mathrm{Deck}(p)$ satisfy $\varphi(y_0) = \psi(y_0)$ for some $y_0 \in Y$, then $\varphi = \psi$.

Proof. Consider the set $A = \{y \in Y : \varphi(y) = \psi(y)\}$. It is closed, and it is open by the same sheet argument used in Theorem 6.7: near any $y \in A$, choose an evenly covered neighborhood U of $p(y)$ and let U_α be the sheet containing y. Both φ and ψ map U_α homeomorphically onto itself because they preserve p. If they agree at y, they agree on U_α. Thus A is open. Since Y is connected and $y_0 \in A$, one has $A = Y$. □

Proposition 6.13. Deck transformations act freely on each fiber

Let $p\colon Y \to X$ be a covering with Y path-connected. If $\varphi \in \mathrm{Deck}(p)$ fixes a point in some fiber $p^{-1}(x)$, then φ is the identity.

Proof. If $\varphi(y) = y$ for some y, then apply Lemma 6.12 with $\psi = \mathrm{id}_Y$. □

Definition 6.14. Normal covering

A path-connected covering $p\colon Y \to X$ is **normal** (or **regular**) if $\mathrm{Deck}(p)$ acts transitively on some, equivalently every, fiber $p^{-1}(x)$.

Theorem 6.15. Deck group of a universal cover

Assume X is good for covering theory and let $\widetilde{p}\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be the universal cover. Then there is a natural isomorphism

$$\mathrm{Deck}(\widetilde{p}) \;\cong\; \pi_1(X, x_0).$$

Proof. For $[\gamma] \in \pi_1(X, x_0)$, define $\varphi_{[\gamma]}\colon \widetilde{X} \to \widetilde{X}$ by transport along γ as in Lecture 5: for $\widetilde{x} \in \widetilde{X}$, let $\widetilde{\gamma}_{\widetilde{x}}$ be the unique lift of γ starting at $\widetilde{x}$, and set

$$\varphi_{[\gamma]}(\widetilde{x}) = \widetilde{\gamma}_{\widetilde{x}}(1).$$

By homotopy lifting, $\varphi_{[\gamma]}$ depends only on the class $[\gamma]$. The map $\varphi_{[\gamma]}$ satisfies $\widetilde{p} \circ \varphi_{[\gamma]} = \widetilde{p}$, so it is a deck transformation. The composition rule for transport from Lecture 5 gives

$$\varphi_{[\gamma\cdot\eta]} = \varphi_{[\eta]} \circ \varphi_{[\gamma]},$$

so $[\gamma] \mapsto \varphi_{[\gamma]}$ is a homomorphism $\pi_1(X, x_0) \to \mathrm{Deck}(\widetilde{p})$.
Injectivity follows from freeness on the fiber: if $\varphi_{[\gamma]}(\widetilde{x}_0) = \widetilde{x}_0$, then the lift of γ starting at $\widetilde{x}_0$ is a loop. Since $\widetilde{X}$ is simply connected, that loop is null-homotopic, hence $[\gamma] = 1$.

For surjectivity, take any $\varphi \in \mathrm{Deck}(\widetilde{p})$. Choose a path $\widetilde{\alpha}$ in $\widetilde{X}$ from $\widetilde{x}_0$ to $\varphi(\widetilde{x}_0)$ and set $\alpha = \widetilde{p} \circ \widetilde{\alpha}$, a loop in X based at x_0. By construction, the lift of α starting at $\widetilde{x}_0$ ends at $\varphi(\widetilde{x}_0)$. Thus $\varphi_{[\alpha]}(\widetilde{x}_0) = \varphi(\widetilde{x}_0)$. By Lemma 6.12, this forces $\varphi_{[\alpha]} = \varphi$. □

Theorem 6.16. Normal coverings and normal subgroups

Assume X is good for covering theory. Let $p\colon (Y, y_0) \to (X, x_0)$ be a path-connected covering and set $H = p_*\pi_1(Y, y_0) \subset \pi_1(X, x_0)$. Then p is normal if and only if H is a normal subgroup of $\pi_1(X, x_0)$.

Proof. Let $\widetilde{p}\colon \widetilde{X} \to X$ be the universal cover and let $q\colon \widetilde{X} \to Y$ be the covering map with $p \circ q = \widetilde{p}$ and $q(\widetilde{x}_0) = y_0$ as in the proof of Theorem 6.10. Using Theorem 6.15, identify $\mathrm{Deck}(\widetilde{p})$ with $\pi_1(X, x_0)$.
Under this identification, the subgroup H consists of those deck transformations φ of $\widetilde{p}$ that preserve fibers of q in the sense that $q \circ \varphi = q$. Equivalently, H is the stabilizer of $\widetilde{x}_0$ for the induced action of $\pi_1(X, x_0)$ on the fiber of p above x_0.
If H is normal, then the quotient group $\pi_1(X, x_0)/H$ acts transitively on the coset space, and this coset space is naturally identified with the fiber $p^{-1}(x_0)$. This gives transitivity of the deck action, hence p is normal.
Conversely, if p is normal, the deck group acts transitively on $p^{-1}(x_0)$. Under the identification of $p^{-1}(x_0)$ with cosets of H, transitivity implies left multiplication by any element of $\pi_1(X, x_0)$ permutes cosets in a way compatible with the deck action. This forces conjugation invariance of H, hence H is normal. □

A more precise relationship between the deck group and the subgroup H uses the normalizer.

Proposition 6.17. Deck group as a normalizer quotient

With notation as in Theorem 6.16, let $N(H) \subset \pi_1(X, x_0)$ be the normalizer

$$N(H) = \{g \in \pi_1(X, x_0) : gHg^{-1} = H\}.$$

Then there is a natural isomorphism

$$\mathrm{Deck}(p) \;\cong\; N(H)/H.$$

In particular, if H is normal then $\mathrm{Deck}(p) \cong \pi_1(X, x_0)/H$.

Proof. Let $\widetilde{X} \to X$ be the universal cover and realize $Y \cong \widetilde{X}/H$ as in Theorem 6.10, where H acts by deck transformations of $\widetilde{X} \to X$. A deck transformation of $Y \to X$ lifts to a deck transformation φ of $\widetilde{X} \to X$ that permutes H-orbits, meaning $\varphi H \varphi^{-1} = H$. Thus lifts correspond to elements of $N(H)$. Two such lifts define the same transformation downstairs exactly when they differ by an element of H. This gives $\mathrm{Deck}(p) \cong N(H)/H$. □

6.4 Examples that will come back later

Examples.

1. **Coverings of the circle.** Let $p\colon \mathbb{R} \to S^1$ be $t \mapsto e^{2\pi i t}$. Then $\mathrm{Deck}(p) \cong \mathbb{Z}$ acts by integer translations on $\mathbb{R}$. For the n-fold covering $S^1 \to S^1$, $z \mapsto z^n$, the associated subgroup is $n\mathbb{Z} \subset \mathbb{Z} \cong \pi_1(S^1)$, and the deck group is $\mathbb{Z}/n\mathbb{Z}$.
2. **Tori and lattices.** The universal cover $\mathbb{R}^n \to T^n = \mathbb{R}^n/\mathbb{Z}^n$ has deck group $\mathbb{Z}^n$ acting by translations. Connected coverings of T^n correspond to subgroups of $\mathbb{Z}^n$, equivalently to sublattices in $\mathbb{Z}^n$.
3. **Graphs and free groups.** If G is a connected finite graph, its universal cover is a tree $\widetilde{G}$. The deck group identifies with $\pi_1(G)$, which is a free group by Lecture 4. Coverings of graphs correspond to subgroups of free groups, and normal coverings correspond to normal subgroups.
4. **Projective space.** The map $S^n \to \mathbb{R}P^n$ is the universal cover for $n \geq 2$. Its deck group is $\mathbb{Z}/2\mathbb{Z}$ generated by the antipodal map.
5. **A lens space hook.** Let $m \geq 2$ and let $\mathbb{Z}/m\mathbb{Z}$ act freely on $S^{2k-1} \subset \mathbb{C}^k$ by

$$[z_1, \ldots, z_k] \cdot \omega = (\omega^{a_1} z_1, \ldots, \omega^{a_k} z_k),$$

where $\omega = e^{2\pi i/m}$ and the integers a_i are chosen so that the action is free. The quotient is a lens space, and the projection $S^{2k-1} \to L$ is a normal covering with deck group $\mathbb{Z}/m\mathbb{Z}$. When $k = 2$ one gets a rich family of 3-manifolds that will later interact naturally with homology, cohomology operations, and

characteristic classes.

7 Chain complexes and singular homology

7.1 Chain complexes, boundaries, and homology

Definition 7.1. Chain complex

A **chain complex** $(C_\bullet, \partial)$ of abelian groups consists of abelian groups C_n for $n \in \mathbb{Z}$ and homomorphisms

$$\partial_n \colon C_n \longrightarrow C_{n-1}$$

such that $\partial_{n-1} \circ \partial_n = 0$ for all n. We often write ∂ for the collection of maps $\{\partial_n\}$.

Definition 7.2. Cycles, boundaries, and homology

Let $(C_\bullet, \partial)$ be a chain complex. Define the group of n**-cycles** and n**-boundaries** by

$$Z_n(C) = \ker(\partial_n) \subset C_n, \qquad B_n(C) = \operatorname{im}(\partial_{n+1}) \subset C_n.$$

Then $B_n(C) \subset Z_n(C)$, and the n**th homology group** of $C_\bullet$ is

$$H_n(C) = Z_n(C)/B_n(C).$$

Lemma 7.3. Boundaries are cycles For any chain complex $(C_\bullet, \partial)$ one has $B_n(C) \subset Z_n(C)$.

Proof. If $c \in B_n(C)$ then $c = \partial_{n+1}(d)$ for some $d \in C_{n+1}$. Then $\partial_n(c) = \partial_n \partial_{n+1}(d) = 0$, so $c \in Z_n(C)$. □

Definition 7.4. Chain map

Let $(C_\bullet, \partial)$ and $(D_\bullet, \partial)$ be chain complexes. A **chain map** $f_\bullet \colon C_\bullet \to D_\bullet$ is a collection of homomorphisms $f_n \colon C_n \to D_n$ such that

$$\partial \circ f_n = f_{n-1} \circ \partial \qquad \text{for all } n,$$

equivalently the diagram

$$\begin{tikzcd} C_n \arrow[r, "f_n"] \arrow[d, "\partial"'] & D_n \arrow[d, "\partial"] \\ C_{n-1} \arrow[r, "f_{n-1}"'] & D_{n-1} \end{tikzcd}$$

commutes for every n.

Proposition 7.5. A chain map induces a map on homology

If $f_\bullet \colon C_\bullet \to D_\bullet$ is a chain map, then for each n there is an induced homomorphism

$$H_n(f) \colon H_n(C) \to H_n(D), \qquad [c] \mapsto [f_n(c)].$$

Proof. If $c \in Z_n(C)$ then $\partial f_n(c) = f_{n-1}\partial(c) = 0$, so $f_n(c) \in Z_n(D)$. If $c = \partial d$ is a boundary, then $f_n(c) = f_n(\partial d) = \partial f_{n+1}(d)$ is a boundary. Thus the assignment $[c] \mapsto [f_n(c)]$ is well-defined on the quotient $Z_n(C)/B_n(C)$ and is a homomorphism. □

Definition 7.6. Chain homotopy

Let $f_\bullet, g_\bullet \colon C_\bullet \to D_\bullet$ be chain maps. A **chain homotopy** from $f_\bullet$ to $g_\bullet$ is a collection of homomorphisms

$$s_n \colon C_n \to D_{n+1}$$

such that for all n,

$$f_n - g_n \;=\; \partial s_n + s_{n-1}\partial.$$

Theorem 7.7. Chain-homotopic maps induce the same map on homology

If $f_\bullet$ and $g_\bullet$ are chain-homotopic chain maps $C_\bullet \to D_\bullet$, then

$$H_n(f) = H_n(g) \qquad \text{for all } n.$$

Proof. Let $s_\bullet$ be a chain homotopy with $f - g = \partial s + s\partial$. Take $c \in Z_n(C)$, so $\partial c = 0$. Then

$$(f_n - g_n)(c) = \partial s_n(c) + s_{n-1}\partial(c) = \partial s_n(c).$$

Thus $(f_n - g_n)(c)$ is a boundary in D_n, so $[f_n(c)] = [g_n(c)]$ in $H_n(D)$. □

Remark. This theorem is the algebraic engine behind homotopy invariance of singular homology: one builds a chain homotopy between the induced chain maps coming from homotopic continuous maps. The geometric content is hidden in the construction of $s_\bullet$, which will be supplied later by a prism operator.

7.2 Δ-complexes and simplicial chains

CW complexes are flexible for building spaces, but when a space comes with a combinatorial decomposition one can compute homology directly from it.

Definition 7.8. Δ-complex

A **Δ-complex** structure on a space X consists of a collection of maps

$$\sigma_\alpha \colon \Delta^n \to X$$

(for various n and indices α) such that:

1. Each σ_α restricts to an embedding on the interior of Δ^n.
2. The images of the interiors form a partition of X.
3. The restriction of σ_α to each face of Δ^n is one of the maps σ_β in the collection.

The maps σ_α are called the **simplices** of the Δ-complex.

Definition 7.9. Simplicial chain groups

Let X be a Δ-complex. For each $n \geq 0$, let $C_n^\Delta(X)$ be the free abelian group on the set of oriented n-simplices of X. Define the boundary homomorphism

$$\partial_n \colon C_n^\Delta(X) \to C_{n-1}^\Delta(X)$$

by sending an oriented simplex $\sigma \colon \Delta^n \to X$ to the alternating sum of its oriented faces:

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \, \sigma \circ \delta_i,$$

where $\delta_i \colon \Delta^{n-1} \hookrightarrow \Delta^n$ is the ith face inclusion. Extend ∂ linearly.

Lemma 7.10. Simplicial boundary squares to zero For a Δ-complex X, the groups $C_n^\Delta(X)$ with the boundary in Definition 7.9 form a chain complex.

Proof. It suffices to show $\partial\partial\sigma = 0$ for each oriented simplex σ. Compute

$$\partial(\partial\sigma) = \sum_{i=0}^{n} \sum_{j=0}^{n-1} (-1)^{i+j} \, \sigma \circ \delta_i \circ \delta_j.$$

For $0 \leq j < i \leq n$, the composite $\delta_i \circ \delta_j$ equals $\delta_j \circ \delta_{i-1}$ as a face inclusion $\Delta^{n-2} \to \Delta^n$. Thus each $(n-2)$-face appears exactly twice with opposite signs: the term indexed by (i, j) cancels the term indexed by $(j, i-1)$. Hence the sum is zero. □

Definition 7.11. Simplicial homology

The homology of the chain complex $(C_\bullet^\Delta(X), \partial)$ is called the **simplicial homology** of the Δ-complex X, denoted $H_n^\Delta(X)$.

Remark. When X is built from finitely many simplices, the groups $C_n^\Delta(X)$ are finitely generated, so $H_n^\Delta(X)$ is in principle computable from an integer matrix. Later we will compare $H_\bullet^\Delta(X)$ with singular homology and explain why they agree for reasonable spaces.

7.3 Singular chains and functoriality

Definition 7.12. Standard simplex

For $n \geq 0$, the **standard n-simplex** is

$$\Delta^n = \Big\{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0,\ \sum_{i=0}^{n} t_i = 1\Big\}.$$

For $0 \leq i \leq n$, let $\delta_i \colon \Delta^{n-1} \to \Delta^n$ denote the inclusion of the ith face, given by inserting 0 in the ith coordinate.

Definition 7.13. Singular simplex and singular chain group

Let X be a space. A **singular n-simplex** in X is a continuous map $\sigma \colon \Delta^n \to X$. Let $C_n(X)$ be the free abelian group on the set of all singular n-simplices in X. Define the boundary homomorphism $\partial_n \colon C_n(X) \to C_{n-1}(X)$ by

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \, \sigma \circ \delta_i,$$

extended $\mathbb{Z}$-linearly. The resulting chain complex $(C_\bullet(X), \partial)$ is called the **singular chain complex** of X.

Lemma 7.14. Singular boundary squares to zero For every space X, the singular chain groups $C_n(X)$ with the boundary in Definition 7.13 form a chain complex.

Proof. The computation is identical to the proof of Lemma 7.10, since it only uses the relations among face maps of the standard simplex. □

Definition 7.15. Singular homology

The **singular homology** groups of X are

$$H_n(X) = H_n(C_\bullet(X)),$$

the homology groups of the singular chain complex.

Definition 7.16. Induced chain map

Let $f \colon X \to Y$ be continuous. Define a homomorphism

$$f_\# \colon C_n(X) \to C_n(Y)$$

on generators by $f_\#(\sigma) = f \circ \sigma$ for each singular simplex $\sigma \colon \Delta^n \to X$, extended linearly.

Proposition 7.17. Functoriality of singular chains

For every continuous map $f \colon X \to Y$, the map $f_\#$ is a chain map

$$f_\# \colon (C_\bullet(X), \partial) \to (C_\bullet(Y), \partial).$$

Moreover,

$$(g \circ f)_\# = g_\# \circ f_\#, \qquad (\mathrm{id}_X)_\# = \mathrm{id}_{C_\bullet(X)}.$$

Proof. We check $f_\#$ commutes with boundaries. For a generator $\sigma \colon \Delta^n \to X$,

$$\partial(f_\#\sigma) = \partial(f \circ \sigma) = \sum_{i=0}^{n} (-1)^i (f \circ \sigma) \circ \delta_i = f \circ \Big(\sum_{i=0}^{n} (-1)^i \sigma \circ \delta_i\Big) = f_\#(\partial\sigma).$$

Thus $\partial \circ f_\# = f_\# \circ \partial$. The identities for composition and identity maps are immediate from associativity of composition of functions. □

Definition 7.18. Induced map on singular homology

For $f\colon X \to Y$, define
$$f_*\colon H_n(X) \to H_n(Y)$$
to be the induced map $H_n(f_\#)$ from Proposition 7.5.

Proposition 7.19. Functoriality of singular homology

Singular homology is a functor: for continuous $f\colon X \to Y$ and $g\colon Y \to Z$,
$$(g \circ f)_* = g_* \circ f_*, \qquad (\mathrm{id}_X)_* = \mathrm{id}_{H_n(X)}.$$

Proof. This follows from Proposition 7.17 and the fact that taking homology is functorial with respect to chain maps (Proposition 7.5). □

7.4 Homotopy invariance, statement and strategy

Theorem 7.20. Homotopy invariance of singular homology

If $f, g\colon X \to Y$ are homotopic, then $f_* = g_*$ on H_n for all n. In particular, if $X \simeq Y$, then $H_n(X) \cong H_n(Y)$ for all n.

Remark. The proof will produce, from a homotopy $H\colon X \times I \to Y$, an explicit chain homotopy between the chain maps $f_\#, g_\#\colon C_\bullet(X) \to C_\bullet(Y)$ induced by $f = H(-,0)$ and $g = H(-,1)$. Once such a chain homotopy is constructed, Theorem 7.7 immediately gives $f_* = g_*$. The missing input is a geometric operator on singular simplices that turns the cylinder $\Delta^n \times I$ into a sum of $(n+1)$-simplices.

Proposition 7.21. Chain homotopy implies equal maps on homology

Let $f, g\colon X \to Y$ be continuous. If there exists a chain homotopy $s_\bullet$ between $f_\#$ and $g_\#$, then $f_* = g_*$ on H_n for all n.

Proof. This is exactly Theorem 7.7 applied to the chain maps $f_\#$ and $g_\#$. □

Remark. In the next lecture we will build the required chain homotopy using a prism decomposition: for each singular simplex $\sigma\colon \Delta^n \to X$ we will triangulate $\Delta^n \times I$ into $(n+1)$-simplices in a canonical way, then compose with the homotopy $H \circ (\sigma \times \mathrm{id}_I)$ to obtain an element of $C_{n+1}(Y)$. The boundary computation will yield the chain-homotopy identity
$$g_\# - f_\# = \partial s + s\partial.$$

8 Exact sequences, relative homology, and excision

8.1 Relative chains and long exact sequence of a pair

Let $A \subset X$ be a subspace.

Definition 8.1. Relative chain complex

For $n \geq 0$, define the **relative singular chain group**
$$C_n(X, A) \;=\; C_n(X)/C_n(A),$$
where $C_n(A) \subset C_n(X)$ is the subgroup generated by singular simplices $\sigma\colon \Delta^n \to X$ whose image lies in A. Let $q\colon C_n(X) \to C_n(X, A)$ be the quotient map. Since $\partial(C_n(A)) \subset C_{n-1}(A)$, the boundary ∂ on $C_\bullet(X)$ induces a boundary on $C_\bullet(X, A)$, still denoted ∂, by
$$\partial(q(c)) = q(\partial c).$$

The homology of this chain complex is the **relative homology** of the pair (X, A):

$$H_n(X, A) = H_n(C_\bullet(X, A)).$$

Lemma 8.2. Short exact sequence of chain complexes For each n there is a short exact sequence of abelian groups

$$0 \longrightarrow C_n(A) \xrightarrow{i_\#} C_n(X) \xrightarrow{q} C_n(X, A) \longrightarrow 0,$$

and these assemble into a short exact sequence of chain complexes

$$0 \longrightarrow C_\bullet(A) \xrightarrow{i_\#} C_\bullet(X) \xrightarrow{q} C_\bullet(X, A) \longrightarrow 0.$$

Proof. Exactness at $C_n(A)$ and $C_n(X)$ is clear because $i_\#$ is the inclusion and q is the quotient by $C_n(A)$. Surjectivity of q is by definition. Compatibility with boundaries follows from $\partial(C_n(A)) \subset C_{n-1}(A)$, which holds because restricting a simplex to a face preserves the condition that its image lies in A. □

Definition 8.3. Maps induced by inclusion

Write $i\colon A \hookrightarrow X$ for inclusion. The chain inclusion $i_\#\colon C_\bullet(A) \to C_\bullet(X)$ induces homomorphisms

$$i_*\colon H_n(A) \to H_n(X).$$

The quotient chain map $q\colon C_\bullet(X) \to C_\bullet(X, A)$ induces homomorphisms

$$j_*\colon H_n(X) \to H_n(X, A).$$

Definition 8.4. Connecting homomorphism

Let $[c] \in H_n(X, A)$, represented by $c \in C_n(X)$ with $q(c)$ a cycle, meaning $\partial(q(c)) = 0$ in $C_{n-1}(X, A)$. Equivalently, $q(\partial c) = 0$, so $\partial c \in C_{n-1}(A)$. Define

$$\partial_*([c]) \;=\; [\partial c] \in H_{n-1}(A).$$

Lemma 8.5. The connecting homomorphism is well-defined The assignment in Definition 8.4 defines a homomorphism

$$\partial_*\colon H_n(X, A) \to H_{n-1}(A).$$

Proof. We must check independence of choices. Suppose $q(c)$ and $q(c')$ represent the same class in $H_n(X, A)$. Then

$$q(c - c') = \partial(q(b))$$

for some $b \in C_{n+1}(X)$, so $q(c - c' - \partial b) = 0$. Hence there exists $a \in C_n(A)$ such that

$$c - c' - \partial b = a.$$

Applying ∂ gives

$$\partial c - \partial c' = \partial a \in B_{n-1}(A),$$

so $[\partial c] = [\partial c']$ in $H_{n-1}(A)$. Additivity is immediate from linearity of ∂. □

Theorem 8.6. Long exact sequence of a pair

For a pair (X, A) there is a long exact sequence

$$\cdots \longrightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial_*} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \longrightarrow \cdots$$
$$\cdots \longrightarrow H_1(X, A) \xrightarrow{\partial_*} H_0(A) \xrightarrow{i_*} H_0(X) \xrightarrow{j_*} H_0(X, A) \longrightarrow 0.$$

Proof. We verify exactness at each term by a direct chase in chains.

Exactness at $H_n(X)$. First, $j_* \circ i_* = 0$ since $q \circ i_\# = 0$ on chains. Now let $[c] \in H_n(X)$ satisfy $j_*([c]) = 0$. Then $q(c)$ is a boundary in $C_\bullet(X, A)$, so there exists $b \in C_{n+1}(X)$ with

$$q(c) = \partial(q(b)) = q(\partial b).$$

Hence $q(c - \partial b) = 0$, so $c - \partial b \in C_n(A)$. Write $c - \partial b = a$ with $a \in C_n(A)$. Then a is a cycle because $\partial a = \partial c - \partial^2 b = 0$. Therefore $[c] = i_*([a])$, so $\ker(j_*) \subset \operatorname{im}(i_*)$. The reverse inclusion $\operatorname{im}(i_*) \subset \ker(j_*)$ was

already noted.

Exactness at $H_n(X, A)$. First, $\partial_* \circ j_* = 0$ since if $[c] \in H_n(X)$ then $\partial c = 0$ and hence $\partial_*([q(c)]) = [\partial c] = 0$. Now let $[q(c)] \in H_n(X, A)$ satisfy $\partial_*([q(c)]) = 0$. Then $\partial c \in C_{n-1}(A)$ is a boundary in A, so $\partial c = \partial a$ for some $a \in C_n(A)$. Then $c - a$ is a cycle in $C_n(X)$, and

$$j_*([c - a]) = [q(c - a)] = [q(c)]$$

because $q(a) = 0$. Thus $[q(c)] \in \operatorname{im}(j_*)$.

Exactness at $H_{n-1}(A)$. First, $i_* \circ \partial_* = 0$ because if $[q(c)] \in H_n(X, A)$ then $\partial c \in C_{n-1}(A)$ and

$$i_*\partial_*([q(c)]) = i_*([\partial c]) = [\partial c] \in H_{n-1}(X),$$

which is zero since it is a boundary in X. Now let $[\alpha] \in H_{n-1}(A)$ satisfy $i_*([\alpha]) = 0$ in $H_{n-1}(X)$. Choose a cycle $a \in Z_{n-1}(A)$ representing $[\alpha]$. The condition $i_*([\alpha]) = 0$ means $a = \partial c$ for some $c \in C_n(X)$. Then $q(c)$ is a relative cycle since $\partial(q(c)) = q(\partial c) = q(a) = 0$. By construction,

$$\partial_*([q(c)]) = [\partial c] = [a] = [\alpha].$$

Thus $\ker(i_*) \subset \operatorname{im}(\partial_*)$.

The remaining exactness statements follow similarly, and exactness terminates at $H_0(X, A)$ because $C_{-1} = 0$. □

Remark. A map of pairs $f\colon (X, A) \to (Y, B)$ induces chain maps $C_\bullet(X, A) \to C_\bullet(Y, B)$ and hence maps on relative homology. With these maps, the long exact sequence is natural.

8.2 Excision statement and geometric meaning

Relative homology is designed to ignore chains supported in A. Excision says that if U sits well inside A, then one can remove U from both X and A without changing relative homology.

Theorem 8.7. Excision

Let $A \subset X$ and let $U \subset A$ be a subspace whose closure in X satisfies

$$\overline{U} \subset \operatorname{int}(A).$$

Then the inclusion of pairs

$$(X \setminus U,\ A \setminus U) \hookrightarrow (X, A)$$

induces an isomorphism on singular homology:

$$H_n(X \setminus U,\ A \setminus U) \xrightarrow{\cong} H_n(X, A) \qquad \text{for all } n \geq 0.$$

Remark. The condition $\overline{U} \subset \operatorname{int}(A)$ guarantees a buffer zone. Chains that intersect U can be subdivided so that the pieces lying near U are entirely contained in $\operatorname{int}(A)$, hence vanish in the relative quotient.

8.3 Proof of excision

We prove excision by reducing to chains built from simplices that are small with respect to an open cover.

Definition 8.8. Small chains with respect to an open cover

Let $\mathcal{U}$ be an open cover of X. Define $C_n^{\mathcal{U}}(X) \subset C_n(X)$ to be the subgroup generated by singular simplices $\sigma\colon \Delta^n \to X$ whose image is contained in a single element of $\mathcal{U}$. The boundary of a $\mathcal{U}$-small simplex is a sum of $\mathcal{U}$-small simplices, so $C_\bullet^{\mathcal{U}}(X)$ is a subcomplex of $C_\bullet(X)$.

Definition 8.9. Barycentric subdivision operator

There exists a natural chain map

$$\operatorname{Sd}\colon C_\bullet(X) \to C_\bullet(X)$$

called **barycentric subdivision**, such that for each n and each singular simplex $\sigma\colon \Delta^n \to X$, the chain $\operatorname{Sd}(\sigma)$ is a signed sum of singular n-simplices whose images lie in $\sigma(\Delta^n)$ and are obtained by subdividing Δ^n into smaller simplices.

Proposition 8.10. Subdivision is chain homotopic to the identity

There exists a natural collection of homomorphisms

$$D_n \colon C_n(X) \to C_{n+1}(X)$$

such that

$$\partial D_n + D_{n-1}\partial = \mathrm{Sd} - \mathrm{id} \qquad \text{for all } n.$$

In particular, Sd induces the identity map on homology.

Proof. The construction is done first for $X = \Delta^n$ using an explicit prism-type triangulation that compares Δ^n with its barycentric subdivision, and then transported to general X by naturality using precomposition with a singular simplex $\sigma \colon \Delta^n \to X$. The identity $\partial D + D\partial = \mathrm{Sd} - \mathrm{id}$ follows from a boundary computation on the prism and linearity. Since Sd is chain homotopic to id, it induces the same map on homology. □

Lemma 8.11. Subdivision eventually makes simplices small Let $\mathcal{U}$ be an open cover of X. For every singular simplex $\sigma \colon \Delta^n \to X$ there exists $N \geq 0$ such that $\mathrm{Sd}^N(\sigma) \in C_n^{\mathcal{U}}(X)$. Consequently, for every chain $c \in C_n(X)$ there exists N such that $\mathrm{Sd}^N(c) \in C_n^{\mathcal{U}}(X)$.

Proof. The image $\sigma(\Delta^n)$ is compact, so $\{\sigma^{-1}(U)\}_{U \in \mathcal{U}}$ is an open cover of the compact space Δ^n. Choose $\epsilon > 0$ such that every subset of Δ^n of diameter $< \epsilon$ is contained in some $\sigma^{-1}(U)$. Barycentric subdivision of Δ^n refines the simplex into pieces of arbitrarily small diameter, so for large N each subsimplex of the N-fold subdivision lies in some $\sigma^{-1}(U)$. Therefore each summand simplex in $\mathrm{Sd}^N(\sigma)$ has image contained in a single $U \in \mathcal{U}$. Linearity gives the statement for a finite sum of simplices, hence for a chain. □

Proposition 8.12. Small chains compute the same homology

Let $\mathcal{U}$ be an open cover of X. The inclusion of chain complexes

$$\iota \colon C_\bullet^{\mathcal{U}}(X) \hookrightarrow C_\bullet(X)$$

induces an isomorphism on homology.

Proof. Define a chain map $r \colon C_\bullet(X) \to C_\bullet^{\mathcal{U}}(X)$ as follows. Given $c \in C_n(X)$, choose N large enough so that $\mathrm{Sd}^N(c) \in C_n^{\mathcal{U}}(X)$ by Lemma 8.11, and set

$$r(c) = \mathrm{Sd}^N(c).$$

To make r a genuine homomorphism independent of choices, fix once and for all a choice $N(c)$ for each basis simplex and extend linearly, or equivalently define r by applying a uniform N on each degree after choosing N large enough for a generating set of chains in that degree. In any case, r is a chain map since Sd is.

Now $\iota \circ r$ equals Sd^N on chains, hence induces the identity on homology by Proposition 8.10. On the other hand, $r \circ \iota$ is the restriction of Sd^N to $C_\bullet^{\mathcal{U}}(X)$, and this also induces the identity on homology of $C_\bullet^{\mathcal{U}}(X)$. Thus ι_* is an isomorphism. □

We now apply these constructions to a pair.

Definition 8.13. Small relative chains

For a pair (X, A) and an open cover $\mathcal{U}$ of X, define

$$C_n^{\mathcal{U}}(A) = C_n(A) \cap C_n^{\mathcal{U}}(X), \qquad C_n^{\mathcal{U}}(X, A) = C_n^{\mathcal{U}}(X)/C_n^{\mathcal{U}}(A),$$

and let $H_n^{\mathcal{U}}(X, A)$ be the homology of $C_\bullet^{\mathcal{U}}(X, A)$.

Lemma 8.14. Small relative chains compute relative homology For any open cover $\mathcal{U}$ of X, the inclusion $C_\bullet^{\mathcal{U}}(X) \hookrightarrow C_\bullet(X)$ induces an isomorphism

$$H_n^{\mathcal{U}}(X, A) \xrightarrow{\cong} H_n(X, A) \qquad \text{for all } n.$$

Proof. Consider the commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & C_\bullet^{\mathcal{U}}(A) \arrow[r] \arrow[d, hook] & C_\bullet^{\mathcal{U}}(X) \arrow[r] \arrow[d, hook] & C_\bullet^{\mathcal{U}}(X,A) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & C_\bullet(A) \arrow[r] & C_\bullet(X) \arrow[r] & C_\bullet(X,A) \arrow[r] & 0. \end{tikzcd}$$

The left and middle vertical maps induce isomorphisms on homology by Proposition 8.12 applied to A and to X. The induced long exact sequences in homology and a diagram chase show that the right vertical map induces an isomorphism on homology as well. □

Proof of Theorem 8.7. Let $U \subset A$ with $\overline{U} \subset \mathrm{int}(A)$. Set

$$\mathcal{U} = \{\, X \setminus U,\ \mathrm{int}(A) \,\},$$

which is an open cover of X. We compare relative chain complexes using $\mathcal{U}$-small chains.

Step 1. A chain-level identification. We claim there is a natural isomorphism of chain complexes

$$C_\bullet^{\mathcal{U}}(X,A) \;\cong\; C_\bullet^{\mathcal{U}}(X \setminus U,\ A \setminus U).$$

Indeed, a $\mathcal{U}$-small simplex σ in X has image contained either in $X \setminus U$ or in $\mathrm{int}(A)$. If $\sigma(\Delta^n) \subset \mathrm{int}(A)$ then σ lies in $C_n(A)$, hence its class is 0 in the quotient $C_n^{\mathcal{U}}(X,A)$. Therefore $C_n^{\mathcal{U}}(X,A)$ is generated by classes of $\mathcal{U}$-small simplices whose images lie in $X \setminus U$. Two such chains represent the same class exactly when their difference lies in $C_n^{\mathcal{U}}(A)$. But a $\mathcal{U}$-small simplex in A that lies in $X \setminus U$ is precisely a simplex in $A \setminus U$. Thus the quotient identifies with

$$C_n^{\mathcal{U}}(X \setminus U)/C_n^{\mathcal{U}}(A \setminus U) \;=\; C_n^{\mathcal{U}}(X \setminus U,\ A \setminus U).$$

This identification respects boundaries since it is induced by inclusions and quotients.

Step 2. Pass to homology. By Lemma 8.14, the natural maps

$$H_n^{\mathcal{U}}(X,A) \xrightarrow{\cong} H_n(X,A), \qquad H_n^{\mathcal{U}}(X \setminus U,\ A \setminus U) \xrightarrow{\cong} H_n(X \setminus U,\ A \setminus U)$$

are isomorphisms. Combining these with the chain-level identification in Step 1 yields the desired isomorphism

$$H_n(X \setminus U,\ A \setminus U) \xrightarrow{\cong} H_n(X,A)$$

for all n. □

8.4 First consequences

Proposition 8.15. Homology of a point and of contractible spaces

If X is contractible, then

$$H_n(X) = 0 \text{ for } n > 0, \qquad H_0(X) \cong \mathbb{Z}.$$

Proof. If X is contractible, then $X \simeq \{*\}$. By homotopy invariance of singular homology, it suffices to compute $H_n(\{*\})$. The chain group $C_0(\{*\}) \cong \mathbb{Z}$ is generated by the constant 0-simplex, and for $n \geq 1$ the groups $C_n(\{*\}) \cong \mathbb{Z}$ are generated by the unique constant n-simplex. For $n \geq 1$, the boundary map $\partial\colon C_n(\{*\}) \to C_{n-1}(\{*\})$ is either 0 or $\pm\mathrm{id}$ depending on n, and one checks that the complex is acyclic in positive degrees. Thus $H_0(\{*\}) \cong \mathbb{Z}$ and $H_n(\{*\}) = 0$ for $n > 0$. □

Proposition 8.16. The group H_0 records path components

For any space X, the group $H_0(X)$ is a free abelian group with a basis indexed by the path components of X. In particular, if X is path-connected then $H_0(X) \cong \mathbb{Z}$.

Proof. A singular 0-simplex is just a point of X, so $C_0(X)$ is the free abelian group on the set X. The subgroup $B_0(X)$ is generated by elements of the form $\partial\sigma$ where $\sigma\colon \Delta^1 \to X$ is a singular 1-simplex, and

$$\partial\sigma = \sigma \circ \delta_0 - \sigma \circ \delta_1$$

is the formal difference of the endpoints of the path σ. Therefore, in $H_0(X) = C_0(X)/B_0(X)$, two points become equal if they can be joined by a path. This identifies $H_0(X)$ with the free abelian group on the set of path components. □

We now compute the homology of spheres.

Theorem 8.17. Homology of spheres

For $n \geq 1$,

$$H_k(S^n) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z} & k = n, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Fix a point $x \in S^n$ and set $A = S^n \setminus \{x\}$. Then $A \cong \mathbb{R}^n$ is contractible, so $H_k(A) = 0$ for $k > 0$ and $H_0(A) \cong \mathbb{Z}$ by Proposition 8.15.

Consider the long exact sequence of the pair (S^n, A) from Theorem 16.5. For $k \geq 2$, we obtain isomorphisms

$$H_k(S^n) \;\cong\; H_k(S^n, A),$$

since $H_k(A) = H_{k-1}(A) = 0$ in that range. For $k = 1$, exactness gives $H_1(S^n) \cong H_1(S^n, A)$ because $H_1(A) = 0$ and the map $H_0(A) \to H_0(S^n)$ is an isomorphism (both spaces are path-connected). Also $H_0(S^n) \cong \mathbb{Z}$ by Proposition 8.16.

It remains to compute $H_k(S^n, A)$. Choose a small open ball $B \subset S^n$ around x and set $U = S^n \setminus \overline{B}$. Then $U \subset A$ and $\overline{U} \subset A = \operatorname{int}(A)$ since A is open. By excision Theorem 8.7, the inclusion

$$(B,\ B \setminus \{x\}) \hookrightarrow (S^n,\ S^n \setminus \{x\})$$

induces isomorphisms on relative homology:

$$H_k(B,\ B \setminus \{x\}) \;\cong\; H_k(S^n, A).$$

Identify B with the open n-ball D^n and $B \setminus \{x\}$ with $D^n \setminus \{0\}$. There is a deformation retraction $D^n \setminus \{0\} \simeq S^{n-1}$ given by radial projection, so

$$H_k(D^n \setminus \{0\}) \;\cong\; H_k(S^{n-1}).$$

Apply the long exact sequence of the pair $(D^n,\ D^n \setminus \{0\})$. Since D^n is contractible, for $k \geq 2$ we get isomorphisms

$$H_k(D^n,\ D^n \setminus \{0\}) \;\cong\; H_{k-1}(D^n \setminus \{0\}) \;\cong\; H_{k-1}(S^{n-1}).$$

For $k = 1$, the relevant exact segment is

$$0 \longrightarrow H_1(D^n,\ D^n \setminus \{0\}) \longrightarrow H_0(D^n \setminus \{0\}) \longrightarrow H_0(D^n) \longrightarrow \cdots$$

If $n \geq 2$, then $D^n \setminus \{0\}$ is path-connected, so $H_0(D^n \setminus \{0\}) \to H_0(D^n)$ is an isomorphism and $H_1(D^n,\ D^n \setminus \{0\}) = 0$. If $n = 1$, then $D^1 \setminus \{0\}$ has two components, so $H_0(D^1 \setminus \{0\}) \cong \mathbb{Z} \oplus \mathbb{Z}$ and the map to $H_0(D^1) \cong \mathbb{Z}$ is addition. Its kernel is $\mathbb{Z}$, hence

$$H_1(D^1,\ D^1 \setminus \{0\}) \cong \mathbb{Z}.$$

Putting this together yields:

1. For $n = 1$, $H_1(S^1) \cong H_1(S^1, S^1 \setminus \{x\}) \cong H_1(D^1, D^1 \setminus \{0\}) \cong \mathbb{Z}$.
2. For $n \geq 2$, $H_1(S^n) \cong H_1(S^n, S^n \setminus \{x\}) \cong H_1(D^n, D^n \setminus \{0\}) = 0$.
3. For $k \geq 2$ and all $n \geq 1$,

$$H_k(S^n) \cong H_k(S^n, A) \cong H_k(D^n, D^n \setminus \{0\}) \cong H_{k-1}(S^{n-1}).$$

This gives an induction on n. Using the initial values $H_0(S^1) \cong \mathbb{Z}$ and $H_1(S^1) \cong \mathbb{Z}$ already obtained, the recursion shows that for $n \geq 1$ the only nonzero homology groups of S^n are $H_0(S^n) \cong \mathbb{Z}$ and $H_n(S^n) \cong \mathbb{Z}$. $\square$

Remark. Excision and the long exact sequence are the local and global mechanisms behind glueing results. The Mayer–Vietoris sequence is a systematic way to extract information from a union $X = U \cup V$ by comparing $(U, U \cap V)$ and $(V, U \cap V)$.

9 Mayer–Vietoris and computations

9.1 Mayer–Vietoris sequence, construction and exactness

Let X be a space and let $U, V \subset X$ be open subsets with $X = U \cup V$. Write $i_U \colon U \hookrightarrow X$, $i_V \colon V \hookrightarrow X$, and $i_{UV} \colon U \cap V \hookrightarrow X$ for inclusions.

Definition 9.1. The U–V subcomplex of singular chains

Let $C_n^{U,V}(X) \subset C_n(X)$ be the subgroup generated by those singular n-simplices $\sigma\colon \Delta^n \to X$ whose image is contained in U or contained in V. Since faces of such a simplex still lie in the same open set, the boundary satisfies

$$\partial(C_n^{U,V}(X)) \subset C_{n-1}^{U,V}(X),$$

so $C_\bullet^{U,V}(X)$ is a subcomplex of $C_\bullet(X)$.

Lemma 9.2. A short exact sequence of chain complexes There is a short exact sequence of chain complexes

$$0 \longrightarrow C_\bullet(U \cap V) \xrightarrow{\alpha} C_\bullet(U) \oplus C_\bullet(V) \xrightarrow{\beta} C_\bullet^{U,V}(X) \longrightarrow 0,$$

where

$$\alpha(c) = (c, -c), \qquad \beta(a, b) = (i_U)_\#(a) + (i_V)_\#(b),$$

and we view $(i_U)_\#(a)$ and $(i_V)_\#(b)$ as chains in X.

Proof. Both maps are chain maps because they are defined using inclusions and linearity, and inclusions commute with the boundary on singular chains.
Exactness at $C_\bullet(U \cap V)$ holds since α is injective.
For exactness at $C_\bullet(U) \oplus C_\bullet(V)$, first note $\beta \circ \alpha = 0$ because

$$\beta(c, -c) = (i_U)_\#(c) - (i_V)_\#(c) = 0$$

as chains in X (the two inclusions agree on $U \cap V$). Conversely, suppose $\beta(a, b) = 0$. Then $(i_U)_\#(a) = -(i_V)_\#(b)$ in $C_\bullet(X)$, so both sides are supported in $U \cap V$. More concretely, the chain $(i_U)_\#(a)$ lies in the subgroup $C_\bullet(U) \subset C_\bullet(X)$, while it equals $-(i_V)_\#(b) \in C_\bullet(V) \subset C_\bullet(X)$. Hence $(i_U)_\#(a) \in C_\bullet(U) \cap C_\bullet(V) = C_\bullet(U \cap V)$. Write $(i_U)_\#(a) = (i_{U\cap V})_\#(c)$ for a unique $c \in C_\bullet(U \cap V)$. Then a is exactly the image of c under inclusion into U, and similarly b is the image of $-c$ under inclusion into V. Thus $(a, b) = \alpha(c)$.
For surjectivity of β, take a generator $\sigma \in C_n^{U,V}(X)$. If $\sigma(\Delta^n) \subset U$, then $\sigma = (i_U)_\#(\sigma)$ lies in the image of β via $(\sigma, 0)$. If $\sigma(\Delta^n) \subset V$, it lies in the image via $(0, \sigma)$. Linearity gives surjectivity. □

Proposition 9.3. The U–V subcomplex computes $H_\bullet(X)$

Let $X = U \cup V$ with U, V open. The inclusion of chain complexes

$$\iota\colon C_\bullet^{U,V}(X) \hookrightarrow C_\bullet(X)$$

induces an isomorphism on homology:

$$\iota_*\colon H_n(C_\bullet^{U,V}(X)) \xrightarrow{\cong} H_n(X) \qquad \text{for all } n.$$

Proof. Let $\mathcal{U} = \{U, V\}$ be an open cover of X. A simplex is $\mathcal{U}$-small exactly when its image lies in U or in V, so $C_\bullet^{U,V}(X) = C_\bullet^{\mathcal{U}}(X)$ in the sense of small chains. Barycentric subdivision eventually makes each singular simplex $\mathcal{U}$-small, and subdivision is chain homotopic to the identity. It follows that the inclusion of $\mathcal{U}$-small chains into all chains induces an isomorphism on homology. □

Theorem 9.4. Mayer–Vietoris long exact sequence

Let $X = U \cup V$ with U, V open. There is a natural long exact sequence

$$\cdots \longrightarrow H_n(U \cap V) \xrightarrow{(j_U)_* \oplus (j_V)_*} H_n(U) \oplus H_n(V) \xrightarrow{(i_U)_* + (i_V)_*} H_n(X) \xrightarrow{\delta} H_{n-1}(U \cap V) \longrightarrow \cdots$$

$$\cdots \longrightarrow H_1(X) \xrightarrow{\delta} H_0(U \cap V) \xrightarrow{(j_U)_* \oplus (j_V)_*} H_0(U) \oplus H_0(V) \xrightarrow{(i_U)_* + (i_V)_*} H_0(X) \longrightarrow 0,$$

where $j_U\colon U \cap V \hookrightarrow U$ and $j_V\colon U \cap V \hookrightarrow V$ are inclusions.

Proof. Apply homology to the short exact sequence of chain complexes in Lemma 9.2. This yields a long exact sequence

$$\cdots \to H_n(U \cap V) \xrightarrow{\alpha_*} H_n(U) \oplus H_n(V) \xrightarrow{\beta_*} H_n(C_\bullet^{U,V}(X)) \xrightarrow{\partial_*} H_{n-1}(U \cap V) \to \cdots$$

with connecting map ∂_* as in the long exact sequence of a short exact sequence of complexes.

Under the identification of α and β in Lemma 9.2, the map α_* equals

$$H_n(U \cap V) \to H_n(U) \oplus H_n(V), \qquad [c] \mapsto ((j_U)_*[c], -(j_V)_*[c]),$$

which differs from $((j_U)_*, (j_V)_*)$ by an automorphism of $H_n(V)$. Replacing $H_n(V)$ by itself via multiplication by -1 gives the stated sign convention.

Next, identify $H_n(C_\bullet^{U,V}(X))$ with $H_n(X)$ using Proposition 9.3. Under this identification, β_* is exactly $(i_U)_* + (i_V)_*$ on homology. Finally, the connecting morphism in the Mayer–Vietoris sequence is the composite

$$H_n(X) \cong H_n(C_\bullet^{U,V}(X)) \xrightarrow{\partial_*} H_{n-1}(U \cap V),$$

and exactness follows from exactness of the long exact sequence associated to Lemma 9.2. □

Remark. A concrete description of the connecting homomorphism δ is useful. Represent $[z] \in H_n(X)$ by a cycle $z \in Z_n(C_\bullet^{U,V}(X))$. Choose chains $a \in C_n(U)$ and $b \in C_n(V)$ with $z = (i_U)_\#(a) + (i_V)_\#(b)$. Then $\partial z = 0$ implies $(i_U)_\#(\partial a) = -(i_V)_\#(\partial b)$, so ∂a and ∂b define the same chain in $C_{n-1}(U \cap V)$ up to sign. The class $\delta([z])$ is represented by ∂a (equivalently by $-\partial b$).

9.2 Degree and applications to spheres

The Mayer–Vietoris sequence gives a clean inductive computation of $H_\bullet(S^n)$ and it provides the natural setting for degree.

Proposition 9.5. Recomputing $H_\bullet(S^n)$ by Mayer–Vietoris

Let $n \geq 1$. Then $H_k(S^n) \cong \mathbb{Z}$ for $k = 0, n$ and $H_k(S^n) = 0$ otherwise.

Proof. Write $S^n = U \cup V$ where U and V are the complements of the south and north poles, so both U and V are homeomorphic to $\mathbb{R}^n$ and hence contractible. Their intersection $U \cap V$ deformation retracts onto an equatorial S^{n-1}.

For $k \geq 2$, the Mayer–Vietoris sequence gives

$$0 = H_k(U) \oplus H_k(V) \longrightarrow H_k(S^n) \xrightarrow{\delta} H_{k-1}(U \cap V) \longrightarrow H_{k-1}(U) \oplus H_{k-1}(V) = 0,$$

so $H_k(S^n) \cong H_{k-1}(U \cap V) \cong H_{k-1}(S^{n-1})$. For $k = 1$, exactness gives $H_1(S^n) \cong \ker(H_0(U \cap V) \to H_0(U) \oplus H_0(V))$. When $n \geq 2$, the space $U \cap V$ is path-connected, so $H_0(U \cap V) \cong \mathbb{Z}$ and the map to $\mathbb{Z} \oplus \mathbb{Z}$ is the diagonal, hence injective, so $H_1(S^n) = 0$. The case $n = 1$ yields $H_1(S^1) \cong \mathbb{Z}$. Induction on n finishes the computation. □

Definition 9.6. Fundamental class of S^n

Fix once and for all a generator

$$[S^n] \in H_n(S^n) \cong \mathbb{Z},$$

called the **fundamental class** of S^n. Changing the generator replaces $[S^n]$ by its negative.

Definition 9.7. Degree

Let $f\colon S^n \to S^n$ be continuous. The induced map

$$f_*\colon H_n(S^n) \to H_n(S^n)$$

is an endomorphism of $\mathbb{Z}$, so there exists a unique integer $\deg(f)$ such that

$$f_*([S^n]) = \deg(f)\,[S^n].$$

This integer is called the **degree** of f.

Proposition 9.8. Basic properties of degree

For continuous maps $f, g\colon S^n \to S^n$:

1. $\deg(\mathrm{id}) = 1$ and $\deg(\text{constant map}) = 0$.
2. If $f \simeq g$, then $\deg(f) = \deg(g)$.
3. $\deg(g \circ f) = \deg(g)\deg(f)$.

Proof. (1) The identity induces the identity on homology. A constant map factors through a point, and $H_n(*) = 0$ for $n > 0$, so the induced map on H_n is zero.
(2) If $f \simeq g$, then $f_* = g_*$ by homotopy invariance of homology, hence they scale $[S^n]$ by the same integer.
(3) Functoriality gives $(g \circ f)_* = g_* \circ f_*$. Applying both sides to $[S^n]$ yields

$$(g \circ f)_*([S^n]) = g_*(f_*([S^n])) = g_*(\deg(f)[S^n]) = \deg(f)\, g_*([S^n]) = \deg(f)\deg(g)[S^n].$$

□

Proposition 9.9. Extension forces degree zero

Let $f\colon S^n \to S^n$ be continuous. If f extends to a continuous map $F\colon D^{n+1} \to S^n$ with $F|_{S^n} = f$, then $\deg(f) = 0$.

Proof. Consider the pair (D^{n+1}, S^n) and its long exact sequence:

$$H_{n+1}(D^{n+1}) \longrightarrow H_{n+1}(D^{n+1}, S^n) \xrightarrow{\partial_*} H_n(S^n) \longrightarrow H_n(D^{n+1}).$$

Since D^{n+1} is contractible, the outer groups vanish, so ∂_* is an isomorphism

$$\partial_*\colon H_{n+1}(D^{n+1}, S^n) \xrightarrow{\cong} H_n(S^n).$$

Now apply the induced map of pairs

$$F\colon (D^{n+1}, S^n) \to (S^n, S^n),$$

whose restriction on the boundary is f. The induced map on relative homology

$$F_*\colon H_{n+1}(D^{n+1}, S^n) \to H_{n+1}(S^n, S^n) = 0$$

is the zero map. By naturality of connecting homomorphisms, the square

$$\begin{tikzcd} H_{n+1}(D^{n+1}, S^n) \arrow[r, "F_*"] \arrow[d, "\partial_*"'] & H_{n+1}(S^n, S^n) \arrow[d, "\partial_*"] \\ H_n(S^n) \arrow[r, "f_*"'] & H_n(S^n) \end{tikzcd}$$

commutes. The top arrow is zero, and the left vertical arrow is an isomorphism, so $f_* = 0$ on $H_n(S^n)$. Thus $\deg(f) = 0$. □

Example. For $n \geq 1$, the reflection $r\colon S^n \to S^n$ given by $r(x_0, x_1, \ldots, x_n) = (-x_0, x_1, \ldots, x_n)$ has degree -1. The antipodal map $a(x) = -x$ is a composition of $(n+1)$ coordinate reflections, hence $\deg(a) = (-1)^{n+1}$.

Remark. For $n = 1$, degree agrees with the winding number, and the map on $H_1(S^1) \cong \mathbb{Z}$ matches the map on $\pi_1(S^1) \cong \mathbb{Z}$ under the Hurewicz map.

9.3 Applications to surfaces and basic cell decompositions

Mayer–Vietoris is especially effective on spaces built by gluing simpler pieces along a common overlap.

Proposition 9.10. Homology of a wedge of circles

Let $X = \bigvee_{i=1}^r S^1$ be a wedge of r circles with $r \geq 1$. Then

$$H_k(X) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z}^r & k = 1, \\ 0 & k \geq 2. \end{cases}$$

Proof. Let x_0 be the wedge point. Choose U to be a small contractible neighborhood of x_0 that intersects each circle in an arc, and let $V = X \setminus \{x_0\}$ thickened slightly so that V is open and deformation retracts onto a disjoint union of r open intervals. Then U is contractible, V deformation retracts to a disjoint union of r points, and $U \cap V$ deformation retracts to a disjoint union of r open intervals, hence also to r points.
From Mayer–Vietoris, for $k \geq 2$ one has $H_k(X) = 0$ since $H_k(U) = H_k(V) = H_{k-1}(U \cap V) = 0$. For $k = 1$, exactness gives

$$0 \to H_1(X) \xrightarrow{\delta} H_0(U \cap V) \to H_0(U) \oplus H_0(V) \to H_0(X) \to 0.$$

Here $H_0(U) \cong \mathbb{Z}$, $H_0(V) \cong \mathbb{Z}^r$, and $H_0(U \cap V) \cong \mathbb{Z}^r$. The map $H_0(U \cap V) \to H_0(U) \oplus H_0(V)$ is the diagonal inclusion into the $\mathbb{Z}^r$ summand together with the fold map to $\mathbb{Z}$, so its kernel is free of rank r. Hence $H_1(X) \cong \mathbb{Z}^r$. Finally X is path-connected, so $H_0(X) \cong \mathbb{Z}$. □

Proposition 9.11. Homology of the torus

Let $T^2 = S^1 \times S^1$. Then

$$H_k(T^2) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z}^2 & k = 1, \\ \mathbb{Z} & k = 2, \\ 0 & k \geq 3. \end{cases}$$

Proof. Realize T^2 as a square with opposite sides identified. Let U be the image of the square with a small open neighborhood of the vertical edges removed, and let V be the image with a small open neighborhood of the horizontal edges removed. Then U and V are open and each deformation retracts onto a wedge of two circles. Their intersection $U \cap V$ deformation retracts onto a wedge of three circles.
Thus

$$H_1(U) \cong \mathbb{Z}^2, \quad H_1(V) \cong \mathbb{Z}^2, \quad H_1(U \cap V) \cong \mathbb{Z}^3,$$

and all higher homology of U, V, and $U \cap V$ vanishes. Mayer–Vietoris in degree 2 gives an exact segment

$$0 = H_2(U) \oplus H_2(V) \to H_2(T^2) \xrightarrow{\delta} H_1(U \cap V) \xrightarrow{\phi} H_1(U) \oplus H_1(V) \to H_1(T^2) \to \cdots$$

so $H_2(T^2) \cong \ker(\phi)$. On generators, ϕ records how the three essential loops in $U \cap V$ map into the two generators in U and the two in V. Two of the loops map to the generators coming from the vertical direction and the horizontal direction, while the third loop maps to the sum of these with opposite sign due to boundary orientation in the overlap. This gives a single relation among the three generators, so $\ker(\phi) \cong \mathbb{Z}$, hence $H_2(T^2) \cong \mathbb{Z}$.
Next, exactness shows $\operatorname{coker}(\phi) \cong H_1(T^2)$. The map $\phi\colon \mathbb{Z}^3 \to \mathbb{Z}^2 \oplus \mathbb{Z}^2$ has rank 2, and its image identifies the corresponding generator in U with the one in V for each of the two directions. Thus the cokernel is $\mathbb{Z}^2$. Finally T^2 is path-connected, so $H_0(T^2) \cong \mathbb{Z}$. □

Theorem 9.12. Homology of a closed orientable surface

Let Σ_g be a closed connected orientable surface of genus $g \geq 1$. Then

$$H_k(\Sigma_g) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z}^{2g} & k = 1, \\ \mathbb{Z} & k = 2, \\ 0 & k \geq 3. \end{cases}$$

Proof. View Σ_g as a connected sum $\Sigma_{g-1} \# T^2$. Choose U and V to be open subsets whose union is Σ_g, where U deformation retracts onto Σ_{g-1} with a disk removed, and V deformation retracts onto T^2 with a disk removed. Then $U \cap V$ deformation retracts onto an annulus, hence onto S^1.
The spaces U and V are connected with boundary, so $H_2(U) = H_2(V) = 0$. Also $H_1(U \cap V) \cong \mathbb{Z}$ and $H_k(U \cap V) = 0$ for $k \geq 2$. Mayer–Vietoris in degree 2 gives

$$0 \to H_2(\Sigma_g) \xrightarrow{\delta} H_1(U \cap V) \cong \mathbb{Z} \xrightarrow{\phi} H_1(U) \oplus H_1(V).$$

The map ϕ sends the core circle of the annulus into the boundary circles of U and V with opposite orientations. Each boundary circle is null-homologous inside the corresponding surface-with-boundary, so $\phi = 0$ on H_1. Hence δ is an isomorphism and $H_2(\Sigma_g) \cong \mathbb{Z}$.
Next, Mayer–Vietoris in degree 1 gives exactness

$$H_1(U \cap V) \xrightarrow{\phi} H_1(U) \oplus H_1(V) \to H_1(\Sigma_g) \to H_0(U \cap V) \to \cdots$$

and $H_0(U \cap V) \to H_0(U) \oplus H_0(V)$ is injective since all three spaces are path-connected. Thus

$$H_1(\Sigma_g) \cong (H_1(U) \oplus H_1(V))/\operatorname{im}(\phi).$$

As above, $\phi = 0$ on H_1, so $H_1(\Sigma_g) \cong H_1(U) \oplus H_1(V)$. Now U deformation retracts onto Σ_{g-1} with a disk removed, which has the same H_1 as Σ_{g-1}, and V deformation retracts onto T^2 with a disk removed, which has the same H_1 as T^2. Thus

$$H_1(\Sigma_g) \cong H_1(\Sigma_{g-1}) \oplus H_1(T^2) \cong \mathbb{Z}^{2(g-1)} \oplus \mathbb{Z}^2 \cong \mathbb{Z}^{2g},$$

by induction starting from $g = 1$ where $\Sigma_1 = T^2$. Finally $H_0(\Sigma_g) \cong \mathbb{Z}$ and higher homology vanishes for dimensional reasons. □

Remark. These computations match the Euler characteristic:

$$\chi(\Sigma_g) = \sum_k (-1)^k \operatorname{rank} H_k(\Sigma_g) = 1 - 2g + 1 = 2 - 2g.$$

A later cellular viewpoint packages this into a chain complex built from cells, and Mayer–Vietoris often serves as the bridge between geometric decompositions and algebraic computations.

9.4 Coefficients viewpoint preview

Homology with coefficients in an abelian group G is obtained by replacing $\mathbb{Z}$-linear combinations of simplices with G-linear combinations.

Definition 9.13. Chains and homology with coefficients

Let G be an abelian group. Define

$$C_n(X; G) = C_n(X) \otimes_{\mathbb{Z}} G, \qquad \partial = \partial \otimes \mathrm{id}_G.$$

The homology of this chain complex is denoted $H_n(X; G)$. Relative groups $H_n(X, A; G)$ are defined by

$$C_n(X, A; G) = C_n(X, A) \otimes_{\mathbb{Z}} G.$$

Proposition 9.14. Formal properties persist with coefficients

For any abelian group G, the constructions of long exact sequences of pairs and Mayer–Vietoris sequences carry over to $H_\bullet(-; G)$. Homotopic maps induce the same map on $H_\bullet(-; G)$.

Proof. All statements follow from applying $\otimes_{\mathbb{Z}} G$ to the relevant short exact sequences of chain complexes and from the fact that chain homotopies remain chain homotopies after tensoring with G. □

Remark. When G is a field, each $H_n(X; G)$ is a vector space. Its dimension is the nth Betti number with coefficients in G, and torsion phenomena in integral homology become visible by comparing different coefficient groups.

10 Cellular homology of CW complexes

10.1 The cellular chain groups

Let X be a CW complex, with skeleta

$$X^{-1} = \varnothing \subset X^0 \subset X^1 \subset \cdots \subset X^n \subset \cdots \subset X.$$

Write $\{e^n_\alpha\}_{\alpha \in I_n}$ for the set of open n-cells of X, with characteristic maps

$$\Phi_\alpha \colon (D^n, S^{n-1}) \longrightarrow (X^n, X^{n-1})$$

whose restrictions $\varphi_\alpha = \Phi_\alpha|_{S^{n-1}} \colon S^{n-1} \to X^{n-1}$ are the attaching maps.

Lemma 10.1. The quotient X^n/X^{n-1} There is a natural homeomorphism

$$X^n/X^{n-1} \cong \bigvee_{\alpha \in I_n} S^n_\alpha,$$

where S^n_α is the quotient $D^n_\alpha / S^{n-1}_\alpha$ coming from the αth cell.

Proof. By definition X^n is obtained from X^{n-1} by attaching the disjoint union of n-disks $\coprod_{\alpha \in I_n} D^n_\alpha$ along $\coprod_{\alpha \in I_n} S^{n-1}_\alpha$. Collapsing X^{n-1} to a point collapses each attaching sphere S^{n-1}_α to that same point, so each disk becomes a sphere $D^n_\alpha / S^{n-1}_\alpha \cong S^n_\alpha$. Since the disks were disjoint before attachment, the resulting spheres meet

only at the collapsed point, which is exactly the wedge. □

Definition 10.2. Cellular chain groups

The **cellular chain group** in degree n is

$$C_n(X;\mathbb{Z}) \;:=\; H_n(X^n, X^{n-1};\mathbb{Z}).$$

If G is an abelian group, we set $C_n(X;G) := C_n(X;\mathbb{Z}) \otimes G$.

Proposition 10.3. $C_n(X;\mathbb{Z})$ is free on the n-cells

For each $n \geq 0$ there is a canonical isomorphism

$$C_n(X;\mathbb{Z}) \;\cong\; \bigoplus_{\alpha \in I_n} \mathbb{Z},$$

and after choosing orientations of the n-cells, a preferred basis element $e^n_\alpha \in C_n(X;\mathbb{Z})$ for each $\alpha \in I_n$.

Proof. Consider the quotient map $q\colon X^n \to X^n/X^{n-1}$. It induces an isomorphism on relative homology

$$q_*\colon H_n(X^n, X^{n-1}) \xrightarrow{\cong} \widetilde{H}_n(X^n/X^{n-1}),$$

since $H_n(X^n, X^{n-1})$ is the reduced homology of the quotient. By Lemma 10.1, the quotient is a wedge of n-spheres, so

$$\widetilde{H}_n(X^n/X^{n-1}) \;\cong\; \bigoplus_{\alpha \in I_n} \widetilde{H}_n(S^n_\alpha) \;\cong\; \bigoplus_{\alpha \in I_n} \mathbb{Z}.$$

Choosing an orientation on the disk D^n_α gives a generator of $H_n(D^n_\alpha, S^{n-1}_\alpha) \cong \mathbb{Z}$, hence a generator of the corresponding summand, which we denote e^n_α. □

10.2 The cellular boundary maps and incidence numbers

The skeleta provide long exact sequences for the pairs (X^n, X^{n-1}) and (X^{n-1}, X^{n-2}). Let

$$\partial_n\colon H_n(X^n, X^{n-1}) \longrightarrow H_{n-1}(X^{n-1})$$

be the connecting homomorphism of the pair (X^n, X^{n-1}) and let

$$j_{n-1}\colon H_{n-1}(X^{n-1}) \longrightarrow H_{n-1}(X^{n-1}, X^{n-2})$$

be induced by inclusion of pairs.

Definition 10.4. Cellular boundary operators

Define

$$d_n \;:=\; j_{n-1} \circ \partial_n\colon C_n(X;\mathbb{Z}) \longrightarrow C_{n-1}(X;\mathbb{Z}).$$

The pair $(C_\bullet(X;\mathbb{Z}), d_\bullet)$ is called the **cellular chain complex** of X.

Lemma 10.5. The cellular boundary squares to zero For each $n \geq 1$ we have $d_{n-1} \circ d_n = 0$.

Proof. Consider the triple (X^n, X^{n-1}, X^{n-2}). The connecting morphisms and inclusion maps in the long exact sequences assemble so that

$$H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} H_{n-1}(X^{n-1}) \xrightarrow{j_{n-1}} H_{n-1}(X^{n-1}, X^{n-2}) \xrightarrow{\partial_{n-1}} H_{n-2}(X^{n-2})$$

is a stretch of two successive maps in an exact sequence (after identifying the relevant groups in the LES of the triple). Exactness forces the composition to be zero, hence $j_{n-2} \circ \partial_{n-1} \circ j_{n-1} \circ \partial_n = 0$. But $j_{n-2} \circ \partial_{n-1} = d_{n-1}$ and $j_{n-1} \circ \partial_n = d_n$ by definition, so $d_{n-1}d_n = 0$. □

The utility of the cellular complex comes from the fact that d_n can be read directly from the attaching maps. Fix orientations of the cells, hence a basis $\{e^n_\alpha\}$ for $C_n(X;\mathbb{Z})$ as in Proposition 10.3.

Definition 10.6. Incidence numbers

For an n-cell e^n_α and an $(n-1)$-cell e^{n-1}_β, define $d_{\alpha\beta} \in \mathbb{Z}$ as the degree of the map

$$\Delta_{\alpha\beta}\colon S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \longrightarrow X^{n-1}/(X^{n-1} - e^{n-1}_\beta) \;\cong\; S^{n-1},$$

where the quotient collapses the complement of the open cell e^{n-1}_β to a point, and the last homeomorphism uses the chosen characteristic map for e^{n-1}_β.

Theorem 10.7. Cellular boundary formula

With respect to the cellular bases, the differential is given by

$$d_n(e^n_\alpha) \;=\; \sum_{\beta \in I_{n-1}} d_{\alpha\beta}\, e^{n-1}_\beta,$$

where only finitely many coefficients $d_{\alpha\beta}$ are nonzero.

Proof. Let $q\colon X^{n-1} \to X^{n-1}/X^{n-2}$ be the quotient, and for each β let $q_\beta\colon X^{n-1}/X^{n-2} \to S^{n-1}_\beta$ be the quotient collapsing all cells but e^{n-1}_β. The map $q_\beta q \varphi_\alpha$ is precisely $\Delta_{\alpha\beta}$.
Choose a generator $[D^n] \in H_n(D^n, S^{n-1})$ compatible with the chosen orientation of e^n_α. Under the identification $C_n(X;\mathbb{Z}) = H_n(X^n, X^{n-1})$, the basis element e^n_α is $\Phi_{\alpha*}[D^n]$. Naturality of the connecting map gives

$$\partial_n(e^n_\alpha) = \partial_n(\Phi_{\alpha*}[D^n]) = (\varphi_\alpha)_*(\partial[D^n]) \;\in\; H_{n-1}(X^{n-1}),$$

and then

$$d_n(e^n_\alpha) = j_{n-1}\partial_n(e^n_\alpha) \;\in\; H_{n-1}(X^{n-1}, X^{n-2}).$$

Projecting to the βth summand of $H_{n-1}(X^{n-1}, X^{n-2}) \cong \bigoplus_\beta \mathbb{Z}\, e^{n-1}_\beta$ amounts to applying $(q_\beta q)_*$, which yields

$$(q_\beta q)_*\, d_n(e^n_\alpha) = (q_\beta q \varphi_\alpha)_*(\partial[D^n]) = \Delta_{\alpha\beta*}(\partial[D^n]).$$

Under the standard identification $H_{n-1}(S^{n-1}) \cong \mathbb{Z}$, the image of $\partial[D^n]$ is a preferred generator, so the last expression equals $\deg(\Delta_{\alpha\beta}) \cdot e^{n-1}_\beta$. This shows the coefficient of e^{n-1}_β is $d_{\alpha\beta}$, hence the formula.
Finiteness follows because $\varphi_\alpha(S^{n-1})$ is compact, so it meets only finitely many open cells of X^{n-1}. □

10.3 Cellular homology equals singular homology

Definition 10.8. Cellular homology

The **cellular homology groups** of X are the homology groups of the cellular chain complex:

$$H^{\text{cell}}_n(X;\mathbb{Z}) \;:=\; H_n(C_\bullet(X;\mathbb{Z}), d_\bullet) \;=\; \ker(d_n)/\mathrm{im}(d_{n+1}).$$

Theorem 10.9. Cellular and singular homology agree on CW complexes

For every CW complex X and each $n \geq 0$, there is a natural isomorphism

$$H^{\text{cell}}_n(X;\mathbb{Z}) \;\cong\; H_n(X;\mathbb{Z}),$$

where $H_n(X;\mathbb{Z})$ denotes singular homology.

Proof. Write $H_n(X)$ for singular homology with $\mathbb{Z}$ coefficients. We use the long exact sequences for the skeleta.
Step 1: stabilization of H_n along the skeleta. For the pair (X^m, X^{m-1}), Proposition 10.3 shows that

$$H_k(X^m, X^{m-1}) = 0 \qquad \text{for } k \neq m.$$

Fix n. For any $m \geq n+2$, the long exact sequence of the pair (X^m, X^{m-1}) contains

$$0 = H_n(X^m, X^{m-1}) \longrightarrow H_n(X^{m-1}) \longrightarrow H_n(X^m) \longrightarrow H_{n-1}(X^m, X^{m-1}) = 0,$$

since $n \neq m$ and $n-1 \neq m$. Hence $H_n(X^{m-1}) \to H_n(X^m)$ is an isomorphism for all $m \geq n+2$. Therefore the direct limit $\mathrm{colim}_m H_n(X^m)$ is already $H_n(X^{n+1})$. Since singular chains have compact image and every compact subset of X lies in some skeleton, the canonical map

$$\mathrm{colim}_m H_n(X^m) \;\longrightarrow\; H_n(X)$$

is an isomorphism. Consequently,

$$H_n(X) \;\cong\; H_n(X^{n+1}).$$

Step 2: $H_n(X^{n+1})$ as a quotient of $H_n(X^n)$. Consider the long exact sequence for the pair (X^{n+1}, X^n):

$$H_{n+1}(X^{n+1}, X^n) \xrightarrow{\partial_{n+1}} H_n(X^n) \longrightarrow H_n(X^{n+1}) \longrightarrow H_n(X^{n+1}, X^n) = 0,$$

using again that $H_n(X^{n+1}, X^n) = 0$ since $n \neq n+1$. Exactness yields

$$H_n(X^{n+1}) \;\cong\; H_n(X^n)/\mathrm{im}(\partial_{n+1}).$$

Combining with Step 1 gives

$$H_n(X) \;\cong\; H_n(X^n)/\mathrm{im}(\partial_{n+1}).$$

Step 3: identify $H_n(X^n)$ with a kernel in the cellular complex. Let

$$j_n \colon H_n(X^n) \longrightarrow H_n(X^n, X^{n-1}) = C_n(X;\mathbb{Z})$$

be induced by inclusion of pairs. In the long exact sequence of (X^n, X^{n-1}),

$$H_n(X^{n-1}) \longrightarrow H_n(X^n) \xrightarrow{j_n} H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} H_{n-1}(X^{n-1}),$$

the group $H_n(X^{n-1})$ vanishes because X^{n-1} has no cells in dimension $\geq n$. Thus j_n is injective. Exactness also gives $\mathrm{im}(j_n) = \ker(\partial_n)$.

Now recall $d_n = j_{n-1} \circ \partial_n$. Since j_{n-1} is injective by the same argument, we have

$$\ker(d_n) = \ker(\partial_n) \subseteq C_n(X;\mathbb{Z}),$$

and hence

$$\ker(d_n) = \mathrm{im}(j_n) \;\cong\; H_n(X^n).$$

Step 4: the quotient by $\mathrm{im}(\partial_{n+1})$ matches the quotient by $\mathrm{im}(d_{n+1})$. Because j_n is injective, it restricts to an isomorphism

$$j_n \colon \mathrm{im}(\partial_{n+1}) \xrightarrow{\cong} \mathrm{im}(j_n \partial_{n+1}) = \mathrm{im}(d_{n+1}) \subseteq C_n(X;\mathbb{Z}).$$

Therefore j_n induces an isomorphism on quotients

$$H_n(X^n)/\mathrm{im}(\partial_{n+1}) \;\xrightarrow{\cong}\; \ker(d_n)/\mathrm{im}(d_{n+1}) = H_n^{\mathrm{cell}}(X;\mathbb{Z}).$$

Combining with Step 2 finishes the proof. □

Remark. This comparison is natural in X. A cellular map $f \colon X \to Y$ preserves skeleta, hence induces maps $H_n(X^n, X^{n-1}) \to H_n(Y^n, Y^{n-1})$ commuting with the differentials d_n.

10.4 Computations: spheres and projective spaces

Example. The sphere S^n. Give S^n the CW structure with one 0-cell and one n-cell. Then $C_k(S^n;\mathbb{Z})$ is $\mathbb{Z}$ for $k = 0, n$ and 0 otherwise, and all differentials vanish. Hence

$$H_k(S^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

Example. Real projective space $\mathbb{R}P^n$. There is a CW structure with one cell e^k in each dimension $0 \leq k \leq n$. Let $d_k \colon C_k(\mathbb{R}P^n) \to C_{k-1}(\mathbb{R}P^n)$ be the cellular differential.

For $k \geq 2$, the attaching map of e^k is the double cover $S^{k-1} \to \mathbb{R}P^{k-1}$. To compute d_k, collapse $\mathbb{R}P^{k-1} - e^{k-1}$ to a point, identifying the quotient with S^{k-1}. The resulting map $S^{k-1} \to S^{k-1}$ has degree $1 + (-1)^k$, since the two hemispheres contribute degrees that differ by the antipodal map on S^{k-1}, whose degree is $(-1)^k$. Thus

$$d_k = \begin{cases} 0, & k \text{ odd}, \\ \times 2, & k \text{ even}, \end{cases} \qquad \text{and} \qquad d_1 = 0$$

(with the sign depending on conventions of orientations, but the absolute value is as above).

Therefore, with integral coefficients,

$$H_q(\mathbb{R}P^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, \\ \mathbb{Z}/2, & q \text{ odd and } 0 < q < n, \\ 0, & q \text{ even and } 0 < q < n, \\ \mathbb{Z}, & q = n \text{ and } n \text{ odd}, \\ 0, & q = n \text{ and } n \text{ even}. \end{cases}$$

Example. Complex projective space $\mathbb{C}P^n$. There is a CW structure with exactly one cell in each even dimension $2k$ for $0 \le k \le n$, and no cells in odd dimensions. Hence $C_{2k}(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$ and $C_{2k+1}(\mathbb{C}P^n;\mathbb{Z}) = 0$. All differentials are zero for degree reasons, so

$$H_q(\mathbb{C}P^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, 2, 4, \dots, 2n, \\ 0, & \text{otherwise.} \end{cases}$$

10.5 Euler characteristic

Definition 10.10. Euler characteristic of a finite CW complex

If X is a finite CW complex and $c_n(X)$ denotes the number of open n-cells, define

$$\chi(X) \;:=\; \sum_{n\ge 0} (-1)^n c_n(X) \;\in\; \mathbb{Z}.$$

Proposition 10.11. Euler characteristic from the cellular chain complex

Let X be a finite CW complex and let F be a field. Then

$$\chi(X) \;=\; \sum_{n\ge 0} (-1)^n \dim_F H_n(X;F).$$

In particular, $\chi(X)$ depends only on the homotopy type of X.

Proof. Over a field F, the cellular chain groups $C_n(X;F) = C_n(X;\mathbb{Z}) \otimes F$ are finite-dimensional vector spaces, and

$$\dim_F C_n(X;F) = c_n(X),$$

since $C_n(X;\mathbb{Z})$ is free of rank $c_n(X)$ by Proposition 10.3. For any finite chain complex of finite-dimensional F-vector spaces, linear algebra gives

$$\sum_n (-1)^n \dim_F C_n \;=\; \sum_n (-1)^n \dim_F H_n(C_\bullet).$$

Applying this to the cellular chain complex and using Theorem 10.9 yields the stated formula.
Homotopy invariance follows because singular homology with field coefficients is a homotopy invariant, hence so is the alternating sum of its dimensions. □

Example. Quick values. From the standard CW structures:

$$\chi(S^n) = 1 + (-1)^n, \qquad \chi(\mathbb{R}P^n) = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \qquad \chi(\mathbb{C}P^n) = n + 1.$$

Remark. The cellular chain complex is the point where CW geometry turns into explicit algebra. Later, when we build cup products and characteristic classes, the same cell structures will keep reappearing, often with the cohomology ring of $\mathbb{C}P^n$ and the mod-2 cohomology of $\mathbb{R}P^n$ as the first serious test cases.

11 Eilenberg–Steenrod axioms and uniqueness on CW complexes

11.1 Axioms and reduced homology

Axiomatics is most efficient on CW complexes, where quotients X/A are again CW complexes and cofibrations are built into the geometry of cell attachments.

Definition 11.1. Reduced homology theory on based CW complexes

A **reduced homology theory** $\widetilde{E}_*$ on based CW complexes assigns to each based CW complex $(X, *)$ a $\mathbb{Z}$-graded family of abelian groups $\widetilde{E}_n(X)$ and to each based cellular map $f\colon X \to Y$ homomorphisms $f_*\colon \widetilde{E}_n(X) \to \widetilde{E}_n(Y)$, functorial in f and homotopy invariant, and satisfying:

(i) Exactness for CW pairs. If $A \subset X$ is a subcomplex and $q\colon X \to X/A$ is the quotient, then for each n there are natural boundary homomorphisms

$$\partial\colon \widetilde{E}_n(X/A) \longrightarrow \widetilde{E}_{n-1}(A)$$

such that the sequence

$$\cdots \longrightarrow \widetilde{E}_n(A) \xrightarrow{i_*} \widetilde{E}_n(X) \xrightarrow{q_*} \widetilde{E}_n(X/A) \xrightarrow{\partial} \widetilde{E}_{n-1}(A) \longrightarrow \cdots$$

is exact.

(ii) Suspension. There are natural isomorphisms

$$\Sigma\colon \widetilde{E}_n(X) \xrightarrow{\cong} \widetilde{E}_{n+1}(\Sigma X)$$

for all based CW complexes X and all $n \in \mathbb{Z}$.

(iii) Additivity. If $X = \bigvee_\alpha X_\alpha$ is a wedge of based CW complexes, then the canonical map

$$\bigoplus_\alpha \widetilde{E}_n(X_\alpha) \longrightarrow \widetilde{E}_n(X)$$

induced by the wedge inclusions is an isomorphism for each n.

(iv) Dimension. One has $\widetilde{E}_n(S^0) = 0$ for $n \neq 0$.

Remark. The groups $\widetilde{E}_0(S^0)$ are not fixed by Definition 11.1; they form the coefficient group of the theory. The dimension axiom Item (iv) only forces triviality outside degree 0.

Definition 11.2. Unreduced groups from reduced groups

Given a reduced theory $\widetilde{E}_*$, define for an unbased CW complex X the **unreduced** groups

$$E_n(X) = \widetilde{E}_n(X_+),$$

where $X_+ = X \sqcup \{*\}$ is X with a disjoint basepoint. For a CW pair (X, A) with $A \subset X$ a subcomplex, define the relative groups

$$E_n(X, A) = \widetilde{E}_n(X/A),$$

where the basepoint of X/A is the image of A.

Proposition 11.3. Long exact sequence of a pair

With the definitions in Definition 11.2, the exactness axiom Item (i) gives a natural long exact sequence

$$\cdots \longrightarrow E_n(A) \xrightarrow{i_*} E_n(X) \xrightarrow{j_*} E_n(X, A) \xrightarrow{\partial} E_{n-1}(A) \longrightarrow \cdots$$

for each CW pair (X, A).

Proof. Apply Item (i) to the based CW pair (X_+, A_+). Since $(X_+)/A_+ \cong X/A$ as based CW complexes, the sequence becomes

$$\cdots \to \widetilde{E}_n(A_+) \to \widetilde{E}_n(X_+) \to \widetilde{E}_n(X/A) \xrightarrow{\partial} \widetilde{E}_{n-1}(A_+) \to \cdots,$$

which is exactly the stated sequence after translating notation using Definition 11.2. □

Definition 11.4. Reduced homology from unreduced homology

Conversely, suppose $E_n(X, A)$ is defined on CW pairs and satisfies the usual Eilenberg–Steenrod axioms (homotopy invariance, exactness, excision, and a dimension axiom). For a based CW complex $(X, *)$ with basepoint a 0-cell, define

$$\widetilde{E}_n(X) = E_n(X, *).$$

Proposition 11.5. Reduced and unreduced formalisms agree on CW complexes

If E_* is an unreduced theory on CW pairs, then Definition 11.4 gives a reduced theory on based CW complexes. If $\widetilde{E}_*$ is a reduced theory, then Definition 11.2 gives an unreduced theory on CW pairs. Moreover, these procedures are inverse to each other up to canonical natural isomorphism.

Proof. Given E_* on pairs, the long exact sequence of the pair $(X, *)$ provides boundary maps and exactness for $\widetilde{E}_n(X) = E_n(X, *)$, and homotopy invariance is inherited. Suspension and additivity may be deduced in the standard way by comparing (CX, X) and $(\bigvee X_\alpha, \bigvee *)$ with quotient pairs.
Conversely, given $\widetilde{E}_*$, the construction Definition 11.2 makes exactness automatic by Item (i). Excision for CW triads reduces to the observation that quotient maps identify the relevant based CW complexes, as explained in Lemma 11.12 below. The two constructions undo each other because $(X_+, *)$ and $(X, \emptyset)$ encode the same quotient data, and $E_n(X, A)$ is designed to be computed from X/A. □

11.2 Suspension and the LES of a cofibration

Cofibration sequences supply the basic exact sequence of reduced theories and produce suspension isomorphisms in a clean way.

Definition 11.6. Reduced cone and reduced suspension

For a based space $(X, *)$, the **reduced cone** CX is the quotient

$$CX = (X \times I)/(X \times \{1\} \cup \{*\} \times I),$$

with basepoint the image of $X \times \{1\} \cup \{*\} \times I$. The **reduced suspension** is

$$\Sigma X = CX/(X \times \{0\}),$$

equivalently $(X \times I)/(\{*\} \times I \cup X \times \{0, 1\})$. If X is a based CW complex with basepoint a vertex, then CX and ΣX inherit natural CW structures.

Lemma 11.7. The reduced cone is contractible For any based CW complex X, the space CX is contractible.

Proof. Let $[x, t] \in CX$ denote the image of $(x, t) \in X \times I$. Define $H\colon CX \times I \to CX$ by $H([x, t], s) = [x, (1 - s)t + s]$. At $s = 0$ this is the identity, and at $s = 1$ it sends every point to the basepoint class $[x, 1]$. This is a contraction. □

Proposition 11.8. Suspension isomorphism from exactness

Let $\widetilde{E}_*$ be a reduced theory on based CW complexes. Then for each n and each based CW complex X, the boundary map in the exact sequence for the CW pair (CX, X) induces a natural isomorphism

$$\widetilde{E}_{n+1}(\Sigma X) \cong \widetilde{E}_n(X).$$

Proof. The quotient CX/X is ΣX by definition. Apply exactness Item (i) to the subcomplex $X \subset CX$ to obtain

$$\cdots \to \widetilde{E}_{n+1}(CX) \to \widetilde{E}_{n+1}(CX/X) \xrightarrow{\partial} \widetilde{E}_n(X) \to \widetilde{E}_n(CX) \to \cdots.$$

By Lemma 11.7 and homotopy invariance, $\widetilde{E}_k(CX) = 0$ for all k. Therefore $\partial\colon \widetilde{E}_{n+1}(\Sigma X) \to \widetilde{E}_n(X)$ is an isomorphism. □

Remark. The suspension axiom Item (ii) can be taken as primitive, or obtained from exactness using cones as in Proposition 11.8. For CW complexes, both viewpoints are compatible.

Exactness in Item (i) is already a cofibration exact sequence for the inclusion of a subcomplex. For more general cofibrations, it is useful to recognize relative groups as reduced groups of a quotient.

Definition 11.9. Cofibration sequence

A **cofibration sequence** of based CW complexes is a sequence

$$A \xrightarrow{i} X \xrightarrow{q} X/A$$

where i is the inclusion of a subcomplex and q is the quotient map collapsing A to the basepoint.

Proposition 11.10. The LES of a cofibration

Let $A \subset X$ be a subcomplex. Then there is a natural long exact sequence

$$\cdots \longrightarrow \widetilde{E}_n(A) \xrightarrow{i_*} \widetilde{E}_n(X) \xrightarrow{q_*} \widetilde{E}_n(X/A) \xrightarrow{\partial} \widetilde{E}_{n-1}(A) \longrightarrow \cdots .$$

Proof. This is the exactness axiom Item (i) in Definition 11.1. □

Example. Let $\widetilde{E}_* = \widetilde{H}_*(-;\mathbb{Z})$ be reduced singular homology. Since $\Sigma S^n \cong S^{n+1}$ and $\widetilde{H}_0(S^0) \cong \mathbb{Z}$, repeated suspension gives

$$\widetilde{H}_k(S^n) \cong \begin{cases} \mathbb{Z} & k = n, \\ 0 & k \neq n. \end{cases}$$

This computation uses only the suspension isomorphism and the value on S^0.

11.3 Deriving Mayer–Vietoris and excision from axioms

The axioms are designed so that standard gluing sequences become consequences of exactness and excision. On CW complexes, excision often becomes a statement about quotients.

Definition 11.11. Excisive CW triad

A **CW triad** $(X; A, B)$ consists of a CW complex X and subcomplexes $A, B \subset X$ with $X = A \cup B$. Set $C = A \cap B$.

Lemma 11.12. Excision via quotient identification in CW triads For a CW triad $(X; A, B)$ with $C = A \cap B$, the canonical map of based CW complexes

$$A/C \longrightarrow X/B$$

induced by the inclusion $A \hookrightarrow X$ is an isomorphism of CW complexes.

Proof. The space X/B is obtained from X by collapsing B to a point. Inside X/B, the image of A is A/C with its subcomplex C collapsed, and no further identifications occur on A. Thus the induced map is a homeomorphism compatible with CW structures. □

Proposition 11.13. A useful isomorphism for triads

Let $(X; A, B)$ be a CW triad and set $C = A \cap B$. Then the map induced by inclusions

$$E_n(A, C) \oplus E_n(B, C) \longrightarrow E_n(X, C)$$

is an isomorphism for every n.

Proof. Using Definition 11.2, the relative group $E_n(A, C)$ is $\widetilde{E}_n(A/C)$, and similarly for (B, C) and (X, C). Since X/C is obtained from A/C and B/C by identifying their basepoints, there is a natural homeomorphism

$$X/C \;\cong\; (A/C) \vee (B/C).$$

Apply additivity Item (iii). □

Theorem 11.14. Mayer–Vietoris sequence for CW triads

Let $(X; A, B)$ be a CW triad and set $C = A \cap B$. There is a natural long exact sequence

$$\cdots \longrightarrow E_n(C) \xrightarrow{\psi} E_n(A) \oplus E_n(B) \xrightarrow{\phi} E_n(X) \xrightarrow{\Delta} E_{n-1}(C) \longrightarrow \cdots$$

where $\psi(c) = (i_{A*}c, i_{B*}c)$ and $\phi(a, b) = j_{A*}a - j_{B*}b$ for the inclusion maps $i_A \colon C \to A$, $i_B \colon C \to B$ and $j_A \colon A \to X$, $j_B \colon B \to X$.

Proof. Consider the long exact sequence of the pair (X, B):

$$\cdots \to E_n(B) \xrightarrow{(j_B)_*} E_n(X) \xrightarrow{p} E_n(X, B) \xrightarrow{\partial} E_{n-1}(B) \to \cdots .$$

By Lemma 11.12, the quotient map induces an isomorphism of based CW complexes

$$X/B \cong A/C,$$

hence an isomorphism

$$E_n(X,B) = \widetilde{E}_n(X/B) \cong \widetilde{E}_n(A/C) = E_n(A,C).$$

Under this identification, the boundary map $E_n(X,B) \to E_{n-1}(B)$ corresponds to the composite

$$E_n(A,C) \xrightarrow{\partial} E_{n-1}(C) \xrightarrow{(i_B)_*} E_{n-1}(B),$$

coming from the long exact sequence of the pair (A,C) followed by inclusion. A parallel statement holds with the roles of A and B exchanged.

Now apply the algebraic Mayer–Vietoris principle to the commutative ladder formed by the long exact sequences of triples

$$(B,C,\emptyset) \subset (X,A,\emptyset) \qquad \text{and} \qquad (A,C,\emptyset) \subset (X,B,\emptyset),$$

together with the isomorphism $E_n(X,B) \cong E_n(A,C)$ and its symmetric counterpart. Exactness of the resulting sequence is a diagram chase. The maps ψ and ϕ are the evident ones induced by inclusions, and the sign in ϕ ensures $\phi \circ \psi = 0$. □

Remark. On spaces covered by open sets U and V, one first replaces $(X;U,V)$ by a CW triad using CW approximation of pairs. The Mayer–Vietoris sequence then follows from Theorem 11.14.

11.4 Uniqueness theorem for homology on CW complexes

Reduced theories satisfying additivity are rigid on CW complexes because the skeleta are built by attaching cells.

Lemma 11.15. Quotients of skeleta are wedges of spheres Let X be a CW complex with skeleta $X^{-1} = \emptyset \subset X^0 \subset X^1 \subset \cdots$. For each $n \geq 0$, the quotient X^n/X^{n-1} is canonically homeomorphic to a wedge of n-spheres, one sphere for each n-cell of X.

Proof. The n-skeleton is obtained from X^{n-1} by attaching n-cells:

$$X^n \;=\; X^{n-1} \cup_\alpha \Big(\bigsqcup_{\lambda \in \Lambda_n} D^n_\lambda \Big),$$

with each attaching map $S^{n-1}_\lambda \to X^{n-1}$. Collapsing X^{n-1} to a point collapses each S^{n-1}_λ to that point as well, so each D^n_λ becomes $D^n_\lambda/S^{n-1}_\lambda \cong S^n$. Different n-cells intersect X^{n-1} only along their attaching spheres, so their images intersect only at the basepoint in the quotient. This identifies X^n/X^{n-1} with a wedge of spheres. □

Lemma 11.16. Suspension propagates the coefficient isomorphism Let $\widetilde{E}_*$ and $\widetilde{F}_*$ be reduced theories on based CW complexes. Suppose $\eta_0 \colon \widetilde{E}_0(S^0) \to \widetilde{F}_0(S^0)$ is an isomorphism. Then for every $n \geq 0$ and every $k \in \mathbb{Z}$, there are induced isomorphisms

$$\eta_k(S^n) \colon \widetilde{E}_k(S^n) \xrightarrow{\cong} \widetilde{F}_k(S^n),$$

natural in S^n.

Proof. Use that $S^n \cong \Sigma^n S^0$ as based CW complexes. Iterating suspension isomorphisms Item (ii) transports η_0 to all degrees on all spheres. □

Lemma 11.17. The five lemma Consider a commutative diagram of abelian groups with exact rows

$$\begin{tikzcd}
A_1 \arrow[r] \arrow[d, "\cong"'] & A_2 \arrow[r] \arrow[d, "\cong"'] & A_3 \arrow[r] \arrow[d] & A_4 \arrow[r] \arrow[d, "\cong"'] & A_5 \arrow[d, "\cong"'] \\
B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5.
\end{tikzcd}$$

If the four indicated vertical maps are isomorphisms, then the middle vertical map $A_3 \to B_3$ is also an isomorphism.

Proof. A standard diagram chase in abelian groups. □

Theorem 11.18. Uniqueness on finite CW complexes

Let $\widetilde{E}_*$ and $\widetilde{F}_*$ be reduced homology theories on based CW complexes. Assume there is a natural transformation $\eta\colon \widetilde{E}_* \Rightarrow \widetilde{F}_*$ commuting with suspension isomorphisms and with the boundary maps in the exact sequences of CW pairs. If $\eta_0\colon \widetilde{E}_0(S^0) \to \widetilde{F}_0(S^0)$ is an isomorphism, then for every finite based CW complex X and every n,

$$\eta_n(X)\colon \widetilde{E}_n(X) \xrightarrow{\cong} \widetilde{F}_n(X)$$

is an isomorphism.

Proof. Proceed by induction on the dimension of X.
If X is 0-dimensional, then X is a finite wedge of copies of S^0. By additivity Item (iii), the groups $\widetilde{E}_n(X)$ and $\widetilde{F}_n(X)$ decompose as finite direct sums of the groups on S^0. Since $\eta(S^0)$ is an isomorphism in degree 0 and both theories vanish on S^0 outside degree 0 by Item (iv), it follows that $\eta(X)$ is an isomorphism in all degrees.
Assume the result holds for all finite CW complexes of dimension at most $n-1$ and let X be finite of dimension n. Apply exactness Item (i) to the CW pair (X^n, X^{n-1}) and the corresponding pair for $\widetilde{F}_*$. Naturality gives a commutative diagram of long exact sequences, of which we isolate the segment

$$\begin{tikzcd}
\widetilde{E}_k(X^{n-1}) \arrow[r] \arrow[d, "\cong"'] & \widetilde{E}_k(X^n) \arrow[r] \arrow[d] & \widetilde{E}_k(X^n/X^{n-1}) \arrow[r] \arrow[d, "\cong"'] & \widetilde{E}_{k-1}(X^{n-1}) \arrow[d, "\cong"'] \\
\widetilde{F}_k(X^{n-1}) \arrow[r] & \widetilde{F}_k(X^n) \arrow[r] & \widetilde{F}_k(X^n/X^{n-1}) \arrow[r] & \widetilde{F}_{k-1}(X^{n-1}).
\end{tikzcd}$$

By the induction hypothesis, the maps on X^{n-1} are isomorphisms. By Lemma 11.15 and additivity Item (iii), the map on X^n/X^{n-1} is a direct sum of maps on spheres. By Lemma 11.16, these are isomorphisms. Applying Lemma 11.17 yields that $\widetilde{E}_k(X^n) \to \widetilde{F}_k(X^n)$ is an isomorphism for each k. Since $X = X^n$, the claim follows. □

Proposition 11.19. Passage to infinite CW complexes

Let $\widetilde{E}_*$ be a reduced homology theory on based CW complexes. If X is a CW complex and $\mathcal{K}$ is the directed system of finite subcomplexes $K \subset X$ containing the basepoint, then for each n the canonical map

$$\varinjlim_{K \in \mathcal{K}} \widetilde{E}_n(K) \longrightarrow \widetilde{E}_n(X)$$

is an isomorphism.

Proof. Write X as an increasing union of its finite subcomplexes. Exactness for pairs and additivity control the effect of attaching cells, and a standard directed-limit argument shows that $\widetilde{E}_n$ commutes with this filtered colimit. □

Theorem 11.20. Uniqueness on CW complexes

Under the hypotheses of Theorem 11.18, the map $\eta_n(X)$ is an isomorphism for every based CW complex X and every n.

Proof. Let X be a CW complex. By Proposition 11.19, both $\widetilde{E}_n(X)$ and $\widetilde{F}_n(X)$ are filtered colimits of the values on finite subcomplexes. The natural transformation η is compatible with these colimits. Since η is an isomorphism on each finite subcomplex by Theorem 11.18, it is an isomorphism on the colimit, hence on X. □

12 Homology with coefficients and the universal coefficient theorem

12.1 Coefficients at the chain level

Singular homology is built from a chain complex of free abelian groups. Concretely, for a space X the singular chain group $C_n(X)$ is the free abelian group on singular n-simplices $\sigma : \Delta^n \to X$, with boundary $\partial : C_n(X) \to C_{n-1}(X)$ defined by the alternating sum of face restrictions. In particular, every $C_n(X)$ is a free $\mathbb{Z}$-module.

Definition 12.1. Homology with coefficients via tensor

Let C_* be a chain complex of abelian groups and let G be an abelian group. Define a new chain complex

$$C_* \otimes G \qquad \text{by} \qquad (C_* \otimes G)_n = C_n \otimes_{\mathbb{Z}} G, \quad d_n \otimes 1 : (C_n \otimes G) \to (C_{n-1} \otimes G).$$

If $C_* = C_*(X)$ is the singular chain complex of a space X, define

$$H_n(X; G) := H_n(C_*(X) \otimes G).$$

Functoriality is immediate. A continuous map $f : X \to Y$ induces a chain map $f_\# : C_*(X) \to C_*(Y)$, hence a chain map $f_\# \otimes 1 : C_*(X) \otimes G \to C_*(Y) \otimes G$, and therefore homomorphisms $f_* : H_n(X; G) \to H_n(Y; G)$.

Remark. When R is a commutative ring and M is an R-module, the same construction works with chain complexes of R-modules and $\otimes_R$. For most of this lecture we stay over $\mathbb{Z}$, so every coefficient group is automatically a $\mathbb{Z}$-module.

A basic structural feature is that short exact sequences of coefficient groups produce long exact sequences in homology. The point is that $C_n(X)$ is free, hence tensoring by $C_n(X)$ preserves exactness.

Proposition 12.2. Long exact sequence from a short exact sequence of coefficients

Let $0 \to G' \xrightarrow{i} G \xrightarrow{p} G'' \to 0$ be a short exact sequence of abelian groups. Then for every space X there is a natural long exact sequence

$$\cdots \to H_n(X; G') \xrightarrow{i_*} H_n(X; G) \xrightarrow{p_*} H_n(X; G'') \xrightarrow{\beta} H_{n-1}(X; G') \to \cdots .$$

Proof. Since $C_n(X)$ is free abelian, it is flat. Therefore tensoring the short exact sequence $0 \to G' \to G \to G'' \to 0$ by $C_n(X)$ yields a short exact sequence

$$0 \to C_n(X) \otimes G' \to C_n(X) \otimes G \to C_n(X) \otimes G'' \to 0$$

for each n. These fit together to a short exact sequence of chain complexes $0 \to C_*(X) \otimes G' \to C_*(X) \otimes G \to C_*(X) \otimes G'' \to 0$. The associated long exact sequence in homology is the one displayed, and naturality follows from functoriality of tensoring and of homology. □

Remark. The connecting homomorphism β is often called a *Bockstein homomorphism* for the coefficient sequence. In practice it is a useful extra operation on mod-m (co)homology coming from $0 \to \mathbb{Z}/m \to \mathbb{Z}/m^2 \to \mathbb{Z}/m \to 0$ or from $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to \mathbb{Z}/m \to 0$.

12.2 Why Tor and Ext appear

The tensor product functor $(-) \otimes G$ is right exact but not left exact. The first derived functor measuring its failure of left exactness is Tor. Dually, the functor $\mathrm{Hom}(-, G)$ is left exact but not right exact, and its first derived functor is Ext.

We will only need the basic features over $\mathbb{Z}$, where every subgroup of a free abelian group is free. This is the reason the story stops at the first derived functor for many topological chain complexes.

Definition 12.3. The functor Tor over $\mathbb{Z}$

Let A and G be abelian groups. Choose a free resolution of A of length 1,

$$0 \to F_1 \xrightarrow{d} F_0 \to A \to 0,$$

with F_0, F_1 free abelian. Define

$$\mathrm{Tor}(A, G) := \ker\big(F_1 \otimes G \xrightarrow{d \otimes 1} F_0 \otimes G\big).$$

Remark. Different free resolutions of A yield canonically isomorphic groups $\mathrm{Tor}(A, G)$, and $\mathrm{Tor}(-, G)$ is functorial in A and in G.

A few computations and rules already go a long way.

Proposition 12.4. Basic computations for Tor

Let A, G be abelian groups.

(1) If A is free abelian, then $\mathrm{Tor}(A, G) = 0$.
(2) For $n \geq 1$, $\mathrm{Tor}(\mathbb{Z}/n, G) \cong \{g \in G \mid ng = 0\}$.
(3) If G is torsionfree, then $\mathrm{Tor}(\mathbb{Z}/n, G) = 0$.
(4) Tor preserves direct sums in the first variable: if $A \cong A_1 \oplus A_2$, then $\mathrm{Tor}(A, G) \cong \mathrm{Tor}(A_1, G) \oplus \mathrm{Tor}(A_2, G)$.

Proof. (1) If A is free, the resolution $0 \to 0 \to A \to A \to 0$ gives $\mathrm{Tor}(A, G) = 0$. (2) Use the standard resolution $0 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \to \mathbb{Z}/n \to 0$ and tensor with G. The kernel of $\times n : G \to G$ is exactly $\{g \in G \mid ng = 0\}$. (3) follows from (2). (4) follows by taking direct sums of free resolutions. □

For cohomology we will later need Ext, but even now it is helpful to know its role and a few computations.

Definition 12.5. The functor Ext over $\mathbb{Z}$

Let A and G be abelian groups. Choose a free resolution $0 \to F_1 \to F_0 \to A \to 0$. Apply $\mathrm{Hom}(-, G)$ to obtain an exact sequence

$$0 \to \mathrm{Hom}(A, G) \to \mathrm{Hom}(F_0, G) \to \mathrm{Hom}(F_1, G) \to \mathrm{Ext}(A, G) \to 0,$$

and define $\mathrm{Ext}(A, G)$ as the cokernel of $\mathrm{Hom}(F_0, G) \to \mathrm{Hom}(F_1, G)$.

Proposition 12.6. Basic computations for Ext

Let A, G be abelian groups.

(1) If A is free abelian, then $\mathrm{Ext}(A, G) = 0$.
(2) For $n \geq 1$, $\mathrm{Ext}(\mathbb{Z}/n, G) \cong G/nG$.
(3) Ext preserves direct sums in the first variable: if $A \cong A_1 \oplus A_2$, then $\mathrm{Ext}(A, G) \cong \mathrm{Ext}(A_1, G) \oplus \mathrm{Ext}(A_2, G)$.

Proof. (1) Use the resolution $0 \to A \to A \to 0$. (2) Dualize the standard resolution $0 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \to \mathbb{Z}/n \to 0$. (3) follows from direct sums of resolutions. □

12.3 Universal coefficient theorem for homology

We now explain the precise relationship between $H_n(X; G)$ and the integral homology groups $H_n(X; \mathbb{Z})$. The correct statement is algebraic and applies to any chain complex of free abelian groups. This covers both singular chains and cellular chains of CW complexes.

Theorem 12.7. Universal coefficient theorem for homology

Let C_* be a chain complex of free abelian groups, and let G be an abelian group. Then for every n there is a natural short exact sequence

$$0 \longrightarrow H_n(C_*) \otimes G \xrightarrow{\alpha} H_n(C_* \otimes G) \xrightarrow{\gamma} \mathrm{Tor}\big(H_{n-1}(C_*), G\big) \longrightarrow 0.$$

The sequence splits, but in general no splitting is natural in C_*.

Proof. Write $d_n : C_n \to C_{n-1}$ for the differential, $Z_n = \ker(d_n)$ for cycles, and $B_{n-1} = \mathrm{im}(d_n)$ for boundaries. Since each C_n is free and $B_{n-1} \subset C_{n-1}$ is a subgroup of a free abelian group, B_{n-1} is also free. Therefore the short exact sequence

$$0 \to Z_n \longrightarrow C_n \xrightarrow{d_n} B_{n-1} \to 0$$

splits. Fix a splitting, so we can write

$$C_n \cong Z_n \oplus B_{n-1}$$

in such a way that d_n is the projection $Z_n \oplus B_{n-1} \to B_{n-1}$ followed by the inclusion $B_{n-1} \hookrightarrow C_{n-1}$. Tensor with G. Because B_{n-1} is free, tensoring preserves exactness here, and we obtain

$$C_n \otimes G \cong (Z_n \otimes G) \oplus (B_{n-1} \otimes G).$$

Under this decomposition, the differential

$$d_n \otimes 1 : C_n \otimes G \to C_{n-1} \otimes G$$

vanishes on $Z_n \otimes G$ and restricts on $B_{n-1} \otimes G$ to the map induced by the inclusion $B_{n-1} \hookrightarrow C_{n-1}$. Since $B_{n-1} \subset Z_{n-1}$, this map factors as

$$B_{n-1} \otimes G \xrightarrow{i_{n-1}\otimes 1} Z_{n-1} \otimes G \hookrightarrow C_{n-1} \otimes G,$$

where $i_{n-1} : B_{n-1} \hookrightarrow Z_{n-1}$ is inclusion.
It follows that the cycles in $C_* \otimes G$ in degree n are

$$Z_n(C_* \otimes G) = (Z_n \otimes G) \ \oplus \ \ker(i_{n-1} \otimes 1).$$

On the other hand, the boundaries in degree n are

$$B_n(C_* \otimes G) = \operatorname{im}(d_{n+1} \otimes 1).$$

Under a splitting $C_{n+1} \cong Z_{n+1} \oplus B_n$, the map $d_{n+1} \otimes 1$ vanishes on $Z_{n+1} \otimes G$ and maps $B_n \otimes G$ into $Z_n \otimes G$ via $i_n \otimes 1 : B_n \otimes G \to Z_n \otimes G$. In particular, $B_n(C_* \otimes G) \subset Z_n \otimes G$ and has zero intersection with $\ker(i_{n-1} \otimes 1)$. Therefore we obtain a short exact sequence

$$0 \to \frac{Z_n \otimes G}{(i_n \otimes 1)(B_n \otimes G)} \longrightarrow H_n(C_* \otimes G) \longrightarrow \ker(i_{n-1} \otimes 1) \to 0.$$

We identify both end terms.
First, the quotient $(Z_n \otimes G)/(i_n \otimes 1)(B_n \otimes G)$ is naturally isomorphic to $H_n(C_*) \otimes G$. Indeed the exact sequence $0 \to B_n \to Z_n \to H_n(C_*) \to 0$ has B_n and Z_n free, so tensoring by G remains exact and gives

$$0 \to B_n \otimes G \xrightarrow{i_n \otimes 1} Z_n \otimes G \to H_n(C_*) \otimes G \to 0.$$

Second, $\ker(i_{n-1} \otimes 1)$ is naturally $\operatorname{Tor}(H_{n-1}(C_*), G)$. To see this, apply $(-) \otimes G$ to $0 \to B_{n-1} \xrightarrow{i_{n-1}} Z_{n-1} \to H_{n-1}(C_*) \to 0$. Because Z_{n-1} is free, $\operatorname{Tor}(Z_{n-1}, G) = 0$, so the associated long exact sequence in Tor yields an exact segment

$$0 \to \operatorname{Tor}(H_{n-1}(C_*), G) \to B_{n-1} \otimes G \xrightarrow{i_{n-1}\otimes 1} Z_{n-1} \otimes G.$$

Hence $\operatorname{Tor}(H_{n-1}(C_*), G) \cong \ker(i_{n-1} \otimes 1)$.
Putting these identifications together gives the desired natural short exact sequence.
For the splitting, choose the splitting $C_n \cong Z_n \oplus B_{n-1}$ for all n, and let $p_n : C_n \to Z_n$ be the projection. Composing p_n with $Z_n \to H_n(C_*)$ gives a map $C_n \to H_n(C_*)$, and these assemble to a chain map $C_* \to H_*(C_*)$ where $H_*(C_*)$ is regarded as a chain complex with zero differential. Tensoring with G and taking homology produces a map $H_n(C_* \otimes G) \to H_n(C_*) \otimes G$ that splits α. The dependence on the chosen projections explains why this splitting is typically not natural. □

Remark. For a CW complex X, the cellular chain complex $C_*^{\text{cell}}(X)$ is a chain complex of free abelian groups. Thus the theorem applies directly to cellular homology as well as singular homology.

12.4 Computational templates

The short exact sequence in the universal coefficient theorem is often easiest to use after decomposing integral homology groups into free and torsion parts. Assume $H_n(X;\mathbb{Z})$ is finitely generated. Then

$$H_n(X;\mathbb{Z}) \cong \mathbb{Z}^{r_n} \oplus T_n$$

where T_n is finite abelian. For many choices of G one can read off $H_n(X;G)$ quickly.

Proposition 12.8. Field coefficients

Let F be a field. Then for every n there is a natural isomorphism

$$H_n(X;F) \cong H_n(X;\mathbb{Z}) \otimes F.$$

In particular, if $H_n(X;\mathbb{Z})$ is finitely generated, then $\dim_F H_n(X;F) = r_n$.

Proof. Since F is a $\mathbb{Z}$-module with no torsion, $\operatorname{Tor}(A, F) = 0$ for every abelian group A. Apply Theorem 12.7. □

The most frequently used case is $F = \mathbb{Q}$ or $F = \mathbb{Z}/p$. Over $\mathbb{Q}$ all torsion is killed by tensoring. Over $\mathbb{Z}/p$, torsion of p-power order contributes in two adjacent degrees, one via tensor and one via Tor.

Proposition 12.9. Mod-p coefficients from integral data

Assume $H_n(X;\mathbb{Z})$ and $H_{n-1}(X;\mathbb{Z})$ are finitely generated, and fix a prime p. Then $H_n(X;\mathbb{Z}/p)$ is a direct sum of copies of $\mathbb{Z}/p$ coming from:

(i) one $\mathbb{Z}/p$ for each $\mathbb{Z}$ summand of $H_n(X;\mathbb{Z})$,

(ii) one $\mathbb{Z}/p$ for each $\mathbb{Z}/p^k$ summand of $H_n(X;\mathbb{Z})$,

(iii) one $\mathbb{Z}/p$ for each $\mathbb{Z}/p^k$ summand of $H_{n-1}(X;\mathbb{Z})$.

Proof. Decompose $H_n(X;\mathbb{Z})$ and $H_{n-1}(X;\mathbb{Z})$ into cyclic summands. Using $\mathbb{Z}\otimes\mathbb{Z}/p \cong \mathbb{Z}/p$ and $(\mathbb{Z}/p^k)\otimes\mathbb{Z}/p \cong \mathbb{Z}/p$, we obtain the contributions in (i) and (ii) from $H_n(X;\mathbb{Z}) \otimes \mathbb{Z}/p$. Using $\mathrm{Tor}(\mathbb{Z}/p^k, \mathbb{Z}/p) \cong \mathbb{Z}/p$ and $\mathrm{Tor}(\mathbb{Z}, \mathbb{Z}/p) = 0$ gives (iii) from $\mathrm{Tor}(H_{n-1}(X;\mathbb{Z}), \mathbb{Z}/p)$. □

Example. Suppose X is path-connected and

$$H_2(X;\mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/4, \qquad H_1(X;\mathbb{Z}) \cong \mathbb{Z}/2.$$

Then

$$H_2(X;\mathbb{Z}/2) \cong (\mathbb{Z} \otimes \mathbb{Z}/2) \ \oplus \ ((\mathbb{Z}/4) \otimes \mathbb{Z}/2) \ \oplus \ \mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong (\mathbb{Z}/2) \oplus (\mathbb{Z}/2) \oplus (\mathbb{Z}/2).$$

The long exact sequence from Proposition 12.2 is often paired with the UCT to track how mod-p information lifts to integral information. The connecting morphism is the Bockstein map for the chosen coefficient sequence.

Example. Consider the short exact sequence $0 \to \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \to \mathbb{Z}/p \to 0$. The associated long exact sequence contains

$$H_n(X;\mathbb{Z}) \xrightarrow{\times p} H_n(X;\mathbb{Z}) \to H_n(X;\mathbb{Z}/p) \xrightarrow{\beta} H_{n-1}(X;\mathbb{Z}).$$

When $H_{n-1}(X;\mathbb{Z})$ has p-torsion, the map β can detect it.

Remark. Later, when cohomology and cup products enter, Bockstein operators become cohomology operations of degree 1 with a derivation property. For the present purposes, it is already useful that they are natural connecting morphisms attached to coefficient sequences.

13 Cohomology groups and universal coefficients

13.1 Cochains and the definition of cohomology

Fix an abelian group G. Let $C_*(X)$ be the singular chain complex of a space X, with boundary maps $\partial_n \colon C_n(X) \to C_{n-1}(X)$.

Definition 13.1. Cochains and coboundary

The group of **n-cochains** on X with coefficients in G is

$$C^n(X;G) := \mathrm{Hom}(C_n(X), G).$$

The **coboundary** $\delta \colon C^n(X;G) \to C^{n+1}(X;G)$ is defined by

$$(\delta\varphi)(c) = \varphi(\partial_{n+1}c), \qquad \varphi \in C^n(X;G),\ c \in C_{n+1}(X).$$

Lemma 13.2. The coboundary squares to zero One has $\delta \circ \delta = 0$.

Proof. Let $\varphi \in C^n(X;G)$ and $c \in C_{n+2}(X)$. Then

$$(\delta(\delta\varphi))(c) = (\delta\varphi)(\partial_{n+2}c) = \varphi(\partial_{n+1}\partial_{n+2}c) = \varphi(0) = 0,$$

since $\partial \circ \partial = 0$ on chains. □

Definition 13.3. Cocycles, coboundaries, and cohomology

Define

$$Z^n(X;G) := \ker(\delta \colon C^n(X;G) \to C^{n+1}(X;G)), \qquad B^n(X;G) := \mathrm{im}(\delta \colon C^{n-1}(X;G) \to C^n(X;G)).$$

The n**th cohomology group** is
$$H^n(X;G) := Z^n(X;G)/B^n(X;G).$$

Definition 13.4. Cohomology of a pair

For a pair (X, A), let $C_*(X, A) = C_*(X)/C_*(A)$. Define
$$C^n(X, A; G) := \mathrm{Hom}(C_n(X, A), G), \qquad H^n(X, A; G) := H^n(C^*(X, A; G)).$$

Proposition 13.5. Long exact sequence of a pair

For a pair (X, A) there is a natural long exact sequence
$$\cdots \to H^n(X, A; G) \xrightarrow{j^*} H^n(X; G) \xrightarrow{i^*} H^n(A; G) \xrightarrow{\Delta} H^{n+1}(X, A; G) \to \cdots.$$

Proof. The short exact sequence of chain complexes
$$0 \to C_*(A) \to C_*(X) \to C_*(X, A) \to 0$$
induces a short exact sequence of cochain complexes after applying $\mathrm{Hom}(-, G)$:
$$0 \to C^*(X, A; G) \to C^*(X; G) \to C^*(A; G) \to 0,$$
where exactness holds degreewise because $\mathrm{Hom}(-, G)$ is left exact and $C_n(A) \to C_n(X)$ is injective for each n. The connecting maps in cohomology give the long exact sequence. Naturality follows from functoriality of singular chains. □

13.2 UCT for cohomology

The cochain complex is obtained from the chain complex by applying $\mathrm{Hom}(-, G)$. The universal coefficient theorem identifies the cohomology of $\mathrm{Hom}(C_*, G)$ in terms of homology of C_*.

Theorem 13.6. Universal coefficient theorem for cohomology

Let C_* be a chain complex of free abelian groups and let G be an abelian group. For each n there is a natural short exact sequence
$$0 \longrightarrow \mathrm{Ext}(H_{n-1}(C_*), G) \longrightarrow H^n(\mathrm{Hom}(C_*, G)) \longrightarrow \mathrm{Hom}(H_n(C_*), G) \longrightarrow 0.$$
The sequence splits, but in general no splitting is natural in C_*.

Proof. Write $d_n \colon C_n \to C_{n-1}$ for the differential, $Z_n = \ker(d_n)$, and $B_n = \mathrm{im}(d_{n+1})$. Since C_n is free, every subgroup is free, so Z_n and B_n are free abelian groups.
Consider the short exact sequences
$$0 \to Z_n \longrightarrow C_n \xrightarrow{d_n} B_{n-1} \to 0, \qquad 0 \to B_n \longrightarrow Z_n \longrightarrow H_n(C_*) \to 0.$$
Because B_{n-1} is free, the first sequence splits, hence $\mathrm{Hom}(C_n, G) \to \mathrm{Hom}(Z_n, G)$ is surjective and
$$\mathrm{Hom}(C_n, G) \cong \mathrm{Hom}(Z_n, G) \oplus \mathrm{Hom}(B_{n-1}, G)$$
noncanonically.
Let $C^n = \mathrm{Hom}(C_n, G)$, with coboundary $\delta \colon C^{n-1} \to C^n$ given by $\delta(\psi) = \psi \circ d_n$. A cochain $\varphi \in C^n$ is a cocycle exactly when $\varphi \circ d_{n+1} = 0$, equivalently φ vanishes on B_n. Thus a cocycle restricts to a homomorphism $Z_n \to G$ that vanishes on B_n, hence factors uniquely through $H_n(C_*) = Z_n/B_n$. Define
$$\rho \colon Z^n(\mathrm{Hom}(C_*, G)) \longrightarrow \mathrm{Hom}(H_n(C_*), G)$$
by sending a cocycle φ to the induced map $\overline{\varphi} \colon H_n(C_*) \to G$.
If φ is a coboundary, say $\varphi = \delta(\psi)$, then φ vanishes on Z_n since $d_n|_{Z_n} = 0$. Hence $\rho(\varphi) = 0$, so ρ descends to a homomorphism
$$\overline{\rho} \colon H^n(\mathrm{Hom}(C_*, G)) \longrightarrow \mathrm{Hom}(H_n(C_*), G).$$
Surjectivity of $\overline{\rho}$. Let $f \colon H_n(C_*) \to G$. Compose $Z_n \to H_n(C_*)$ with f to obtain a homomorphism $Z_n \to G$ vanishing on B_n. Using the splitting $C_n \cong Z_n \oplus B_{n-1}$, extend this map by zero on B_{n-1} to a cochain $\varphi \in \mathrm{Hom}(C_n, G)$. Then φ vanishes on $B_n \subset Z_n$, hence $\delta\varphi = 0$. By construction, $\overline{\rho}([\varphi]) = f$.

Identifying the kernel of $\overline{\rho}$. A class $[\varphi] \in H^n(\mathrm{Hom}(C_*, G))$ lies in $\ker(\overline{\rho})$ exactly when the restriction $\varphi|_{Z_n}$ vanishes on Z_n modulo coboundaries. Concretely, $\overline{\rho}([\varphi]) = 0$ means φ vanishes on Z_n. Thus φ factors through the quotient $C_n/Z_n \cong B_{n-1}$, so such cocycles correspond to elements of $\mathrm{Hom}(B_{n-1}, G)$.
Two such cocycles define the same cohomology class precisely when they differ by a coboundary $\delta(\psi)$. For $\delta(\psi)$, its effect on $B_{n-1} \subset C_n$ is given by restricting $\psi \in \mathrm{Hom}(C_{n-1}, G)$ along $B_{n-1} \hookrightarrow C_{n-1}$. Because $C_{n-1} \to Z_{n-1}$ is surjective after splitting $C_{n-1} \cong Z_{n-1} \oplus B_{n-2}$, the image of

$$\mathrm{Hom}(C_{n-1}, G) \longrightarrow \mathrm{Hom}(B_{n-1}, G)$$

coincides with the image of the restriction map $\mathrm{Hom}(Z_{n-1}, G) \to \mathrm{Hom}(B_{n-1}, G)$.
Apply $\mathrm{Hom}(-, G)$ to the short exact sequence

$$0 \to B_{n-1} \longrightarrow Z_{n-1} \longrightarrow H_{n-1}(C_*) \to 0.$$

Since Z_{n-1} is free, $\mathrm{Ext}(Z_{n-1}, G) = 0$, hence there is an exact sequence

$$\mathrm{Hom}(Z_{n-1}, G) \longrightarrow \mathrm{Hom}(B_{n-1}, G) \longrightarrow \mathrm{Ext}(H_{n-1}(C_*), G) \to 0,$$

so the cokernel of the restriction map is $\mathrm{Ext}(H_{n-1}(C_*), G)$. This cokernel is exactly $\ker(\overline{\rho})$. Therefore we obtain a short exact sequence

$$0 \to \mathrm{Ext}(H_{n-1}(C_*), G) \to H^n(\mathrm{Hom}(C_*, G)) \xrightarrow{\overline{\rho}} \mathrm{Hom}(H_n(C_*), G) \to 0.$$

The existence of a splitting follows by choosing splittings $C_n \cong Z_n \oplus B_{n-1}$ and extending maps $H_n \to G$ to cocycles as above. The dependence on these choices explains why a splitting is usually not natural. □

Corollary 13.7 (UCT for singular cohomology)**.** For every space X and abelian group G, there is a natural short exact sequence

$$0 \longrightarrow \mathrm{Ext}(H_{n-1}(X;\mathbb{Z}), G) \longrightarrow H^n(X;G) \longrightarrow \mathrm{Hom}(H_n(X;\mathbb{Z}), G) \longrightarrow 0.$$

Proof. Apply Theorem 13.6 to the singular chain complex $C_*(X)$, which is free in each degree. □

Remark. When $H_n(X;\mathbb{Z})$ is free abelian, the group $\mathrm{Ext}(H_{n-1}(X;\mathbb{Z}), G)$ is the entire obstruction to viewing $H^n(X;G)$ as $\mathrm{Hom}(H_n(X;\mathbb{Z}), G)$.

13.3 Naturality and functorial viewpoints

Cohomology is contravariant in the space and covariant in the coefficient group.

Definition 13.8. Pullback on cochains and cohomology

Let $f\colon X \to Y$ be continuous. The induced chain map $f_\#\colon C_*(X) \to C_*(Y)$ gives a homomorphism on cochains

$$f^*\colon C^n(Y;G) \to C^n(X;G), \qquad (f^*\varphi)(c) = \varphi(f_\#(c)).$$

This commutes with δ, hence induces

$$f^*\colon H^n(Y;G) \to H^n(X;G).$$

Proposition 13.9. Functoriality

For maps $X \xrightarrow{f} Y \xrightarrow{g} Z$ one has $(g \circ f)^* = f^* \circ g^*$ and $(\mathrm{id}_X)^* = \mathrm{id}$ on cohomology.

Proof. The corresponding statements hold on the chain level for $(g \circ f)_\# = g_\# \circ f_\#$, and the cochain pullback is defined by precomposition. □

Definition 13.10. Change of coefficients

A homomorphism of abelian groups $\lambda\colon G \to G'$ induces a cochain map

$$\lambda_*\colon C^n(X;G) \to C^n(X;G'), \qquad \varphi \mapsto \lambda \circ \varphi,$$

hence homomorphisms $\lambda_*\colon H^n(X;G) \to H^n(X;G')$.

Proposition 13.11. Naturality of the UCT sequence

The short exact sequence in Corollary 13.7 is natural in X and in G. In particular, for $f\colon X \to Y$ the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Ext}(H_{n-1}(Y), G) \arrow[r] \arrow[d, "f^*"'] & H^n(Y;G) \arrow[r] \arrow[d, "f^*"'] & \mathrm{Hom}(H_n(Y), G) \arrow[r] \arrow[d, "f^*"'] & 0 \\
0 \arrow[r] & \mathrm{Ext}(H_{n-1}(X), G) \arrow[r] & H^n(X;G) \arrow[r] & \mathrm{Hom}(H_n(X), G) \arrow[r] & 0
\end{tikzcd}$$

commutes, where f^* on $\mathrm{Hom}(H_n(-), G)$ is precomposition with $f_*\colon H_n(X) \to H_n(Y)$.

Proof. Every map in Theorem 13.6 is defined from the functorial constructions $C_* \mapsto \mathrm{Hom}(C_*, G)$, taking cocycles, and passing to homology and cohomology. The map to $\mathrm{Hom}(H_n, G)$ is induced by evaluation on cycles, hence is compatible with chain maps. The identification of the kernel with $\mathrm{Ext}(H_{n-1}, G)$ comes from the exact sequence obtained by applying $\mathrm{Hom}(-, G)$ to $0 \to B_{n-1} \to Z_{n-1} \to H_{n-1} \to 0$, which is functorial. □

The map $H^n(X;G) \to \mathrm{Hom}(H_n(X;\mathbb{Z}), G)$ in Corollary 13.7 is the evaluation pairing.

Definition 13.12. Evaluation pairing

Let $[\varphi] \in H^n(X;G)$ and let $[z] \in H_n(X;\mathbb{Z})$. Choose a cocycle representative $\varphi \in Z^n(X;G)$ and a cycle representative $z \in Z_n(X)$. Define

$$\langle [\varphi], [z] \rangle := \varphi(z) \in G.$$

Lemma 13.13. Well-definedness and naturality of evaluation The pairing in Definition 13.12 is well-defined and satisfies

$$\langle f^* a, b \rangle = \langle a, f_* b \rangle$$

for $a \in H^n(Y;G)$, $b \in H_n(X;\mathbb{Z})$, and $f\colon X \to Y$.

Proof. If φ is changed by a coboundary, its value on cycles changes by zero. If z is changed by a boundary, a cocycle evaluates to zero on it. For naturality, compute using the definitions:

$$\langle f^*[\varphi], [z] \rangle = (f^*\varphi)(z) = \varphi(f_\# z) = \langle [\varphi], f_*[z] \rangle.$$

□

13.4 Computations via cellular cochains

Let X be a CW complex. Write $C_n^{\mathrm{cell}}(X)$ for the cellular chain group, the free abelian group generated by the n-cells, with boundary maps determined by degrees of attaching maps. The cellular chain complex is free in each degree.

Definition 13.14. Cellular cochains

Define the **cellular cochain group** by

$$C^n_{\mathrm{cell}}(X;G) := \mathrm{Hom}(C_n^{\mathrm{cell}}(X), G).$$

The coboundary $\delta_{\mathrm{cell}}\colon C^n_{\mathrm{cell}}(X;G) \to C^{n+1}_{\mathrm{cell}}(X;G)$ is precomposition with the cellular boundary $\partial^{\mathrm{cell}}_{n+1}\colon C^{\mathrm{cell}}_{n+1}(X) \to C^{\mathrm{cell}}_n(X)$. Define

$$H^n_{\mathrm{cell}}(X;G) := H^n(C^*_{\mathrm{cell}}(X;G)).$$

Proposition 13.15. Cellular cochains as functions on cells

If Λ_n denotes the set of n-cells of X, then there is a canonical identification

$$C^n_{\mathrm{cell}}(X;G) \cong \prod_{\lambda \in \Lambda_n} G,$$

sending a cochain to its values on the basis of n-cells. Under this identification, δ_{cell} is given by an incidence formula

$$(\delta_{\mathrm{cell}} a)(e^{n+1}_\mu) = \sum_{\lambda \in \Lambda_n} [e^{n+1}_\mu : e^n_\lambda]\ a(e^n_\lambda),$$

where $[e_\mu^{n+1} : e_\lambda^n] \in \mathbb{Z}$ are the cellular incidence numbers.

Proof. Since $C_n^{\text{cell}}(X) \cong \bigoplus_{\lambda\in\Lambda_n} \mathbb{Z}$ is free on the n-cells,

$$\operatorname{Hom}\Big(\bigoplus_{\lambda\in\Lambda_n} \mathbb{Z},\, G\Big) \cong \prod_{\lambda\in\Lambda_n} \operatorname{Hom}(\mathbb{Z}, G) \cong \prod_{\lambda\in\Lambda_n} G.$$

The formula for δ_{cell} is the dual of the cellular boundary formula. □

Theorem 13.16. Cellular cohomology agrees with singular cohomology

For every CW complex X and every abelian group G, there is a natural isomorphism

$$H^n_{\text{cell}}(X; G) \cong H^n(X; G).$$

Proof. Both $H^*_{\text{cell}}(-; G)$ and $H^*(-; G)$ are reduced cohomology theories on CW complexes. They are homotopy invariant, satisfy long exact sequences of pairs for subcomplex inclusions, and take wedges to products. On spheres, the cellular cochain complex has one generator in degrees 0 and n and gives the same groups as singular cohomology. The uniqueness argument on CW complexes, proved previously for homology and adapting verbatim for cohomology using the long exact sequences of skeleta and the five lemma, yields the result. □

Example. Let $X = S^n$ with the CW structure having one 0-cell and one n-cell. Then $C_k^{\text{cell}}(S^n) \cong \mathbb{Z}$ for $k = 0, n$ and is 0 otherwise, and all cellular differentials are zero. Hence

$$H^k(S^n; G) \cong \begin{cases} G & k = 0, n, \\ 0 & \text{otherwise.} \end{cases}$$

Example. Let $X = \mathbb{R}P^2$ with one cell e^k in each dimension $k = 0, 1, 2$. The cellular boundary map $\partial_2^{\text{cell}} \colon C_2^{\text{cell}} \to C_1^{\text{cell}}$ is multiplication by 2, and $\partial_1^{\text{cell}} = 0$. Thus $C^0_{\text{cell}}(X; \mathbb{Z}) \cong \mathbb{Z}$, $C^1_{\text{cell}}(X; \mathbb{Z}) \cong \mathbb{Z}$, $C^2_{\text{cell}}(X; \mathbb{Z}) \cong \mathbb{Z}$, with $\delta^0_{\text{cell}} = 0$ and δ^1_{cell} multiplication by 2. Therefore

$$H^0(\mathbb{R}P^2; \mathbb{Z}) \cong \mathbb{Z}, \qquad H^1(\mathbb{R}P^2; \mathbb{Z}) = 0, \qquad H^2(\mathbb{R}P^2; \mathbb{Z}) \cong \mathbb{Z}/2.$$

With $G = \mathbb{Z}/2$, the map δ^1_{cell} becomes zero, giving

$$H^k(\mathbb{R}P^2; \mathbb{Z}/2) \cong \mathbb{Z}/2 \quad \text{for } k = 0, 1, 2.$$

Remark. For finite CW complexes, Corollary 13.7 is often the fastest route from $H_*(X; \mathbb{Z})$ to $H^*(X; G)$. Cellular cochains give an independent computation, and the comparison is frequently a good way to locate torsion contributions coming from Ext.

14 Cup product and the cohomology ring

14.1 Cross product and cup product definitions

Fix a commutative coefficient ring R. Write $S_*(X) = C_*(X; \mathbb{Z})$ for singular chains with integer coefficients. We use $C^n(X; R) = \operatorname{Hom}(S_n(X), R)$ with coboundary $\delta(\varphi) = \varphi \circ \partial$ and $H^n(X; R) = H^n(C^*(X; R))$.

The basic multiplicative structure comes from two chain-level ingredients.

Definition 14.1. The shuffle (Eilenberg–Zilber) map

There is a natural chain map

$$\mathrm{EZ} \colon S_*(X) \otimes S_*(Y) \longrightarrow S_*(X \times Y)$$

which on a pair of singular simplices $\sigma \colon \Delta^p \to X$ and $\tau \colon \Delta^q \to Y$ is given by a signed sum over (p, q)-shuffles, producing a singular $(p + q)$-chain in $X \times Y$.

Remark. The map EZ is strictly associative at the chain level when constructed via shuffles. It is part of the Eilenberg–Zilber theorem, together with the Alexander–Whitney map in the opposite direction, and these maps are chain homotopy inverses.

Dually, one may use a diagonal approximation on a single space.

Definition 14.2. Diagonal approximation and the Alexander–Whitney map

An **approximation of the diagonal** is a natural chain map

$$D\colon S_*(X) \longrightarrow S_*(X) \otimes S_*(X)$$

which in degree 0 sends a point x to $x \otimes x$. A standard choice is the **Alexander–Whitney** diagonal: for a singular n-simplex $\sigma\colon \Delta^n \to X$ and a decomposition $n = p + q$ define $\sigma_p^1 = \sigma \circ a_p$ and $\sigma_q^2 = \sigma \circ b_q$ using the front and back face maps $a_p\colon \Delta^p \to \Delta^n$ and $b_q\colon \Delta^q \to \Delta^n$, and set

$$D(\sigma) = \sum_{p+q=n} \sigma_p^1 \otimes \sigma_q^2,$$

extended linearly.

Remark. Any two natural diagonal approximations are naturally chain homotopic. Consequently, the induced product on cohomology does not depend on which diagonal approximation one uses.

Definition 14.3. Cross product on cochains

Let $\varphi \in C^p(X; R)$ and $\psi \in C^q(Y; R)$. Define $\varphi \times \psi \in C^{p+q}(X \times Y; R)$ by

$$(\varphi \times \psi)(c) := (\varphi \otimes \psi)(\mathrm{AW}(c)),$$

where $\mathrm{AW}\colon S_*(X \times Y) \to S_*(X) \otimes S_*(Y)$ is the Alexander–Whitney chain map. Equivalently, one can define a chain-level cross product $S_p(X) \otimes S_q(Y) \to S_{p+q}(X \times Y)$ using shuffles and then dualize.

Definition 14.4. Cup product

Let $\varphi \in C^p(X; R)$ and $\psi \in C^q(X; R)$. Define a cochain $\varphi \smile \psi \in C^{p+q}(X; R)$ by

$$\varphi \smile \psi := (\varphi \otimes \psi) \circ D,$$

where $D\colon S_{p+q}(X) \to \bigoplus_{i+j=p+q} S_i(X) \otimes S_j(X)$ is a chosen diagonal approximation. For the Alexander–Whitney diagonal, if $\sigma\colon \Delta^{p+q} \to X$ then

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma|[0, \dots, p]) \cdot \psi(\sigma|[p, \dots, p+q]),$$

where $\sigma|[0, \dots, p]$ and $\sigma|[p, \dots, p+q]$ denote the front and back faces.

Proposition 14.5. Leibniz rule for the coboundary

For $\varphi \in C^p(X; R)$ and $\psi \in C^q(X; R)$ one has

$$\delta(\varphi \smile \psi) = (\delta\varphi) \smile \psi + (-1)^p \varphi \smile (\delta\psi).$$

In particular, cup product descends to a bilinear pairing

$$H^p(X; R) \otimes_R H^q(X; R) \longrightarrow H^{p+q}(X; R).$$

Proof. Since D is a chain map, $(\partial \otimes 1 + 1 \otimes \partial) \circ D = D \circ \partial$. Evaluate $\delta(\varphi \smile \psi)$ on a chain $c \in S_{p+q+1}(X)$:

$$\delta(\varphi \smile \psi)(c) = (\varphi \otimes \psi)(D(\partial c)) = (\varphi \otimes \psi)((\partial \otimes 1 + 1 \otimes \partial)D(c)).$$

The term $(\partial \otimes 1)$ produces $(\delta\varphi) \smile \psi$. For the term $(1 \otimes \partial)$, moving ∂ past a degree-p cochain contributes the sign $(-1)^p$. This is the usual Koszul sign rule for graded Hom. □

Proposition 14.6. Unit and functoriality

The class $1 \in H^0(X; R)$ represented by the constant cochain $S_0(X) \to R$ is a two-sided unit for $\smile$. For a continuous map $f\colon X \to Y$,

$$f^*(a \smile b) = f^*a \smile f^*b$$

for all $a, b \in H^*(Y; R)$.

Proof. For the unit, D agrees with $x \mapsto x \otimes x$ in degree 0, and the Alexander–Whitney formula shows that 1 evaluates to $1 \in R$ on every 0-simplex, so $1 \smile \varphi = \varphi = \varphi \smile 1$ at the cochain level. For functoriality, naturality

of D means $(f_\# \otimes f_\#) \circ D = D \circ f_\#$. Taking $\mathrm{Hom}(-, R)$ and using the definition of $\smile$ gives the identity. □

Remark. Associativity holds on cohomology because two natural ways of iterating a diagonal approximation are naturally chain homotopic. With the Alexander–Whitney diagonal, associativity holds already at the cochain level.

14.2 Graded commutativity and functoriality

Let $p, q \geq 0$ and $\varphi \in C^p(X; R)$, $\psi \in C^q(X; R)$. The relation $\varphi \smile \psi = (-1)^{pq}\psi \smile \varphi$ is generally false on the cochain level. The correct statement is that the difference is a coboundary when φ and ψ are cocycles. A convenient chain-level witness is the *cup$_1$ product*.

Definition 14.7. The cup$_1$ product

There is a bilinear operation

$$\smile_1 \colon C^p(X; R) \otimes C^q(X; R) \longrightarrow C^{p+q-1}(X; R)$$

defined by an explicit alternating sum over overlaps of front and back faces of a simplex. It is natural in X.

Proposition 14.8. A coboundary formula for cup$_1$

For $\varphi \in C^p(X; R)$ and $\psi \in C^q(X; R)$ one has

$$\delta(\varphi \smile_1 \psi) = \varphi \smile \psi - (-1)^{pq}\psi \smile \varphi \; - \; (\delta\varphi) \smile_1 \psi \; - \; (-1)^p \, \varphi \smile_1 (\delta\psi).$$

Proof. This is a direct computation from the simplicial formula for $\smile$ and $\smile_1$ and the definition of δ on cochains. The signs are arranged so that the terms produced by δ cancel in pairs, leaving precisely the stated combination. □

Theorem 14.9. Graded commutativity

If $a \in H^p(X; R)$ and $b \in H^q(X; R)$, then

$$a \smile b = (-1)^{pq} \, b \smile a \qquad \text{in } H^{p+q}(X; R).$$

Proof. Choose cocycle representatives φ and ψ for a and b. Then $\delta\varphi = \delta\psi = 0$, and Proposition 14.8 reduces to

$$\varphi \smile \psi - (-1)^{pq}\psi \smile \varphi = \delta(\varphi \smile_1 \psi),$$

so the difference is a coboundary and vanishes in cohomology. □

Remark. If $R = \mathbb{Z}/2$, then $(-1)^{pq} = 1$ and the cup product is strictly commutative on cohomology. Over rings where $2 \neq 0$, graded commutativity is the correct replacement.

14.3 Künneth formula in cohomology, usable form

The cross product and the cup product interact well, and under mild hypotheses the cross product computes the cohomology of a product space.

Proposition 14.10. Compatibility of $\times$ and $\smile$

Let $a_1 \in H^p(X; R)$, $a_2 \in H^{p'}(X; R)$, $b_1 \in H^q(Y; R)$, and $b_2 \in H^{q'}(Y; R)$. Then in $H^{p+p'+q+q'}(X \times Y; R)$ one has

$$(a_1 \times b_1) \smile (a_2 \times b_2) = (-1)^{qp'} \, (a_1 \smile a_2) \times (b_1 \smile b_2).$$

Proof. At the cochain level, the diagonal on $X \times Y$ is naturally chain homotopic to the composite

$$S_*(X \times Y) \xrightarrow{D} S_*(X \times Y) \otimes S_*(X \times Y) \xrightarrow{\mathrm{AW} \otimes \mathrm{AW}} (S_*(X) \otimes S_*(Y)) \otimes (S_*(X) \otimes S_*(Y)),$$

followed by regrouping factors. Swapping the middle factors $S_*(Y) \otimes S_*(X)$ produces the sign $(-1)^{qp'}$ in degree (q, p'). Evaluating against $(a_1 \otimes b_1) \otimes (a_2 \otimes b_2)$ yields the identity in cohomology. □

Theorem 14.11. Künneth theorem for cohomology, a common hypothesis

Let X and Y be CW complexes, and let R be a principal ideal domain. Assume $H^k(Y;R)$ is a finitely generated free R-module for all k. Then the cross product induces an isomorphism of graded rings

$$H^*(X;R) \,\widehat{\otimes}_R\, H^*(Y;R) \xrightarrow{\cong} H^*(X \times Y;R),$$

where $\widehat{\otimes}$ denotes the graded tensor product, incorporating the Koszul sign rule.

Proof. Under the stated freeness hypothesis, the cohomological Künneth map is an isomorphism on groups. The ring structure on $H^*(X \times Y;R)$ is determined by those on $H^*(X;R)$ and $H^*(Y;R)$ via Proposition 14.10. □

Corollary 14.12 (Field coefficients)**.** If F is a field, then for CW complexes X and Y the cross product is an isomorphism of graded F-algebras

$$H^*(X;F) \otimes_F H^*(Y;F) \xrightarrow{\cong} H^*(X \times Y;F).$$

Proof. Over a field, all vector spaces are free and Tor terms vanish in the Künneth spectral sequence. The cross product gives an isomorphism of graded vector spaces, and Proposition 14.10 identifies the algebra structure. □

Example. Let $k_1, \dots, k_n$ be odd. Over $\mathbb{Z}$, the Künneth theorem yields

$$H^*(S^{k_1} \times \cdots \times S^{k_n};\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha_1, \dots, \alpha_n], \qquad |\alpha_i| = k_i,$$

an exterior algebra on odd-degree generators. If some k_i are even, then $H^*(S^{k_i};\mathbb{Z})$ is a truncated polynomial ring $\mathbb{Z}[\alpha]/(\alpha^2)$ with $|\alpha| = k_i$, and graded tensor products mix the two behaviors.

14.4 Computations: projective spaces and basic H-spaces hints

Projective spaces are the first family where the ring structure carries real information beyond the groups.

Theorem 14.13. Cohomology rings of projective spaces

Let $n \geq 1$.

(1) As a graded ring,

$$H^*(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{n+1}), \qquad |\alpha| = 2.$$

(2) With $\mathbb{Z}/2$ coefficients,

$$H^*(\mathbb{R}P^n;\mathbb{Z}/2) \cong (\mathbb{Z}/2)[\beta]/(\beta^{n+1}), \qquad |\beta| = 1.$$

(3) As a graded ring,

$$H^*(\mathbb{H}P^n;\mathbb{Z}) \cong \mathbb{Z}[\gamma]/(\gamma^{n+1}), \qquad |\gamma| = 4.$$

Proof. (1) The CW structure on $\mathbb{C}P^n$ has one cell in each even dimension $0, 2, \dots, 2n$. Hence additively $H^{2i}(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$ for $0 \leq i \leq n$ and $H^k = 0$ for odd k. Let $\alpha \in H^2(\mathbb{C}P^n;\mathbb{Z})$ be a generator whose restriction to $\mathbb{C}P^1 \cong S^2$ is the standard generator. The inclusion $\mathbb{C}P^{n-1} \hookrightarrow \mathbb{C}P^n$ induces an isomorphism on H^{2i} for $2i \leq 2n-2$, so by induction α^i generates H^{2i} for $i < n$. The top group $H^{2n}(\mathbb{C}P^n;\mathbb{Z})$ is also infinite cyclic. Since multiplication by α maps H^{2n-2} to H^{2n}, there is an integer m such that $\alpha \smile (m\alpha^{n-1}) = m\alpha^n$ is a generator of H^{2n}. This forces $m = \pm 1$, so α^n generates the top group and the only relation is $\alpha^{n+1} = 0$ for dimensional reasons. Thus the ring is $\mathbb{Z}[\alpha]/(\alpha^{n+1})$.

(2) The CW structure on $\mathbb{R}P^n$ has one cell in every dimension $0, 1, \dots, n$. With $\mathbb{Z}/2$ coefficients, additively $H^k(\mathbb{R}P^n;\mathbb{Z}/2) \cong \mathbb{Z}/2$ for $0 \leq k \leq n$. Let $\beta \in H^1(\mathbb{R}P^n;\mathbb{Z}/2)$ be the nonzero class. The same inductive argument using the skeletal inclusion $\mathbb{R}P^{n-1} \hookrightarrow \mathbb{R}P^n$ shows that β^k is nonzero for $k \leq n$ and generates H^k, while $\beta^{n+1} = 0$. Hence the truncated polynomial description.

(3) The CW structure on $\mathbb{H}P^n$ has one cell in each dimension $0, 4, \dots, 4n$. The argument from (1) applies with $|\gamma| = 4$. □

Example. For $\mathbb{C}P^\infty$ one obtains

$$H^*(\mathbb{C}P^\infty;\mathbb{Z}) \cong \mathbb{Z}[\alpha], \qquad |\alpha| = 2,$$

since there is no top cell imposing a truncation relation.

Remark. Later, $\alpha \in H^2(\mathbb{C}P^\infty;\mathbb{Z})$ will be realized as the first Chern class of the tautological line bundle, and the polynomial structure becomes the algebraic home for characteristic classes.

A different source of structure appears when a space admits a multiplication.

Definition 14.14. H-space

An H**-space** is a based space (X, e) together with a based map $\mu\colon X \times X \to X$ such that $\mu(e, x)$ and $\mu(x, e)$ are based homotopic to x.

Remark. The induced map $\mu^*\colon H^*(X;R) \to H^*(X \times X;R)$ interacts with the cup product and, via the Künneth isomorphism when available, can be viewed as a coproduct. This makes $H^*(X;R)$ into a kind of Hopf algebra object. For connected compact Lie groups, the rational cohomology ring is an exterior algebra on odd-degree generators, mirroring the behavior of products of odd spheres at the level of cohomology.

Example. If G is a topological group, then G is an H-space via multiplication. For instance, $SU(2) \cong S^3$, and more generally $U(n)$ and $SU(n)$ have the same cohomology groups as products of odd spheres $S^1 \times S^3 \times \cdots \times S^{2n-1}$ and $S^3 \times \cdots \times S^{2n-1}$, including their cup product structures.

15 Cap product and Poincaré duality

15.1 Orientations and fundamental classes

Let M be a (Hausdorff, second countable) topological n-manifold, and fix a commutative ring R. For $x \in M$ consider the pair $(M, M-\{x\})$. Choose a coordinate chart $U \cong \mathbb{R}^n$ with $x \in U$. Excision and homotopy invariance give

$$H_i(M, M-\{x\};R) \cong H_i(U, U-\{x\};R) \cong \widetilde{H}_{i-1}(U-\{x\};R) \cong \widetilde{H}_{i-1}(S^{n-1};R).$$

Hence $H_i(M, M-\{x\};R) = 0$ for $i \neq n$ and

$$H_n(M, M-\{x\};R) \cong R,$$

canonically up to the choice of a generator.

Definition 15.1. Local orientation and R-orientability

An R**-orientation at** $x \in M$ is a choice of generator

$$\mu_x \in H_n(M, M-\{x\};R) \cong R.$$

An R**-orientation of** M is a choice of generators μ_x for all $x \in M$ such that for every x there is a neighborhood U of x with the following property: for all $y \in U$, the images of μ_x and μ_y in $H_n(M, M-(\{x\}\cup\{y\});R)$ agree under the maps induced by inclusions. If such an orientation exists, M is R**-orientable**.

Definition 15.2. Fundamental classes

Let $X \subset M$ be a subspace. An element

$$z \in H_n(M, M-X;R)$$

is an R**-fundamental class of** M **at** X if for every $x \in X$ its image under

$$H_n(M, M-X;R) \longrightarrow H_n(M, M-\{x\};R)$$

is a generator. If $X = M$, we call $z \in H_n(M;R)$ a **fundamental class** and write $[M] = z$.

For connected manifolds, R-orientability implies the top-dimensional homology is rank one.

Proposition 15.3. Top homology of a compact orientable manifold

Let M be a connected compact R-oriented n-manifold without boundary. Then $H_n(M;R) \cong R$, generated by the fundamental class $[M]$.

Proof. Choose a finite triangulation of M (or a finite CW structure). Let K be the corresponding finite simplicial complex. Local orientations determine a coherent choice of orientation on n-simplices, hence a

simplicial n-cycle z whose image in $H_n(M, M - \{x\}; R)$ is a generator for every x. This cycle represents a nonzero class in $H_n(M; R)$.
Now use a good cover argument (or induction over the skeleta) to show that $H_n(M; R)$ is generated by any cycle with the local generator property: cover M by finitely many open sets each homeomorphic to $\mathbb{R}^n$ and apply Mayer–Vietoris repeatedly. On $\mathbb{R}^n$ the group $H_n(\mathbb{R}^n; R) = 0$, so the only contribution to H_n comes from the global gluing captured by local generator data. The local generator property forces the top class to be unique up to multiplication by a unit of R, hence $H_n(M; R) \cong R$. □

Remark. When $R = \mathbb{Z}$, the condition says exactly that M is orientable in the usual sense and that $[M] \in H_n(M; \mathbb{Z})$ is the usual fundamental class.

15.2 Cap product definition and properties

We work with singular chains $S_*(X) = C_*(X; \mathbb{Z})$. For a ring R we set $C_*(X; R) = S_*(X) \otimes R$ and $C^*(X; R) = \mathrm{Hom}(S_*(X), R)$. Let δ be the coboundary on cochains and ∂ the boundary on chains.

Definition 15.4. Cap product on the chain level

Let $p, q \geq 0$. For $\varphi \in C^p(X; R)$ and a singular simplex $\sigma\colon \Delta^{p+q} \to X$, define a (q)-chain

$$\sigma \cap \varphi \in C_q(X; R)$$

by the Alexander–Whitney formula

$$\sigma \cap \varphi := \varphi(\sigma|[0, \ldots, p]) \cdot \sigma|[p, \ldots, p+q],$$

extended R-linearly to chains. For $x \in C_{p+q}(X; R)$, define $x \cap \varphi$ by linearity.

Remark. One may also define $\cap$ for a pair (X, A) and more generally for triads using relative diagonal approximations. We will use the relative version in the boundary case below.

The cap product is characterized by its interaction with ∂ and δ.

Proposition 15.5. Boundary formula and naturality

Let $x \in C_{p+q}(X; R)$ and $\varphi \in C^p(X; R)$. Then

$$\partial(x \cap \varphi) = (-1)^p(\partial x \cap \varphi - x \cap \delta\varphi).$$

If $f\colon X \to Y$ is continuous, then for $x \in C_*(X; R)$ and $\psi \in C^*(Y; R)$,

$$f_*(x \cap f^*\psi) = f_*x \cap \psi.$$

Proof. It suffices to check the boundary identity on a simplex $\sigma\colon \Delta^{p+q} \to X$. Write $\sigma_F = \sigma|[0, \ldots, p]$ and $\sigma_B = \sigma|[p, \ldots, p+q]$. Expanding $\partial(\sigma \cap \varphi)$ gives an alternating sum of faces of σ_B multiplied by the scalar $\varphi(\sigma_F)$. Expanding $\partial\sigma \cap \varphi$ gives a sum over faces of σ. Those faces that meet only the back vertices contribute $\partial(\sigma_B)$ with coefficient $\varphi(\sigma_F)$. The remaining faces shift one vertex from the front block to the back block and correspond exactly to $-\sigma \cap \delta\varphi$, with the overall sign $(-1)^p$ coming from the position of the omitted vertex. Collecting terms yields the formula.
For naturality, on a simplex σ one has $(f \circ \sigma) \cap \psi = \sigma \cap f^*\psi$ by the defining formula, hence $f_*(\sigma \cap f^*\psi) = (f \circ \sigma) \cap \psi$. Extend linearly. □

Proposition 15.6. Associativity with cup product

Let $x \in H_{p+q+r}(X; R)$, $\alpha \in H^p(X; R)$, and $\beta \in H^q(X; R)$. Then

$$(x \cap \alpha) \cap \beta = x \cap (\alpha \smile \beta) \qquad \text{in } H_r(X; R).$$

Proof. Choose cocycle representatives a, b for α, β and a cycle representative c for x. Using the Alexander–Whitney formulas for $\smile$ and $\cap$ on a simplex σ, both expressions evaluate by first restricting σ to the front p-face for a, then to the next q-face for b, and keeping the remaining r-face as the resulting chain. The difference between the two constructions is a coboundary at the cochain level, hence yields equal classes in homology. □

The cap product is adjoint to the cup product pairing in the presence of evaluation.

Proposition 15.7. Fundamental identity

Let $\alpha \in H^p(X;R)$, $\beta \in H^q(X;R)$, and $x \in H_{p+q}(X;R)$. Then

$$\langle \alpha \smile \beta, x\rangle = \langle \beta, x \cap \alpha\rangle.$$

Proof. Choose cocycle representatives a, b and a cycle representative c. By definition, $(a \smile b)(c)$ evaluates a on the front p-faces and b on the back q-faces using the Alexander–Whitney diagonal. On the other hand, $c \cap a$ is a q-chain obtained by weighting each back face by the scalar a evaluates on the corresponding front face. Evaluating b on $c \cap a$ produces the same sum. □

15.3 Poincaré duality theorem, statement and proof

We now specialize to manifolds. Throughout this subsection, M is a connected compact R-oriented n-manifold without boundary, with fundamental class $[M] \in H_n(M;R)$.

Theorem 15.8. Poincaré duality

For each k the map

$$D\colon H^k(M;R) \longrightarrow H_{n-k}(M;R), \qquad D(\alpha) = [M] \cap \alpha,$$

is an isomorphism.

Proof. Step 1: Reduce to simplicial (co)chains. Choose a finite triangulation of M by a finite simplicial complex K. Simplicial homology and cohomology with coefficients in R agree with the singular theories for $|K| \cong M$, and the isomorphisms respect cap products. Thus it suffices to prove the theorem for simplicial chains and cochains on K.

Write $C_*(K;R)$ and $C^*(K;R) = \mathrm{Hom}(C_*(K;\mathbb{Z}), R)$ for simplicial chains and cochains. Choose an oriented fundamental cycle

$$z = \sum_{\sigma^n} \varepsilon(\sigma)\,\sigma \in C_n(K;R)$$

representing $[M]$.

Step 2: Cap with z is a chain map. Define

$$\Phi\colon C^k(K;R) \longrightarrow C_{n-k}(K;R), \qquad \Phi(\varphi) := z \cap \varphi.$$

By Proposition 15.5, for $\varphi \in C^k$,

$$\partial(z \cap \varphi) = (-1)^k(\partial z \cap \varphi - z \cap \delta\varphi) = (-1)^{k+1} z \cap \delta\varphi,$$

since $\partial z = 0$. Hence Φ is a chain map from $(C^*(K;R), \delta)$ to $(C_{n-*}(K;R), (-1)^{*+1}\partial)$, and induces

$$H^k(K;R) \longrightarrow H_{n-k}(K;R),$$

which is exactly $\alpha \mapsto [M] \cap \alpha$.

Step 3: Dual cell decomposition and a chain-level duality isomorphism. Let K' be the barycentric subdivision of K. For each simplex σ of K of dimension k, there is an associated **dual cell** σ^* in K': it is the union of simplices of K' whose vertices are barycenters of a strictly increasing flag

$$\sigma = \sigma_0 < \sigma_1 < \cdots < \sigma_{n-k}$$

of simplices in K. These dual cells form a CW decomposition K^* of $|K|$ (one cell σ^* for each simplex σ), of dimension n. Moreover $\dim(\sigma^*) = n - k$.

Let $C_*^{\mathrm{cell}}(K^*;R)$ be the cellular chain complex of K^* with coefficients in R. Define an R-linear map on basis elements

$$\Psi\colon C^k(K;R) \longrightarrow C_{n-k}^{\mathrm{cell}}(K^*;R)$$

as follows. For each k-simplex σ in K, let $\sigma^\vee \in C^k(K;R)$ be the cochain taking value 1 on σ (with the chosen orientation) and 0 on other k-simplices. Orient each dual cell σ^* by requiring that, at points of $\sigma \cap \sigma^*$, the product orientation

$$(\text{orientation of } \sigma) \cdot (\text{orientation of } \sigma^*)$$

agree with the given orientation of M. Set $\Psi(\sigma^\vee) = \sigma^*$ and extend R-linearly.

We claim Ψ is a chain isomorphism, i.e.

$$\partial_{\mathrm{cell}} \circ \Psi = \Psi \circ \delta.$$

Indeed, the cellular boundary of τ^* for a $(k+1)$-simplex τ is a sum of σ^* over its k-faces $\sigma < \tau$, with coefficients given by the incidence numbers of the triangulation. On the cochain side, $\delta(\sigma^\vee)$ is the cochain that, evaluated on a $(k+1)$-simplex τ, returns exactly the same incidence number. Thus the matrices describing ∂_{cell} in the dual basis and δ in the primal basis coincide. Therefore Ψ is a chain isomorphism, hence induces isomorphisms

$$H^k(K;R) \cong H^{\text{cell}}_{n-k}(K^*;R).$$

Step 4: Identify $H^{\text{cell}}_{n-k}(K^*;R)$ with $H_{n-k}(K;R)$ and compare with Φ. Since K^* is a CW decomposition of the same underlying space $|K| \cong M$, cellular homology computes singular homology:

$$H^{\text{cell}}_{n-k}(K^*;R) \cong H_{n-k}(M;R) \cong H_{n-k}(K;R).$$

Composing with Ψ gives an isomorphism

$$H^k(M;R) \cong H_{n-k}(M;R).$$

It remains to see that this isomorphism is the cap product map $\alpha \mapsto [M] \cap \alpha$. On a basis cochain $\sigma^\vee$, the chain $z \cap \sigma^\vee$ is supported on the simplices of K whose first k vertices form σ; geometrically, this is a subdivision chain carried by the dual cell σ^*, with coefficient 1 and the same orientation convention. Hence $\Phi(\sigma^\vee)$ and the cellular chain σ^* represent the same class in $H_{n-k}(M;R)$. By linearity, Φ induces the same isomorphism on homology as Ψ. Therefore D is an isomorphism. □

Corollary 15.9 (Perfect pairing modulo torsion (integer coefficients))**.** Let M be a connected closed orientable n-manifold. The pairing

$$H^k(M;\mathbb{Z})/\text{tors} \otimes H^{n-k}(M;\mathbb{Z})/\text{tors} \longrightarrow \mathbb{Z}, \qquad (\alpha,\beta) \mapsto \langle \alpha \smile \beta, [M] \rangle,$$

is nonsingular.

Proof. By Proposition 15.7,

$$\langle \alpha \smile \beta, [M] \rangle = \langle \beta, [M] \cap \alpha \rangle.$$

The map $H^k(M;\mathbb{Z}) \to H_{n-k}(M;\mathbb{Z})$ given by capping with $[M]$ is an isomorphism by Theorem 15.8. Using the universal coefficient theorem for cohomology identifies the free part of H^k with Hom of the free part of H_k, and the pairing becomes evaluation between a free module and its dual. □

15.4 Manifolds with boundary and relative duality

Let M now be a compact R-oriented n-manifold with (possibly empty) boundary ∂M. There is a relative version of the fundamental class.

Definition 15.10. Relative fundamental class

An element $[M, \partial M] \in H_n(M, \partial M; R)$ is a **relative fundamental class** if for every $x \in \text{int}(M)$ its image in $H_n(M, M - \{x\}; R)$ is a generator.

Proposition 15.11. Boundary of the fundamental class

If M is R-oriented with relative fundamental class $[M, \partial M]$, then

$$\partial[M, \partial M] \in H_{n-1}(\partial M; R)$$

is a fundamental class of ∂M (with the induced orientation).

Proof. Take $x \in \partial M$ and a collar neighborhood $U \cong [0,\varepsilon) \times \mathbb{R}^{n-1}$ with x corresponding to $(0,0)$. Excision identifies $(M, \partial M)$ near x with $([0,\varepsilon) \times \mathbb{R}^{n-1}, \{0\} \times \mathbb{R}^{n-1})$. Under this identification, the boundary map in the long exact sequence of the pair sends the local generator in $H_n([0,\varepsilon) \times \mathbb{R}^{n-1}, \{0\} \times \mathbb{R}^{n-1}; R)$ to the local generator of $H_{n-1}(\mathbb{R}^{n-1}, \mathbb{R}^{n-1} - \{0\}; R)$, which is precisely the orientation class of the boundary. □

Relative cap products are defined for pairs and yield the correct target groups.

Definition 15.12. Relative cap products

For a pair (X, A) there are natural products

$$\cap\colon H^k(X;R) \otimes H_n(X,A;R) \to H_{n-k}(X,A;R), \qquad \cap\colon H^k(X,A;R) \otimes H_n(X,A;R) \to H_{n-k}(X;R),$$

constructed by a relative diagonal approximation and evaluation on cochains.

Theorem 15.13. Lefschetz duality

Let M be a compact R-oriented n-manifold with boundary, and let $[M, \partial M] \in H_n(M, \partial M; R)$ be a relative fundamental class. Then for each k the maps

$$D\colon H^k(M; R) \longrightarrow H_{n-k}(M, \partial M; R), \qquad D(\alpha) = [M, \partial M] \cap \alpha,$$

and

$$D'\colon H^k(M, \partial M; R) \longrightarrow H_{n-k}(M; R), \qquad D'(\alpha) = [M, \partial M] \cap \alpha,$$

are isomorphisms.

Proof. Triangulate $(M, \partial M)$ by a finite simplicial pair (K, L) with $|K| \cong M$ and $|L| \cong \partial M$. Form the dual cell decomposition K^* inside $|K|$. Dual cells corresponding to simplices of L lie in ∂M, and the remaining dual cells form a CW decomposition of the interior. On the chain level, the same construction as in the proof of Theorem 15.8 gives chain isomorphisms

$$C^k(K; R) \cong C^{\text{cell}}_{n-k}(K^*, (K^*)|_{\partial M}; R), \qquad C^k(K, L; R) \cong C^{\text{cell}}_{n-k}(K^*; R),$$

intertwining coboundary with cellular boundary. Passing to (co)homology yields

$$H^k(M; R) \cong H_{n-k}(M, \partial M; R), \qquad H^k(M, \partial M; R) \cong H_{n-k}(M; R).$$

As before, the dual-cell isomorphism agrees with capping by the relative fundamental cycle representing $[M, \partial M]$, so the induced isomorphisms are exactly D and D'. □

16 Homotopy groups and long exact sequences

16.1 Homotopy groups and basic properties

Fix a based space (X, x_0). For $n \geq 0$, the nth homotopy group $\pi_n(X, x_0)$ is built out of based maps from an n-sphere into X. Concretely, it is often more convenient to use cubes.

Definition 16.1. The homotopy groups

For $n \geq 0$, define

$$\pi_n(X, x_0) := [(S^n, s_0), (X, x_0)]$$

the set of based homotopy classes of based maps. Equivalently,

$$\pi_n(X, x_0) \cong [(I^n, \partial I^n), (X, x_0)],$$

where maps $I^n \to X$ are required to send ∂I^n to x_0 and homotopies are taken relative to ∂I^n.

The cube model makes the group law completely explicit.

Proposition 16.2. Group structure and functoriality

For $n \geq 1$, the set $\pi_n(X, x_0)$ is a group. For $n \geq 2$, it is abelian. Moreover, every based map $f : (X, x_0) \to (Y, y_0)$ induces a homomorphism

$$f_* : \pi_n(X, x_0) \longrightarrow \pi_n(Y, y_0),$$

and $(g \circ f)_* = g_* \circ f_*$, $(\mathrm{id}_X)_* = \mathrm{id}$.

Proof. We use the cube model. Let $[a], [b] \in \pi_n(X, x_0)$ be represented by maps

$$a, b : (I^n, \partial I^n) \to (X, x_0).$$

Define the product $a \cdot b : (I^n, \partial I^n) \to (X, x_0)$ by concatenation in the last coordinate:

$$(a \cdot b)(t_1, \ldots, t_{n-1}, t_n) = \begin{cases} a(t_1, \ldots, t_{n-1}, 2t_n) & 0 \leq t_n \leq \frac{1}{2}, \\ b(t_1, \ldots, t_{n-1}, 2t_n - 1) & \frac{1}{2} \leq t_n \leq 1. \end{cases}$$

Since both a and b send ∂I^n to x_0, this formula is continuous and still sends ∂I^n to x_0. The constant map

$c(t) = x_0$ is a two sided identity. The inverse of $[a]$ is represented by reversing the last coordinate:

$$a^{-1}(t_1, \dots, t_{n-1}, t_n) := a(t_1, \dots, t_{n-1}, 1 - t_n).$$

Associativity holds up to a homotopy rel ∂I^n obtained by linearly reparametrizing the last coordinate to compare $(a \cdot b) \cdot c$ with $a \cdot (b \cdot c)$. This produces a well-defined group structure on homotopy classes.
Functoriality is immediate from $(f \circ a) \cdot (f \circ b) = f \circ (a \cdot b)$, and similarly for inverses and identities, so f_* is a homomorphism and respects composition.
For commutativity when $n \geq 2$, define a second product $a \star b$ by concatenation in the first coordinate:

$$(a \star b)(t_1, \dots, t_n) = \begin{cases} a(2t_1, t_2, \dots, t_n) & 0 \leq t_1 \leq \frac{1}{2}, \\ b(2t_1 - 1, t_2, \dots, t_n) & \frac{1}{2} \leq t_1 \leq 1. \end{cases}$$

Both $\cdot$ and $\star$ have the same unit class $[c]$. They also satisfy an interchange law on representatives,

$$(a \cdot b) \star (a' \cdot b') \simeq (a \star a') \cdot (b \star b') \quad \text{rel } \partial I^n,$$

since when $n \geq 2$ the two concatenations take place in independent coordinates and can be arranged by a homotopy of the square subdivision of the (t_1, t_n)-plane. The Eckmann–Hilton argument then applies: any two unital operations with such an interchange law must coincide and be commutative. Hence $\pi_n(X, x_0)$ is abelian for $n \geq 2$. □

Loop spaces package the shift in degree. Let ΩX be the based loop space with basepoint the constant loop at x_0.

Lemma 16.3. Loop space adjunction For $n \geq 1$, there is a natural bijection

$$\pi_n(X, x_0) \cong \pi_{n-1}(\Omega X, c_{x_0}),$$

hence iteratively $\pi_n(X) \cong \pi_0(\Omega^n X)$.

Proof. A based map $S^n \to X$ is the same data as a based map $S^{n-1} \to \Omega X$ by adjunction, since $S^n \cong \Sigma S^{n-1}$ and based maps $\Sigma A \to X$ correspond to based maps $A \to \Omega X$. Passing to based homotopy classes gives the stated bijection, and naturality is clear. □

16.2 LES of a pair and relative homotopy groups

Relative homotopy groups measure the failure of a map to compress into a subspace.

Definition 16.4. Relative homotopy groups

Let (X, A, x_0) be a based pair with $x_0 \in A \subset X$. For $n \geq 1$, define

$$\pi_n(X, A, x_0) := [(D^n, S^{n-1}, s_0), (X, A, x_0)],$$

the set of homotopy classes of maps $f : D^n \to X$ such that $f(S^{n-1}) \subset A$ and $f(s_0) = x_0$, where homotopies are through such maps and are fixed on s_0.

There are three basic maps relating absolute and relative groups.

- The inclusion $i : (A, x_0) \hookrightarrow (X, x_0)$ induces $i_* : \pi_n(A, x_0) \to \pi_n(X, x_0)$.
- Any $[g] \in \pi_n(X, x_0)$ can be represented by a map $g : D^n \to X$ sending S^{n-1} to $x_0 \in A$, hence defines an element of $\pi_n(X, A, x_0)$. This gives a natural map $j_* : \pi_n(X, x_0) \to \pi_n(X, A, x_0)$.
- Restriction to the boundary gives the boundary map

  $$\partial : \pi_n(X, A, x_0) \longrightarrow \pi_{n-1}(A, x_0), \qquad \partial[f] = [\, f|_{S^{n-1}} \,].$$

  For $n \geq 2$ this is a homomorphism.

Theorem 16.5. Long exact sequence of a pair

For every based pair (X, A, x_0), there is a long exact sequence

$$\cdots \longrightarrow \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \longrightarrow \cdots$$

continuing down to π_1 and then to pointed sets at π_0.

Proof. We prove exactness at the three consecutive terms

$$\pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0).$$

Exactness at $\pi_n(X, x_0)$. First, $j_* \circ i_* = 0$ since if $a : D^n \to A$ represents an element of $\pi_n(A, x_0)$, then regarded as a map into X it already lands in A, so it represents the trivial class in the relative group $\pi_n(X, A, x_0)$.
Conversely, suppose $[g] \in \pi_n(X, x_0)$ lies in $\ker(j_*)$. Choose a representative $g : D^n \to X$ with $g(S^{n-1}) = x_0$. The hypothesis $j_*[g] = 0$ means that g, viewed as a relative map $(D^n, S^{n-1}) \to (X, A)$, is homotopic through relative maps to the constant relative map. Unwinding the definition, there exists a homotopy

$$H : D^n \times I \to X$$

with $H(-, 0) = g$, $H(-, 1)$ landing in A, and $H(S^{n-1} \times I) \subset A$ (in fact it is constant at x_0 along $S^{n-1} \times I$ in the cube model). Then $H(-, 1) : D^n \to A$ represents an element of $\pi_n(A, x_0)$ whose image under i_* is $[g]$. Hence $\ker(j_*) \subset \operatorname{im}(i_*)$.

Exactness at $\pi_n(X, A, x_0)$. First, $\partial \circ j_* = 0$ since if g represents an element of $\pi_n(X, x_0)$ then as a relative map it sends S^{n-1} to x_0, so its restriction to S^{n-1} is null in $\pi_{n-1}(A, x_0)$.
Conversely, let $[f] \in \pi_n(X, A, x_0)$ satisfy $\partial[f] = 0$. Choose a representative $f : D^n \to X$ with $f(S^{n-1}) \subset A$ and $f(s_0) = x_0$. The condition $\partial[f] = 0$ means that the boundary map

$$f|_{S^{n-1}} : (S^{n-1}, s_0) \to (A, x_0)$$

is nullhomotopic in A. Let

$$K : S^{n-1} \times I \to A$$

be a based nullhomotopy from $f|_{S^{n-1}}$ to the constant map at x_0.
Now glue f to the homotopy K along the boundary to build a map $g : S^n \to X$ as follows. Regard S^n as the union of two n-disks $D^n_+ \cup D^n_-$ along their boundary S^{n-1}. Put $g|_{D^n_+} = f$. On D^n_- define g by collapsing radially onto S^{n-1} and then using the homotopy parameter to run K from $f|_{S^{n-1}}$ to the constant map. By construction, g is continuous, based, and its restriction to the equator is homotopic inside A to the constant. When we view g as an element of $\pi_n(X, x_0)$ and apply j_*, the resulting relative class is represented by f. Hence $[f] \in \operatorname{im}(j_*)$. Therefore $\ker(\partial) \subset \operatorname{im}(j_*)$.

Exactness at $\pi_{n-1}(A, x_0)$. We have $i_* \circ \partial = 0$ since if $f : D^n \to X$ is relative, then $f|_{S^{n-1}}$ extends to D^n in X by f, so it represents the trivial element after applying i_*. Conversely, suppose $[h] \in \pi_{n-1}(A, x_0)$ maps to 0 in $\pi_{n-1}(X, x_0)$. Choose a representative $h : S^{n-1} \to A$. Being nullhomotopic in X means h extends to a map $f : D^n \to X$. Since h lands in A, this extension represents an element of $\pi_n(X, A, x_0)$ with boundary $\partial[f] = [h]$. Thus $\ker(i_*) \subset \operatorname{im}(\partial)$.
Putting the three verifications together gives exactness at the indicated terms. The remaining terms in the long exact sequence are handled similarly. □

16.3 Fibrations and the LES of homotopy groups

Long exact sequences become especially powerful once we know how to produce them from maps that behave like projections.

Definition 16.6. Homotopy lifting property and fibrations

A map $p : E \to B$ has the **homotopy lifting property** with respect to a space Y if for every homotopy $G : Y \times I \to B$ and every lift $g_0 : Y \to E$ of $G(-, 0)$, there exists a homotopy $\widetilde{G} : Y \times I \to E$ with

$$p \circ \widetilde{G} = G \qquad \text{and} \qquad \widetilde{G}(-, 0) = g_0.$$

A **fibration** is a map $p : E \to B$ having the homotopy lifting property with respect to all spaces Y. A **Serre fibration** is a map having the homotopy lifting property with respect to all disks D^k.

Examples.

1. The projection $\mathrm{pr}_1 : B \times F \to B$ is a fibration, since one can lift a homotopy by keeping the F coordinate fixed.
2. A covering map $p : \widetilde{B} \to B$ is a fibration, with the familiar path and homotopy lifting.

3. For any based space (X, x_0), let PX be the based path space
$$PX = \{\gamma : I \to X \mid \gamma(0) = x_0\},$$
and let $\mathrm{ev}_1 : PX \to X$ send γ to $\gamma(1)$. This is a fibration, and the fiber over x_0 is ΩX.

Fix basepoints $b_0 \in B$ and $e_0 \in E$ with $p(e_0) = b_0$, and let $F = p^{-1}(b_0)$ be the fiber with basepoint $e_0 \in F$.

Theorem 16.7. Relative isomorphism for Serre fibrations

Assume that $p : E \to B$ has the homotopy lifting property with respect to disks D^k for all $k \geq 0$. Then for all $n \geq 1$ the induced map
$$p_* : \pi_n(E, F, e_0) \longrightarrow \pi_n(B, b_0)$$
is an isomorphism.

Proof. We show surjectivity and injectivity.

Surjectivity. Let $[\alpha] \in \pi_n(B, b_0)$ be represented by a based map $\alpha : (D^n, S^{n-1}) \to (B, b_0)$. Consider the constant lift of the boundary, namely $S^{n-1} \to F \subset E$ sending everything to e_0. The homotopy lifting property for (D^n, S^{n-1}) yields a lift $\widetilde{\alpha} : D^n \to E$ extending this boundary lift and satisfying $p \circ \widetilde{\alpha} = \alpha$. Then $\widetilde{\alpha}$ defines a class in $\pi_n(E, F, e_0)$ mapping to $[\alpha]$.

Injectivity. Suppose $[f] \in \pi_n(E, F, e_0)$ maps to $0 \in \pi_n(B, b_0)$. Choose a representative $f : (D^n, S^{n-1}) \to (E, F)$ with $f(s_0) = e_0$. The condition $p_*[f] = 0$ means that $p \circ f$ is homotopic rel S^{n-1} to the constant map at b_0. Let
$$H : D^n \times I \to B$$
be such a nullhomotopy, with $H(-, 0) = p \circ f$, $H(-, 1) = c_{b_0}$, and $H(S^{n-1} \times I) = b_0$.
Apply the homotopy lifting property to lift H starting from the given lift f at time 0, and keeping the boundary in F throughout. This produces $\widetilde{H} : D^n \times I \to E$ with $\widetilde{H}(-, 0) = f$, $p \circ \widetilde{H} = H$, and $\widetilde{H}(S^{n-1} \times I) \subset F$. At time 1 we obtain a map $\widetilde{H}(-, 1) : D^n \to E$ landing in the fiber F since $H(-, 1)$ is constantly b_0. Hence f is homotopic rel S^{n-1} to a map with image in F, which means $[f] = 0$ in $\pi_n(E, F, e_0)$. □

Combining Theorem 16.7 with the long exact sequence of the pair (E, F) gives the long exact sequence of a fibration.

Theorem 16.8. Long exact sequence of a fibration

Let $p : E \to B$ be a Serre fibration with fiber $F = p^{-1}(b_0)$, and assume B is path-connected. There is a natural long exact sequence of groups
$$\cdots \to \pi_n(F, e_0) \to \pi_n(E, e_0) \xrightarrow{p_*} \pi_n(B, b_0) \xrightarrow{\delta} \pi_{n-1}(F, e_0) \to \cdots \to \pi_1(B, b_0) \to \pi_0(F) \to \pi_0(E) \to \pi_0(B) \to 0.$$

Proof. Apply Theorem 16.5 to the pair (E, F, e_0) to obtain
$$\cdots \to \pi_n(F) \to \pi_n(E) \to \pi_n(E, F) \xrightarrow{\partial} \pi_{n-1}(F) \to \cdots$$
and then use the isomorphism $p_* : \pi_n(E, F) \cong \pi_n(B)$ from Theorem 16.7 to identify $\pi_n(E, F)$ with $\pi_n(B)$. The connecting morphism δ is the composite
$$\pi_n(B) \xrightarrow{(p_*)^{-1}} \pi_n(E, F) \xrightarrow{\partial} \pi_{n-1}(F).$$
Exactness follows from exactness of the pair sequence and the fact that $(p_*)^{-1}$ is an isomorphism. □

It is useful to keep a geometric picture of δ in mind. Represent a class in $\pi_n(B, b_0)$ by a based map $\alpha : (D^n, S^{n-1}) \to (B, b_0)$. Lift α to $\widetilde{\alpha} : D^n \to E$ with $\widetilde{\alpha}(S^{n-1}) \subset F$ and $\widetilde{\alpha}(s_0) = e_0$. Then
$$\delta[\alpha] = [\widetilde{\alpha}|_{S^{n-1}}] \in \pi_{n-1}(F, e_0).$$
Different choices of lifts and representatives lead to the same class because any two choices differ by a lifted homotopy relative to the boundary.

16.4 First computations and applications

Proposition 16.9. Products

For based spaces (X, x_0) and (Y, y_0), the projection maps induce an isomorphism

$$\pi_n(X \times Y, (x_0, y_0)) \cong \pi_n(X, x_0) \times \pi_n(Y, y_0)$$

for all $n \geq 0$.

Proof. A based map $(S^n, s_0) \to (X \times Y, (x_0, y_0))$ is equivalent to a pair of based maps $S^n \to X$ and $S^n \to Y$. Homotopies correspond componentwise, and the group structures match because concatenation is defined coordinatewise in a product. □

Proposition 16.10. Higher homotopy of the circle

For $n \geq 2$, one has $\pi_n(S^1) = 0$.

Proof. Let $p : \mathbb{R} \to S^1$ be the universal covering map, based at $0 \in \mathbb{R}$ and $1 \in S^1$. This is a fibration with fiber $F = p^{-1}(1) \cong \mathbb{Z}$, a discrete space. For $n \geq 2$ we have $\pi_n(\mathbb{R}) = 0$ since $\mathbb{R}$ is contractible, and also $\pi_{n-1}(F) = 0$ since F is discrete and $n - 1 \geq 1$. The long exact sequence of the fibration gives

$$0 = \pi_n(\mathbb{R}) \longrightarrow \pi_n(S^1) \longrightarrow \pi_{n-1}(F) = 0,$$

so $\pi_n(S^1) = 0$. □

The path space fibration turns loop spaces into a degree shift on homotopy groups.

Proposition 16.11. The path loop fibration and degree shifting

Let $\mathrm{ev}_1 : PX \to X$ be the endpoint evaluation fibration. The total space PX is contractible, and the fiber over x_0 is ΩX. For $n \geq 1$ the long exact sequence yields natural isomorphisms

$$\pi_n(\Omega X, c_{x_0}) \cong \pi_{n+1}(X, x_0).$$

Proof. A contraction of PX is given by shrinking paths toward the constant path, namely

$$H : PX \times I \to PX, \qquad H(\gamma, s)(t) = \gamma(st).$$

At $s = 0$ this is the constant path, and at $s = 1$ it is γ. The fiber of ev_1 over x_0 is exactly the based loop space. Since PX is contractible, $\pi_k(PX) = 0$ for all $k \geq 1$. The long exact sequence of the fibration $\Omega X \to PX \to X$ therefore contains, for each $n \geq 1$, an isomorphism

$$\pi_{n+1}(X) \xrightarrow{\cong} \pi_n(\Omega X),$$

coming from exactness at $\pi_{n+1}(X) \to \pi_n(\Omega X) \to \pi_n(PX) = 0$ and $\pi_{n+1}(PX) = 0 \to \pi_{n+1}(X)$. □

The fibration long exact sequence is often most effective when combined with the pair long exact sequence and the fact that many familiar maps can be replaced by fibrations up to homotopy.

17 CW approximation, cellular approximation, and Whitehead's theorem

17.1 CW approximation and weak equivalences

The phrase "good space" will mean "a space for which CW methods control homotopy classes of maps and homotopy invariants." The first step is to separate two notions that coincide on CW complexes but not on arbitrary spaces.

Definition 17.1. Weak homotopy equivalence and n-equivalence

Let $f\colon X \to Y$ be a continuous map and fix basepoints when needed.

1. We say that f is a **weak homotopy equivalence** if for every $x_0 \in X$ and every $n \geq 0$ the induced map
$$\pi_n(f)\colon \pi_n(X, x_0) \longrightarrow \pi_n(Y, f(x_0))$$
is an isomorphism (bijective for $n = 0$ and group isomorphism for $n \geq 1$).
2. For an integer $n \geq 1$, we say that f is an **n-equivalence** if $\pi_i(f)$ is an isomorphism for $0 \leq i < n$ and a surjection for $i = n$.

Definition 17.2. CW approximation

A **CW approximation** of a space X is a CW complex Z together with a map $p\colon Z \to X$ that is a weak homotopy equivalence.

On CW complexes, weak equivalences are as strong as homotopy equivalences. Before proving that, it is useful to know that every space admits a CW approximation. The construction is explicit and it is the reason CW complexes are the default language in homotopy theory.

Theorem 17.3. Existence of CW approximations

For every space X there exists a CW complex Z and a weak homotopy equivalence $p\colon Z \to X$.

Proof. We build Z as an increasing union
$$Z^{(0)} \subset Z^{(1)} \subset Z^{(2)} \subset \cdots$$
of CW complexes, together with maps $p_k\colon Z^{(k)} \to X$ that are compatible under inclusion. The colimit $Z = \operatorname{colim}_k Z^{(k)}$ is then a CW complex and the maps assemble to $p\colon Z \to X$.

Step 0. Let $\pi_0(X)$ be the set of path components. Choose one point $x_\gamma \in X$ in each component and let $Z^{(0)}$ be the discrete set of 0-cells $\{z_\gamma\}$ indexed by components. Define $p_0(z_\gamma) = x_\gamma$. Then $\pi_0(p_0)$ is bijective.

Induction hypothesis. Suppose $Z^{(k-1)}$ has been constructed, with chosen basepoints $z_\gamma \in Z^{(k-1)}$ mapping to $x_\gamma \in X$, such that
$$\pi_i(p_{k-1})\colon \pi_i(Z^{(k-1)}, z_\gamma) \to \pi_i(X, x_\gamma)$$
is an isomorphism for $i < k-1$, and a surjection for $i = k-1$.

Step k has two parts.

(A) Kill the kernel in degree $k-1$. For each component γ, consider the kernel
$$\ker(\pi_{k-1}(p_{k-1})\colon \pi_{k-1}(Z^{(k-1)}, z_\gamma) \to \pi_{k-1}(X, x_\gamma)).$$
Choose a set of based maps
$$\varphi_\alpha\colon (S^{k-1}, s_0) \to (Z^{(k-1)}, z_\gamma)$$
representing all nontrivial elements in these kernels, ranging over all components. Attach a k-cell e^k_α for each such φ_α via the attaching map φ_α. Call the resulting space $Z^{(k-1)}_+$ and extend p_{k-1} over each new k-cell by choosing a nullhomotopy of $p_{k-1} \circ \varphi_\alpha$ in X, which exists by the definition of the kernel. Denote the extended map by $p_{k-1,+}\colon Z^{(k-1)}_+ \to X$.

Attaching a k-cell along φ_α forces the class $[\varphi_\alpha]$ to vanish in π_{k-1} of the new space. Thus the map on π_{k-1} becomes injective after all such attachments. More precisely, one uses the long exact sequence of the pair $(Z^{(k-1)}_+, Z^{(k-1)})$ to check that the kernel of $\pi_{k-1}(p_{k-1,+})$ is now trivial.

(B) Force surjectivity in degree k. For each component γ, choose based maps
$$\psi_\beta\colon (S^k, s_0) \to (X, x_\gamma)$$
representing all elements of $\pi_k(X, x_\gamma)$. Now attach a k-cell e^k_β for each ψ_β along the constant attaching map $S^{k-1} \to \{z_\gamma\} \subset Z^{(k-1)}_+$ so that the boundary of e^k_β is collapsed to the basepoint z_γ. The resulting quotient of the attaching sphere is S^k, so we may extend $p_{k-1,+}$ over each e^k_β by sending its characteristic S^k to ψ_β. Let $Z^{(k)}$ be the CW complex after these attachments and let $p_k\colon Z^{(k)} \to X$ be the extension.

By construction, every class in $\pi_k(X, x_\gamma)$ is hit by $\pi_k(p_k)$. Hence $\pi_k(p_k)$ is surjective.

Verification of the induction. Attaching k-cells does not affect π_i for $i < k-1$, and the previous stage already had $\pi_i(p_{k-1})$ an isomorphism there. Part (A) makes $\pi_{k-1}(p_k)$ injective and the induction hypothesis already gives surjectivity of $\pi_{k-1}(p_{k-1})$, hence $\pi_{k-1}(p_k)$ is an isomorphism. Part (B) makes $\pi_k(p_k)$ surjective, as required.

Pass to the colimit. Set $Z = \mathrm{colim}_k Z^{(k)}$ and $p = \mathrm{colim}_k p_k$. For a fixed n, choose $k \geq n$. Since $Z^{(k)} \to Z$ is an inclusion of a skeleton, the induced map $\pi_n(Z^{(k)}, z_\gamma) \to \pi_n(Z, z_\gamma)$ is an isomorphism. The construction ensures that $\pi_n(p_k)$ is an isomorphism, so $\pi_n(p)$ is an isomorphism as well. Hence p is a weak homotopy equivalence. □

Remark. CW approximations are far from unique as spaces, but they are unique up to homotopy equivalence. This will fall out quickly once weak equivalences between CW complexes are shown to be homotopy equivalences.

17.2 Connectivity of skeleta and cellular approximation

To control maps cell-by-cell, the key fact is that high-dimensional cells do not contribute to low-dimensional relative homotopy groups.

Definition 17.4. Connectivity for spaces and pairs

Let $n \geq 0$.

1. A space X is **n-connected** if $\pi_q(X, x) = 0$ for all $0 \leq q \leq n$ and all basepoints $x \in X$.
2. A pair (X, A) is **n-connected** if $\pi_0(A) \to \pi_0(X)$ is surjective and $\pi_q(X, A, a) = 0$ for all $1 \leq q \leq n$ and all $a \in A$.

Equivalently, the inclusion $A \hookrightarrow X$ is an n-equivalence.

Lemma 17.5. Relative CW pairs with no low-dimensional cells are highly connected Let (X, A) be a relative CW complex, meaning X is obtained from A by attaching cells. If there are no relative m-cells for $m \leq n$, then (X, A) is n-connected. In particular, for any CW complex Y, the pair (Y, Y^n) is n-connected.

Proof. We show $\pi_q(X, A, a) = 0$ for $1 \leq q \leq n$. Represent an element of $\pi_q(X, A, a)$ by a map of pairs

$$f\colon (D^q, S^{q-1}) \to (X, A)$$

sending S^{q-1} to A and a chosen basepoint of S^{q-1} to a.
The image $f(D^q)$ is compact. A compact subset of a CW complex meets only finitely many cells, so $f(D^q)$ lies in a finite subcomplex of X obtained from a finite subcomplex of A by attaching finitely many cells, all of dimension $> n \geq q$.
Thus it suffices to treat the case when $X = A \cup_\phi D^r$ is obtained from A by attaching a single r-cell with $r > n \geq q$. Let $e^r \subset X$ be the open cell. By a standard approximation argument on the compact domain D^q, we can homotope f relative to S^{q-1} so that its image misses a chosen point $p \in e^r$. (One can triangulate D^q and use a simplicial approximation in a chart on $e^r \cong \mathbb{R}^r$, then perturb to avoid p since $r > q$.)
Now $X - \{p\}$ deformation retracts onto A since removing a point from the interior of a cell collapses that cell onto its boundary sphere, and the boundary sphere is already attached to A. Therefore f is homotopic relative S^{q-1} to a map landing in A. This shows the class of f in $\pi_q(X, A, a)$ is trivial. □

Cellular approximation is the main practical consequence.

Theorem 17.6. Cellular approximation theorem

Let (X, A) and (Y, B) be relative CW complexes and let $f\colon (X, A) \to (Y, B)$ be a map of pairs. Then f is homotopic relative to A to a cellular map g, meaning $g(X^n) \subset Y^n$ for all n. If f is already cellular on a subcomplex $A \subset X$, the homotopy may be taken stationary on A.

Proof. We build g skeleton-by-skeleton, producing a homotopy from f to g that is fixed on A.
For $n = 0$, each point of Y is path connected to Y^0. For each 0-cell of X not in A, choose a path in Y from its image under f to some 0-cell in Y^0, and use these paths to homotope $f|_{X^0}$ into Y^0 while keeping A^0 fixed.

Assume inductively that we have constructed a map

$$g_{n-1}\colon X^{n-1} \to Y^{n-1}$$

and a homotopy $H_{n-1}\colon X^{n-1} \times I \to Y$ relative to A^{n-1} from $f|_{X^{n-1}}$ to g_{n-1}.
Consider an n-cell $e^n \subset X$ not contained in A, with characteristic map $\chi\colon (D^n, S^{n-1}) \to (X^n, X^{n-1})$. The boundary map lands in X^{n-1}, so the composite

$$g_{n-1} \circ \chi|_{S^{n-1}}\colon S^{n-1} \to Y^{n-1}$$

is already cellular. We would like to extend over D^n with image in Y^n.
By Lemma 17.5, the pair (Y, Y^n) is n-connected, hence the inclusion $Y^n \hookrightarrow Y$ is an n-equivalence. Apply this to the map of pairs

$$f \circ \chi\colon (D^n, S^{n-1}) \to (Y, Y)$$

together with the boundary data given by $g_{n-1} \circ \chi|_{S^{n-1}}\colon S^{n-1} \to Y^{n-1} \subset Y^n$. The n-connectedness implies that $f \circ \chi$ is homotopic relative to S^{n-1} to a map

$$\widetilde{\chi}\colon (D^n, S^{n-1}) \to (Y^n, Y^{n-1}).$$

Doing this simultaneously for all n-cells not in A produces a map $g_n\colon X^n \to Y^n$ that agrees with g_{n-1} on X^{n-1}.
Finally we glue the cellwise homotopies to obtain a homotopy on X^n relative $X^{n-1} \cup A^n$ from $f|_{X^n}$ to the composite $X^n \xrightarrow{g_n} Y^n \hookrightarrow Y$. This gluing uses the homotopy extension property for the CW pair $(X^n, X^{n-1} \cup A^n)$, which holds because CW inclusions are cofibrations.
Passing to the colimit over n gives a map $g\colon X \to Y$ that is cellular and a homotopy $f \simeq g$ fixed on A. □

Corollary 17.7 (A basic vanishing range for spheres)**.** For the standard CW structures on spheres, $\pi_n(S^k) = 0$ when $n < k$.

Proof. Let $n < k$ and consider a based map $S^n \to S^k$. By Theorem 17.6, it is homotopic through based maps to a cellular map. Since S^k has no cells in dimensions strictly between 0 and k, a cellular map must carry the n-skeleton of S^n into the n-skeleton of S^k, which is just the basepoint. Hence the map is nullhomotopic. □

17.3 HELP and Whitehead's theorem

Whitehead's theorem is easiest to prove once we know that n-equivalences have good lifting properties against CW pairs.

Theorem 17.8. Homotopy extension and lifting property (HELP)

Let (X, A) be a relative CW complex of dimension $\leq n$, and let $e\colon Y \to Z$ be an n-equivalence between path-connected spaces. Given maps

$$f\colon X \to Z, \qquad g\colon A \to Y,$$

and a homotopy $h\colon A \times I \to Z$ such that

$$h(-,0) = f|_A, \qquad h(-,1) = e \circ g,$$

there exist a map $\widetilde{g}\colon X \to Y$ extending g and a homotopy $\widetilde{h}\colon X \times I \to Z$ extending h such that

$$\widetilde{h}(-,0) = f, \qquad \widetilde{h}(-,1) = e \circ \widetilde{g}.$$

Proof. We construct $\widetilde{g}$ and $\widetilde{h}$ by induction over the cells of $X - A$.
Start with $X^{(-1)} = A$ and set $\widetilde{g} = g$ and $\widetilde{h} = h$ on $A \times I$.
Suppose we have defined $\widetilde{g}$ on the $(k-1)$-skeleton of X and $\widetilde{h}$ on the corresponding cylinder, extending the required data, with $k \leq n$. Let $e^k \subset X - A$ be a k-cell with characteristic map $\chi\colon (D^k, S^{k-1}) \to (X^{(k)}, X^{(k-1)})$. The restriction of $\widetilde{g}$ to S^{k-1} is already defined, and the restriction of $\widetilde{h}$ to $S^{k-1} \times I$ is already defined, yielding

a commutative boundary diagram

$$\begin{tikzcd}
S^{k-1} \arrow[r, "\widetilde{g}\circ\chi"] \arrow[d, hook] & Y \arrow[d, "e"] \\
S^{k-1}\times I \arrow[r, "\widetilde{h}\circ(\chi\times\mathrm{id})"] \arrow[d, hook] & Z \arrow[d, equal] \\
D^k \arrow[r, "f\circ\chi"'] & Z
\end{tikzcd}$$

where the left vertical arrows are the inclusions $S^{k-1} \hookrightarrow S^{k-1} \times I$ at $t = 1$ and $S^{k-1} \hookrightarrow D^k$.
The obstruction to extending $\widetilde{g} \circ \chi$ to D^k so that $e \circ$ (extension) matches the endpoint $t = 1$ of the homotopy is measured by a relative homotopy class in $\pi_k(Z, Y)$. Since e is an n-equivalence and $k \leq n$, we have $\pi_k(Z, Y) = 0$. Concretely, the condition $\pi_{k-1}(Y) \to \pi_{k-1}(Z)$ is an isomorphism and $\pi_k(Y) \to \pi_k(Z)$ is surjective is equivalent to $\pi_k(Z, Y) = 0$ by the long exact sequence of the pair and the definition of relative homotopy groups.
Hence the boundary data extends across D^k. This produces an extension of $\widetilde{g}$ over the cell e^k and an extension of $\widetilde{h}$ over $e^k \times I$. Doing this for all k-cells not in A and using the CW topology gives the required global extensions $\widetilde{g}$ and $\widetilde{h}$. □

A clean consequence is that n-equivalences control homotopy classes of maps out of low-dimensional CW complexes.

Theorem 17.9. Maps out of CW complexes detect n-equivalences

Let $e\colon Y \to Z$ be an n-equivalence and let X be a CW complex.

1. If $\dim X < n$, then
$$e_*\colon [X, Y] \to [X, Z]$$
is a bijection.
2. If $\dim X = n$, then e_* is surjective.

Here $[X, W]$ denotes unbased homotopy classes of maps $X \to W$.

Proof. Surjectivity when $\dim X = n$ follows by applying HELP to the pair $(X, \varnothing)$ with h the constant homotopy and $f\colon X \to Z$ an arbitrary map. This produces $\widetilde{g}\colon X \to Y$ with $e \circ \widetilde{g} \simeq f$.
Injectivity when $\dim X < n$ follows similarly by applying HELP to the pair $(X \times I, X \times \partial I)$, again with a constant homotopy. If $e \circ f_0 \simeq e \circ f_1$ via a homotopy $H\colon X \times I \to Z$, then HELP lifts H to a homotopy in Y between f_0 and f_1. □

Now Whitehead's theorem is immediate.

Theorem 17.10. Whitehead theorem for CW complexes

Let $e\colon Y \to Z$ be a weak homotopy equivalence between CW complexes. Then e is a homotopy equivalence. More generally, if e is an n-equivalence and $\dim Y, \dim Z < n$, then e is a homotopy equivalence.

Proof. Assume first that e is a weak equivalence. Apply Theorem 17.9 with $X = Z$ and n arbitrarily large to conclude that
$$e_*\colon [Z, Y] \to [Z, Z]$$
is a bijection. Choose $f\colon Z \to Y$ such that $e \circ f \simeq \mathrm{id}_Z$.
Then $e \circ f \circ e \simeq e$ as maps $Y \to Z$. Apply Theorem 17.9 with $X = Y$ to see that
$$e_*\colon [Y, Y] \to [Y, Z]$$
is also a bijection. Since $e_*([f \circ e]) = [e] = e_*([\mathrm{id}_Y])$, injectivity gives $f \circ e \simeq \mathrm{id}_Y$. Thus f is a homotopy inverse of e.
For the n-equivalence statement, use Theorem 17.9 with $X = Z$ and $X = Y$ and note that the dimension bound puts us in the bijective range. □

17.4 Consequences for classification problems

The results above combine into a robust pattern.

Proposition 17.11. Weak equivalence is an equivalence relation on CW complexes

On the class of CW complexes, weak homotopy equivalence coincides with homotopy equivalence. In particular, it is symmetric and transitive on CW complexes.

Proof. If $f\colon X \to Y$ is a weak equivalence between CW complexes, Theorem 17.10 gives a homotopy inverse $g\colon Y \to X$. Then g is also a weak equivalence, giving symmetry. Transitivity follows because the composite of homotopy equivalences is a homotopy equivalence. □

Proposition 17.12. Uniqueness of CW approximations up to homotopy

If $p\colon Z \to X$ and $p'\colon Z' \to X$ are CW approximations of the same space, then Z and Z' are homotopy equivalent.

Proof. Consider the map $Z \xrightarrow{p} X$. Since p' is a weak equivalence and Z is a CW complex, Theorem 17.9 applied to $e = p'$ shows that

$$(p')_*\colon [Z, Z'] \to [Z, X]$$

is a bijection. Hence there exists $u\colon Z \to Z'$ with $p' \circ u \simeq p$. Similarly there exists $v\colon Z' \to Z$ with $p \circ v \simeq p'$. The composites $v \circ u$ and id_Z become equal in $[Z, X]$, so by injectivity again we have $v \circ u \simeq \mathrm{id}_Z$. Likewise $u \circ v \simeq \mathrm{id}_{Z'}$. □

Theorem 17.9 is also a practical test for building maps from partial information. If $e\colon Y \to Z$ is an n-equivalence and X is a CW complex with $\dim X \leq n$, then a map $X \to Z$ can be lifted up to homotopy to a map $X \to Y$. The dimension bound keeps the extension problems inside the range where relative homotopy groups vanish, and CW induction turns those vanishing statements into actual extensions.

18 Hurewicz theorem, basic computations, and the road ahead

18.1 Hurewicz homomorphism and the first nontrivial case

Fix a based space (X, x_0). For $n \geq 0$ choose a generator

$$i_n \in \widetilde{H}_n(S^n; \mathbb{Z})$$

characterized inductively by the suspension convention $\Sigma i_n = i_{n+1}$. (For $n > 0$ and path-connected X, one has $\widetilde{H}_n(X) \cong H_n(X)$.)

Definition 18.1. Hurewicz homomorphism

For $n \geq 1$, the **Hurewicz homomorphism** is

$$h\colon \pi_n(X, x_0) \longrightarrow \widetilde{H}_n(X; \mathbb{Z}), \qquad h([f]) := f_*(i_n),$$

where $f\colon (S^n, s_0) \to (X, x_0)$ represents the class $[f]$.

Lemma 18.2. Homomorphism property For $n \geq 1$, the map h is a group homomorphism.

Proof. Write $p\colon S^n \to S^n \vee S^n$ for the pinch map and $\nabla\colon X \vee X \to X$ for the fold map. For based maps $f, g\colon S^n \to X$, the sum $[f] + [g] \in \pi_n(X)$ is represented by the composite

$$S^n \xrightarrow{p} S^n \vee S^n \xrightarrow{f \vee g} X \vee X \xrightarrow{\nabla} X.$$

On reduced homology, $p_*(i_n) = i_n \oplus i_n$ under the identification $\widetilde{H}_n(S^n \vee S^n) \cong \widetilde{H}_n(S^n) \oplus \widetilde{H}_n(S^n)$, and ∇_* adds the two summands. Therefore

$$h([f] + [g]) = (\nabla \circ (f \vee g) \circ p)_*(i_n) = \nabla_*((f \vee g)_*(i_n \oplus i_n)) = f_*(i_n) + g_*(i_n) = h([f]) + h([g]).$$

□

Lemma 18.3. Naturality For a based map $\varphi\colon (X, x_0) \to (Y, y_0)$ and $n \geq 1$, the square

$$\begin{tikzcd} \pi_n(X, x_0) \arrow[r, "h"] \arrow[d, "\varphi_*"'] & \widetilde{H}_n(X) \arrow[d, "\varphi_*"] \\ \pi_n(Y, y_0) \arrow[r, "h"'] & \widetilde{H}_n(Y) \end{tikzcd}$$

commutes.

Proof. If $f\colon S^n \to X$ represents a class in $\pi_n(X)$, then

$$(\varphi_* \circ h)([f]) = \varphi_*(f_*(i_n)) = (\varphi \circ f)_*(i_n) = h(\varphi_*[f]).$$

□

The $n = 1$ case is already informative.

Proposition 18.4. The case $n = 1$

If X is path-connected, then $h\colon \pi_1(X, x_0) \to H_1(X; \mathbb{Z})$ is the abelianization map.

Proof. Singular 1-cycles are generated by loops, and boundaries force commutators to vanish, so $H_1(X)$ is the maximal abelian quotient of $\pi_1(X)$. The map h sends a loop class to its singular 1-cycle class, hence coincides with abelianization. □

18.2 Higher connectivity and relative Hurewicz

There is also a relative version. For a based pair (X, A, x_0) with $x_0 \in A \subset X$, define

$$h\colon \pi_n(X, A, x_0) \longrightarrow H_n(X, A; \mathbb{Z}), \qquad h([f]) := f_*([D^n, S^{n-1}]),$$

where $f\colon (D^n, S^{n-1}, s_0) \to (X, A, x_0)$ and $[D^n, S^{n-1}]$ is the preferred generator of $H_n(D^n, S^{n-1}) \cong \mathbb{Z}$.

The cleanest statement is for simply connected data, and a small correction is needed in general because π_1 acts on higher homotopy groups.

Theorem 18.5. Relative Hurewicz, simply connected version

Let (X, A) be an $(n-1)$-connected pair of path-connected spaces with $n \geq 2$, and assume A is simply connected. Then

$$H_i(X, A) = 0 \quad \text{for } i < n, \qquad h\colon \pi_n(X, A, x_0) \xrightarrow{\cong} H_n(X, A)$$

is an isomorphism.

Remark. When A is not simply connected, the kernel of the relative Hurewicz map contains elements of the form $[\gamma] \cdot [f] - [f]$ coming from the $\pi_1(A)$-action. One obtains a correct statement by quotienting $\pi_n(X, A)$ by the subgroup generated by such elements (and taking the normal closure when $n = 2$). In that form, Hurewicz becomes an isomorphism for $(n-1)$-connected pairs.

18.3 Applications and sample computations

The absolute Hurewicz theorem identifies the first nonzero homotopy group of a simply connected space with its first nonzero homology group.

Theorem 18.6. Hurewicz theorem

Let (X, x_0) be an $(n-1)$-connected based space, with $n \geq 2$. Then

$$\widetilde{H}_i(X) = 0 \quad \text{for } i < n, \qquad h\colon \pi_n(X, x_0) \xrightarrow{\cong} \widetilde{H}_n(X)$$

is an isomorphism.

Proof. Step 1: reduce to a CW complex with no cells below dimension n. Take a CW approximation $p\colon Z \to X$, so p is a weak homotopy equivalence and hence induces isomorphisms on π_n and on $\widetilde{H}_n$. By naturality of h (Lemma 18.3), it suffices to prove the theorem for Z.

Thus assume X is a CW complex and $(n-1)$-connected. By cellular approximation, the inclusion $X^{n-1} \hookrightarrow X$ induces isomorphisms on π_i for $i < n-1$ and a surjection on π_{n-1}. Since $\pi_i(X) = 0$ for $i \leq n-1$, it follows that $\pi_i(X^{n-1}) = 0$ for all $i \geq 0$. As X^{n-1} is a CW complex, Whitehead's theorem implies X^{n-1} is contractible. Collapsing X^{n-1} to a point produces a CW complex X' homotopy equivalent to X with a single 0-cell and no q-cells for $1 \leq q < n$. Replace X by this X'.

Step 2: identify the n-skeleton. With this CW structure, the n-skeleton X^n is a wedge of n-spheres, one summand for each n-cell:

$$X^n \cong \bigvee_{\alpha \in I} S^n_\alpha.$$

For $n \geq 2$, $\pi_n(X^n)$ is a free abelian group on the inclusions of the wedge summands, and $\widetilde{H}_n(X^n)$ is a free abelian group on the fundamental classes of those spheres. The Hurewicz map sends each inclusion generator to the corresponding homology generator, hence

$$h\colon \pi_n(X^n) \xrightarrow{\cong} \widetilde{H}_n(X^n)$$

is an isomorphism.

Step 3: compare X^n with X. Consider the pair (X, X^n). As a relative CW complex, it is obtained from X^n by attaching cells of dimension $\geq n+1$. Therefore (X, X^n) is n-connected, meaning

$$\pi_k(X, X^n) = 0 \quad \text{for } k \leq n.$$

One way to see this is to represent an element of $\pi_k(X, X^n)$ by a map of pairs

$$f\colon (D^k, S^{k-1}) \to (X, X^n),$$

note that the compact image $f(D^k)$ meets only finitely many cells, and then homotope f rel S^{k-1} off each cell of dimension $> n$ one at a time since $k \leq n$. After removing all such cells, the map lands in X^n, hence is trivial in the relative group.

The long exact sequence of the pair then gives that the inclusion $j\colon X^n \hookrightarrow X$ induces an isomorphism

$$j_*\colon \pi_n(X^n) \xrightarrow{\cong} \pi_n(X).$$

The same skeletal argument applies to homology. Since the relative cellular chain complex $C_*(X, X^n)$ has no generators in degrees $\leq n$, we have

$$H_k(X, X^n) = 0 \quad \text{for } k \leq n,$$

and the long exact sequence of the pair implies that

$$j_*\colon \widetilde{H}_n(X^n) \xrightarrow{\cong} \widetilde{H}_n(X).$$

Step 4: conclude by naturality. The inclusions fit into the commutative square

$$\begin{tikzcd}
\pi_n(X^n) \arrow[r, "h"] \arrow[d, "j_*"'] & \widetilde{H}_n(X^n) \arrow[d, "j_*"] \\
\pi_n(X) \arrow[r, "h"'] & \widetilde{H}_n(X),
\end{tikzcd}$$

and the three other sides are isomorphisms. Hence $h\colon \pi_n(X) \to \widetilde{H}_n(X)$ is an isomorphism.

Finally, since X has no cells in degrees $1, \ldots, n-1$, its reduced homology vanishes in those degrees, so $\widetilde{H}_i(X) = 0$ for $i < n$. □

A few quick consequences are worth recording in a form that you can use immediately.

Corollary 18.7 (The first nonzero homotopy group is homological)**.** Let X be simply connected and let $n \geq 2$ be minimal such that $\widetilde{H}_n(X) \neq 0$. Then $\pi_i(X) = 0$ for $i < n$ and

$$\pi_n(X) \cong \widetilde{H}_n(X)$$

via the Hurewicz map.

Proof. If $\widetilde{H}_i(X) = 0$ for $i < n$, then by the Hurewicz theorem applied to the Postnikov truncation viewpoint one concludes $\pi_i(X) = 0$ for $i < n$ and h is an isomorphism in degree n. Equivalently, one can apply Theorem 18.6 directly once $(n-1)$-connectedness has been established by induction on n using the exact sequences of pairs associated to skeleta. □

Corollary 18.8 (Spheres in their own dimension)**.** For $n \geq 2$,

$$\pi_n(S^n) \cong H_n(S^n) \cong \mathbb{Z}.$$

A generator is represented by the identity map $\mathrm{id}\colon S^n \to S^n$.

Proof. The sphere S^n is $(n-1)$-connected and $\widetilde{H}_n(S^n) \cong \mathbb{Z}$. The Hurewicz map sends $[\mathrm{id}]$ to i_n, which generates $\widetilde{H}_n(S^n)$, so it is an isomorphism. □

Example. Let $\mathbb{C}P^m$ be complex projective space with its standard CW structure having one cell in each even dimension $0, 2, 4, \ldots, 2m$. It is simply connected, hence 1-connected. Therefore $h\colon \pi_2(\mathbb{C}P^m) \to H_2(\mathbb{C}P^m)$ is an isomorphism. Since $H_2(\mathbb{C}P^m) \cong \mathbb{Z}$, we obtain $\pi_2(\mathbb{C}P^m) \cong \mathbb{Z}$.

The theorem also explains why attaching maps control the *first* place where homology can be nontrivial. This shows up concretely in the cellular boundary maps.

Proposition 18.9. Cellular boundaries from attaching maps

Let X be a CW complex and write $C_n(X)$ for the cellular chain groups. For $n > 1$, the cellular boundary

$$d_n\colon C_n(X) \to C_{n-1}(X)$$

is determined by the attaching maps of the n-cells by a degree formula: if e^n_α is an n-cell and e^{n-1}_β is an $(n-1)$-cell, then the coefficient of e^{n-1}_β in $d_n(e^n_\alpha)$ is the degree of the composite

$$S^{n-1}_\alpha \xrightarrow{\varphi_\alpha} X^{n-1} \longrightarrow X^{n-1}/(X^{n-1} - e^{n-1}_\beta) \cong S^{n-1}_\beta,$$

where φ_α is the attaching map of e^n_α.

Proof. The cellular boundary is the connecting map in the long exact sequence of the triple (X^n, X^{n-1}, X^{n-2}). Using the identifications

$$H_n(X^n, X^{n-1}) \cong \bigoplus_\alpha H_n(D^n_\alpha, S^{n-1}_\alpha) \cong \bigoplus_\alpha \mathbb{Z}, \qquad H_{n-1}(X^{n-1}, X^{n-2}) \cong \bigoplus_\beta \mathbb{Z},$$

one tracks a generator through the boundary map and then projects to the β-summand by collapsing complements of the cell e^{n-1}_β. The resulting map $S^{n-1}_\alpha \to S^{n-1}_\beta$ has an integer degree, and these degrees are exactly the matrix entries of d_n in the cellular bases. □

This degree description, together with Hurewicz, makes the "first nontrivial information" in π_* visible in the same combinatorial data that computes H_*.